

HYDRA MAPS, NUMENS, AND (p, q) -ADIC ANALYSIS: A CRITICAL EXPOSITORY EDITION OF THE MATHEMATICAL WORK OF MAXWELL C. SIEGEL

MATHEMATICAL WORKS BY MAXWELL C. SIEGEL

ABSTRACT. This critical expository edition reconstructs the mathematical programme of Maxwell C. Siegel from his author-controlled and arXiv corpus. Its organizing object is a residue-class piecewise-affine arithmetic dynamical system, called a Hydra map, together with the digit-recursive function called its numen. The defining recursion is derived from affine word composition, its extension from non-negative integers to p -adic integers is separated from the place-dependent convergence hypotheses, and the Correspondence Principle is stated with all of the hypotheses used in the latest source formulation. The analytic developments are then placed around that core: singularity conservation and dreamcatchers; Dirichlet series, Mellin–Perron inversion and the contour formulation of weak Collatz; (p, q) -adic Fourier analysis and Wiener–Tauberian cyclicity; rising-continuous functions and variable topologies of convergence; and the later algebra of products of F-series.

This is an edition of Siegel’s corpus, not a new joint research article. Siegel retains authorship of the source mathematics and of his terms *Hydra map*, *numen*, *dreamcatcher*, and *rising-continuity*. Source forms are preserved; any running notation is placed beside them by an explicit crosswalk. Published, current-preprint, superseded, and withdrawn witnesses remain distinct, and source defects are recorded before a corrected statement is proved.

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The critical apparatus, typed reconstructions, counterexamples, and edition-supplied proofs are components of this edition and are not attributed to Siegel as source text.

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1. EDITORIAL METHOD, ATTRIBUTION, AND MATHEMATICAL STATUS

1.1. What this edition is. The primary sources are the mathematical works written by Maxwell Charles Siegel and distributed through arXiv or through his author-controlled site.

The purpose of the present text is narrower and more exact. It fixes a canonical witness for each work, states how older versions relate to it, places any running notation beside the typed source notation through an explicit crosswalk, and arranges the results by mathematical dependence rather than by upload date. The result is a modern recasting of the corpus: neither a facsimile anthology nor a survey that replaces formulas by descriptions.

1.2. Four statement statuses. The edition uses the following four dispositions. A source-issue or status environment states its disposition locally; Chapter 10 records the source identity and disposition of each theorem-like item admitted to its result table. A theorem-style heading by itself neither assigns authorship nor certifies that a source proof has been completed.

- (i) A *reconstructed theorem* is a statement in Siegel's corpus whose source hypotheses and conclusion are preserved and placed beside any running notation through an explicit typed crosswalk, and whose proof is reproduced or reconstructed at the level needed by the subsequent chain.
- (ii) A *source theorem* is stated as a theorem of the named source. If this edition gives only its proof mechanism, that limitation is said explicitly; the label does not mean that this edition supplies an independent proof of every auxiliary assertion in the source.
- (iii) A *source issue* records a typing error, a changed convention, an omitted hypothesis, a withdrawn paper, or a point at which the source itself supplies only a sketch. The issue is not repaired by pretending that the altered statement was what the source always said.
- (iv) An *editorial proposition* is a formal consequence proved in this edition in order to join two source formulations. It is marked as editorial and is not attributed to Siegel as an original theorem.

Extraction is not verification. Conversely, a typographical defect does not by itself refute the intended theorem. The edition therefore distinguishes the exact source string, the mathematically typed reconstruction, and the status of the proof.

1.3. Notation policy. The source convention $x \stackrel{a}{\equiv} y$ is rendered as $x \equiv y \pmod{a}$. The source symbol $[x]_a$ is retained for the standard representative when it is part of a digit formula. Iverson brackets are written $\mathbf{1}_P$. Composition is read from right to left. Thus, for a word $\mathbf{j} = (j_0, \dots, j_{m-1})$, the ordered branch composition used below is

$$H_{\mathbf{j}} = H_{j_0} \circ H_{j_1} \circ \dots \circ H_{j_{m-1}}.$$

The source sometimes indexes a word from 1 and sometimes from 0; the edition uses 0-based digit indices and states the convention whenever an order-sensitive identity is used.

The name (p, q) -*adic analysis* is retained as Siegel’s name for analysis of functions whose domain carries a p -adic topology and whose values lie in a field with a distinct q -adic (or, more generally, place-dependent) topology. This is not the standard adjective “adic” applied to a single topological ring; the domain and codomain places must always be displayed.

The name *rising-continuity* is retained for the source condition

$$f(z) = \lim_{n \rightarrow \infty} f([z]_{p^n}),$$

where the limit is taken in the specified codomain. In standard language it is continuity along the canonical truncation sequence, not automatically topological continuity with respect to every sequence converging to z . The distinction is used in section 10.

1.4. Version precedence. The latest non-withdrawn arXiv version controls a work unless an author-controlled later witness explicitly corrects it. A later technical manual controls a definition when it declares that earlier definitions are inconsistent and fixes a forward convention. Under this rule the 2026 Hydra/Numen manual controls the basic definitions of Hydra maps, numens, and the general Correspondence Principle; the 2022 dissertation remains the largest source for the analytic development. A withdrawn record remains in the historical corpus but cannot control a theorem contradicted by its own withdrawal notice.

Source issue 1.1 (Canonical does not mean infallible). The word *canonical* refers to source identity, version precedence, and the stable presentation adopted by this edition. It does not convert an author claim into a proved theorem. In particular, the 2026 paper labels its Correspondence Principle proof a sketch; this edition preserves that fact and separates the fully checked affine-word calculation from the remaining dynamical implications.

1.5. The exactness standard. Each comparison is attached to an explicit object. For example, the chain used for a characteristic distribution is the following typed sequence, not the sentence “Fourier analysis detects periodic points.” Starting from a centered admissible pre-Hydra

$$\mathcal{H} = (K, \Lambda, (H_j)_{j \in \mathcal{A}_p}) \in \{(K, \Lambda, \mathbf{H}) : H_j \in \text{Aff}_{\mathbb{F}}(K)\},$$

affine word multiplication gives

$$(M_H, X_H) \in \text{Map}(\Sigma_p^*, \text{GL}_{\mathbb{F}}(K)) \times \text{Map}(\Sigma_p^*, K).$$

Centeredness proves that the second component descends along $\text{Dig}_p : \Sigma_p^* \rightarrow \mathbb{N}_0$ to $X_H : \mathbb{N}_0 \rightarrow K$. At a place ℓ let

$$D_{H,\ell} = \{z \in \mathbb{Z}_p^{\text{dig}} : (X_H([z]_{p^N}))_{N \geq 0} \text{ converges in } K_\ell\}.$$

The truncation limit is a map

$$X_{H,\ell} : D_{H,\ell} \longrightarrow K_\ell.$$

These measurability assertions are automatic. Each truncation map $S_N(z) = X_H([z]_{p^N})$ is locally constant. Writing $d_\ell(x, y) = |x - y|_\ell$, completeness of K_ℓ gives

$$D_{H,\ell} = \bigcap_{r \geq 1} \bigcup_{N_0 \geq 0} \bigcap_{m, n \geq N_0} \{z : d_\ell(S_m(z), S_n(z)) < 1/r\},$$

so $D_{H,\ell}$ is Borel. On that Borel subspace, $X_{H,\ell} = \lim_N S_N$ is Borel as the pointwise limit of metric-space-valued Borel maps. Thus only conullness remains a hypothesis. If $m_p(D_{H,\ell}) = 1$, then the pushforward is the Borel probability

$$\nu_{H,\ell} = (X_{H,\ell})_*(m_p|_{D_{H,\ell}}) \in \text{Prob}(K_\ell).$$

Its Fourier–Stieltjes transform is the bounded continuous function

$$\widehat{\nu}_{H,\ell} : \widehat{K}_\ell \longrightarrow \mathbb{C}, \quad \widehat{\nu}_{H,\ell}(\gamma) = \int_{K_\ell} \gamma(-x) d\nu_{H,\ell}(x).$$

If a cyclic-translate space is subsequently considered, translations are indexed by the character group, not by K_ℓ : for $\eta \in \widehat{K_\ell}$, $(\tau_\eta f)(\gamma) = f(\gamma + \eta)$, and the asserted closure must name its ambient function space and norm. No self-duality of K_ℓ is being silently chosen here. Likewise, the numen is not merely “a function encoding the dynamics”: it is the translation term of an ordered affine branch composition and satisfies a digit recursion with a specified initial condition. This level of explicitness is the governing editorial standard throughout the edition.

2. THE CANONICAL SIEGEL CORPUS

2.1. Logical works and version witnesses. The primary research corpus occupies ten arXiv lineages as of 23 August 2026. A lineage is a bibliographic container, not necessarily a homogeneous sequence of minor revisions. In particular, 2007.15936v1, 2111.07883v1, and 2208.11082v1 differ substantially from the later versions filed under the same identifiers and are retained in the apparatus as distinct substantive witnesses. A withdrawal stub is a bibliographic event, not a replacement source for the withdrawn mathematics. The table records the controlling witness used by this edition.

arXiv	Work	Witness	Editorial status
1909.09733	<i>Conservation of Singularities in Functional Equations Associated to Collatz-Type Dynamical Systems; or, Dreamcatchers for Hydra Maps</i>	v3	Current historical source for dreamcatchers and singularity conservation.
2002.12762	<i>Fourier Analysis of the Parity-Vector Parameterization of the Generalized Collatz $px+1$ Maps</i>	v1	Withdrawn in v2 as obsolete; retained as a superseded stage, not controlling.
2007.15936	<i>The Collatz Conjecture & Non-Archimedean Spectral Theory: Part I</i>	v3	Published core for the shortened $qx+1$ numen and Correspondence Principle.
2111.07882	<i>Functional Equations Associated to Collatz-Type Maps on Integer Rings of Algebraic Number Fields</i>	v1	Withdrawn in v2 for an error in the section on quotient rings; historical only.
2111.07883	<i>Non-Archimedean Spectral Theory, Part I.5: How to Write the (Weak) Collatz Conjecture as a Contour Integral</i>	v2	Current source for the Dirichlet/Mellin/contour formulation.
2208.11082	<i>Non-Archimedean Spectral Theory, Part II: (p, q)-Adic Fourier Analysis and Wiener’s Tauberian Theorem</i>	v2	Current and published analytic source.
2307.00436	<i>Infinite Series Whose Topology of Convergence Varies From Point to Point</i>	v1	Published source for frames, rising-continuity, and point-dependent convergence.
2412.02902	<i>(p, q)-Adic Analysis and the Collatz Conjecture</i>	v1	ArXiv witness of the 2022 dissertation; principal synthetic source.
2507.13358	<i>Algebras of p-Adic Distributions Induced by Pointwise Products of F-Series</i>	v1	Current extension to products, frames, distributions, and breakdown varieties.

arXiv	Work	Witness	Editorial status
2601.17030	<i>The Hydra Map and Numen Formalisms for Collatz-Type Problems</i>	v3	Latest terminology and foundational manual; v3 corrects Proposition 7.6.

The author site also supplies mathematically substantive version witnesses. Its numbering must not be confused with the numbering of the papers:

Witness	Author-site title	Canonical relation
Web install- ment I	<i>The Numen of a Hydra Map</i>	Expository/version witness to paper Part I, 2007.15936v3; it does not control a conflicting formula.
Web install- ment II	<i>The Correspondence Principle</i>	Continuation of Web I and likewise a witness to paper Part I. It is <i>not</i> paper Part II. Its Dirichlet–Perron coda is a precursor to 2111.07883v2.
Web install- ment III	<i>(p, q)-adic Fourier Analysis & Tauberian Spectral Theory</i>	Evolving exposition to be collated against paper Part II, 2208.11082v2. A later webpage modification date alone gives no theorem precedence.
Web install- ment IV	<i>A Fourier Analysis of the Numen</i>	Site-only application/exposition, collated chiefly against the one-dimensional dissertation material.

Web installments I and II were recovered as current WordPress snapshots during the audit; the public API does not expose their revision histories. The site also contains a March 2025 talk note on arithmetic properties of F-series and companion expository material. Those witnesses are part of the author-controlled corpus but do not supersede a later arXiv theorem merely because their prose is newer.

The authority order used when witnesses conflict is

2601 definitions $\longrightarrow$ 2307 F-series/frame definitions
 $\longrightarrow$ latest specialized paper $\longrightarrow$ dissertation
 $\longrightarrow$ web exposition $\longrightarrow$ historical or withdrawn witness.

This order is claim-specific: it does not erase unique material in an older witness, and it does not turn a later unproved claim into a theorem.

2.2. Mathematical dependency order. The upload chronology obscures the mathematical order. The canonical reading order used in this edition is:

residue-class affine dynamics $\longrightarrow$ word composition $\longrightarrow$ numen recursion
 $\longrightarrow$ p -adic extension $\longrightarrow$ Correspondence Principle,
 set-series/permutation operators $\longrightarrow$ dreamcatchers $\longrightarrow$ singularity conservation,
 Haar integration and Fourier series $\longrightarrow$ Wiener–Tauberian cyclicity $\longrightarrow$ spectral reformulation,
 rising-continuity and frames $\longrightarrow$ F-series products $\longrightarrow$ distribution algebras and breakdown loci.

The contour/Mellin formulation branches from the one-dimensional numen before returning to the spectral programme through its Dirichlet series.

2.3. Supersession and withdrawal rules. The 2026 manual explicitly warns that earlier definitions of Hydra maps may be inconsistent with its definitions. Accordingly, this edition does not splice the 2019, dissertation, and 2026 definitions into an artificial single quotation. It gives the 2026 object first and then records the older specializations.

The withdrawn 2111.07882 paper is not used to certify the number-ring generalization. The later 2601.17030 manual supplies a new formulation on $\mathcal{O}_K$ and is treated independently. Similarly, 2002.12762v1 is a historical predecessor to the numen formulation, not a coequal canonical statement of the current theory.

Status record 2.1 (Coverage of the edition). All ten arXiv lineages, the materially distinct earlier versions, all four numbered Web installments, and the three public dissertation witnesses are represented in the corpus register or apparatus. The mathematical exposition gives full definitions and the principal theorem chains of the non-withdrawn research strands. The two withdrawn works are described only to the point needed to identify what was superseded or invalidated. Author-site posts are used as explicit version or expository witnesses, not as silent replacements for formal proofs. General teaching notes lie outside this research edition; the specialized p -adic background notes are apparatus rather than independent core works.

3. ARITHMETIC DYNAMICS BEFORE THE ENCODING

The numen formalism does not begin with a transform. It begins with an ordinary discrete dynamical system and with the elementary distinction between a bounded recurrent trajectory and a trajectory which escapes every bounded subset. This section fixes those notions before any coding is introduced.

3.1. Forward orbits and eventual repetition. Let Y be a set and let $T : Y \rightarrow Y$. Write $T^{\circ 0} = \text{id}_Y$ and $T^{\circ(n+1)} = T \circ T^{\circ n}$.

Definition 3.1. For $y \in Y$, the *forward orbit* is

$$\mathcal{O}_T^+(y) = (T^{\circ n}(y))_{n \geq 0}.$$

The point y is *periodic* if $T^{\circ m}(y) = y$ for some $m \geq 1$, and *preperiodic* if $T^{\circ a}(y)$ is periodic for some $a \geq 0$. It is *strictly preperiodic* if it is preperiodic but not periodic. The least positive period of a periodic point is its *exact period*.

Lemma 3.2 (eventual repetition). *For $y \in Y$, the following are equivalent.*

- (i) y is preperiodic;
- (ii) $T^{\circ m}(y) = T^{\circ n}(y)$ for some $0 \leq m < n$;
- (iii) the set $\{T^{\circ n}(y) : n \geq 0\}$ is finite.

Proof. If $T^{\circ m}(y) = T^{\circ n}(y)$ with $m < n$, then $T^{\circ m}(y)$ is periodic of period dividing $n - m$. This proves (ii) $\Rightarrow$ (i). A preperiodic orbit consists of a finite initial tail followed by a finite cycle, proving (i) $\Rightarrow$ (iii). Finally, an infinite sequence taking values in a finite set repeats a value, so (iii) $\Rightarrow$ (ii). $\square$

The relation used in Siegel's latest manual is

$$x \sim_T y \iff T^{\circ m}(x) = T^{\circ n}(y) \text{ for some } m, n \geq 0.$$

It is an equivalence relation: symmetry and reflexivity are immediate, and transitivity follows by iterating the two common-forward-image equalities until their middle iterates agree. Its equivalence classes are the weak connected components of the functional graph of T .

3.2. The compact-bounded dichotomy. Suppose now that Y is contained in a normed finite-dimensional real vector space V . In the arithmetic applications the unconditional locally finite model is the image of the ring of integers of a number field under its full Minkowski embedding. The image under one archimedean embedding need not be locally finite when the field has degree greater than one. A subset $Y \subset V$ is *locally finite* when $Y \cap B$ is finite for every bounded $B \subset V$.

Proposition 3.3 (the full Minkowski image is a lattice). *Let K be a number field of degree $d = r_1 + 2r_2$. Choose its real embeddings $\sigma_1, \dots, \sigma_{r_1}$ and one embedding τ_j from each pair of complex-conjugate embeddings. Then*

$$\iota_K : K \longrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^d, \quad x \longmapsto (\sigma_1(x), \dots, \sigma_{r_1}(x), \tau_1(x), \dots, \tau_{r_2}(x))$$

maps $\mathcal{O}_K$ onto a full lattice. In particular, $\iota_K(\mathcal{O}_K)$ is locally finite.

Proof. Choose an integral basis $\omega_1, \dots, \omega_d$ of $\mathcal{O}_K$. The matrix whose columns are the images of these basis elements under all d embeddings has nonzero determinant: its squared determinant is the nonzero field discriminant of the basis. Replacing each conjugate pair of complex coordinates by its real and imaginary parts is an invertible real-linear change of coordinates. Hence the real-linear map

$$A : \mathbb{R}^d \longrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}, \quad e_j \longmapsto \iota_K(\omega_j),$$

is an isomorphism, and $\iota_K(\mathcal{O}_K) = A(\mathbb{Z}^d)$. If B is bounded, then $A^{-1}(B)$ is bounded and therefore meets $\mathbb{Z}^d$ in finitely many points. Thus B meets $\iota_K(\mathcal{O}_K)$ in finitely many points. $\square$

Proposition 3.4 (preperiodic or escaping). *Let $Y \subset V$ be locally finite and let $T : Y \rightarrow Y$. For every $y \in Y$ exactly one of the following occurs:*

- (i) y is preperiodic;
- (ii) $\|T^{on}(y)\| \rightarrow \infty$ as $n \rightarrow \infty$.

Proof. The alternatives are disjoint because a preperiodic orbit is finite. If y is not preperiodic, theorem 3.2 says that all members of its forward orbit are distinct. Fix $R > 0$. The set $Y \cap \{v : \|v\| \leq R\}$ is finite, so only finitely many of those distinct orbit points can lie in it. Hence there is N_R such that $\|T^{on}(y)\| > R$ for all $n \geq N_R$. Since R was arbitrary, the norms tend to infinity. $\square$

Source issue 3.5 (the discreteness hypothesis in the 2026 statement). The source states theorem 3.4 for a uniformly discrete subset of an arbitrary valued field and infers that every bounded ball meets that subset in finitely many points. Uniform separation alone does not imply this in a non-proper metric space. The proof is valid under the locally finite hypothesis above; in particular it is valid for the full Minkowski lattice. It is not automatically valid after projection to one archimedean embedding, where bounded sets may contain infinitely many algebraic integers. This edition does not silently retain either the more general but insufficient source hypothesis or that single-embedding specialization without an added discreteness assumption.

3.3. The Collatz and shortened $qx + 1$ maps. The classical Collatz map on $\mathbb{Z}$ is

$$C(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ 3n + 1, & n \equiv 1 \pmod{2}. \end{cases}$$

Its odd branch is followed by an even value. Compressing that forced even step gives the shortened map

$$T_3(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n + 1)/2, & n \equiv 1 \pmod{2}. \end{cases}$$

More generally, for an odd integer q , put

$$T_q(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (qn + 1)/2, & n \equiv 1 \pmod{2}. \end{cases} \tag{1}$$

The restriction to odd q is structural: it makes the second expression an integer exactly on the odd residue class. The parity itinerary of n is

$$\varepsilon_k(n) = [T_q^{\circ k}(n)]_2 \in \{0, 1\}, \quad k \geq 0,$$

where $[a]_2$ denotes the standard residue representative. For a finite orbit segment of length m we package the itinerary as

$$\varepsilon_m(n) = (\varepsilon_0(n), \dots, \varepsilon_{m-1}(n)).$$

There are two orders in the corpus which must not be conflated. The entries of an itinerary are chronological: the branch indexed by ε_0 is applied first. The numen word $\mathbf{j} = (j_0, \dots, j_{m-1})$ is used in the ordered affine composition

$$H_{\mathbf{j}} = H_{j_0} \circ \dots \circ H_{j_{m-1}},$$

so the rightmost branch acts first. Consequently the numen word attached to a chronological orbit segment is its reversal unless a statement explicitly adopts the opposite composition convention. Every order-sensitive formula below is derived rather than inferred from prose.

3.4. What the coding is required to preserve. The goal is not merely to replace an integer orbit by a digit string. A usable coding must simultaneously retain:

- (a) the residue class which makes each branch the genuine branch of the piecewise map;
- (b) the linear part of a branch word;
- (c) the translation part of that branch word;
- (d) convergence at a specified completion when an infinite digit string is substituted; and
- (e) enough branch correctness to turn a fixed point of a formal branch composition back into a periodic point of the original map.

Hydra maps provide (a), the pair (M_H, X_H) provides (b)–(c), the extension lemma provides (d), and integrality supplies the principal mechanism for (e).

4. HYDRA MAPS AND THEIR AFFINE WORD CALCULUS

The 2026 manual is the controlling source for the word *Hydra*. Earlier papers use related residue-class maps, but not always the same formal data. We first type the current object, then isolate an internal inconsistency in the source definition and state the convention used in the rest of the edition.

4.1. Global-field data. Let $\mathbb{F}$ be either $\mathbb{Q}$ or a rational function field $\mathbb{F}_q(T)$, and let K be a finite extension of $\mathbb{F}$. Write $\mathcal{O}_K$ for the corresponding ring of integers and $d = [K : \mathbb{F}]$. If ℓ is a place of K , let K_ℓ be the completion and $|\cdot|_\ell$ a fixed representative absolute value. For $A \in \text{End}_{\mathbb{F}}(K)$ put

$$\|A\|_\ell = \sup_{x \in K, x \neq 0} \frac{|Ax|_\ell}{|x|_\ell} \in [0, \infty]. \quad (2)$$

This quantity is finite exactly when A is continuous for the ℓ -adic topology, in which case A extends uniquely to a bounded operator on K_ℓ . All placewise operator statements below include this finiteness hypothesis. A general $\mathbb{F}$ -linear endomorphism need not be continuous at a chosen single place: for example, a field automorphism can interchange two places over the same place of $\mathbb{F}$. Continuity is automatic when the map is already typed on the selected local factor, as for scalar multiplication.

Let $\Lambda \subsetneq \mathcal{O}_K$ be a nonzero ideal. Its quotient is finite; set

$$p = |\mathcal{O}_K/\Lambda| \geq 2.$$

Choose an enumeration of the cosets, with the zero coset indexed by 0:

$$\mathcal{O}_K/\Lambda = \{\Lambda_0, \dots, \Lambda_{p-1}\}, \quad \Lambda_0 = \Lambda.$$

This enumeration is additional data. It supplies a bijection from residue classes to the digit alphabet $\mathcal{A}_p = \{0, \dots, p-1\}$; there is no canonical digit order for a general ideal quotient.

Definition 4.1 (affine $\mathbb{F}$ -automorphism). The affine group used by Siegel is the group of affine $\mathbb{F}$ -linear automorphisms of the affine space underlying K :

$$\text{Aff}_{\mathbb{F}}(K) = \text{GL}_{\mathbb{F}}(K) \ltimes K = \text{End}_{\mathbb{F}}(K)^{\times} \ltimes K.$$

An element (r, c) acts by $x \mapsto rx + c$, where $r \in \text{End}_{\mathbb{F}}(K)^{\times}$ and $c \in K$. Its multiplication law is

$$(r, c)(s, d) = (r \circ s, r(d) + c), \quad (3)$$

and its inverse is $(r, c)^{-1} = (r^{-1}, -r^{-1}c)$.

The use of $\text{End}_{\mathbb{F}}(K)$ rather than scalar multiplication by elements of K is essential. When $[K : \mathbb{F}] > 1$, an $\mathbb{F}$ -linear map need not be K -linear. The controlling source calls these maps “affine lattice morphisms.” That name is not used as a type assertion here: an arbitrary element of $\text{Aff}_{\mathbb{F}}(K)$ need not preserve $\mathcal{O}_K$, Λ , or any lattice in K . Lattice preservation is always stated separately when it is needed.

Definition 4.2 (pre-Hydra). A *pre-Hydra* is a tuple

$$\mathcal{H} = (K, \Lambda, (H_j)_{j \in \mathcal{A}_p}), \quad H_j = (r_j, c_j) \in \text{Aff}_{\mathbb{F}}(K).$$

The H_j are its branches. The associated piecewise expression, whenever it is integer-valued on the selected residue classes, is

$$H(x) = \sum_{j=0}^{p-1} \mathbf{1}_{x \in \Lambda_j} (r_j x + c_j), \quad x \in \mathcal{O}_K. \quad (4)$$

Definition 4.3 (admissible and exact-integral). A pre-Hydra is *admissible* when

$$H_j(\Lambda_j) \subseteq \mathcal{O}_K \quad (0 \leq j < p). \quad (5)$$

It is *exact-integral* when, more strongly,

$$H_j(x) \in \mathcal{O}_K \iff x \in \Lambda_j \quad (x \in \mathcal{O}_K, 0 \leq j < p). \quad (6)$$

This is the domain used in Definition 4.5 of the controlling source. No field-wide predicate over $x \in K$ is substituted for it. An admissible pre-Hydra defines a map $H : \mathcal{O}_K \rightarrow \mathcal{O}_K$ by equation (4); this is the Hydra map used below.

Source issue 4.4 (the two integrality clauses in the manual). In Definition 4.3 of arXiv:2601.17030v3, the displayed Hydra is required to satisfy $r_j z + c_j \in \mathcal{O}_K$ for every $z \in \mathcal{O}_K$ and every j . Definition 4.5 then calls a Hydra integral when $H_j(z) \in \mathcal{O}_K$ *if and only if* $z \in \Lambda_j$, with $z \in \mathcal{O}_K$. For $p \geq 2$ these conditions cannot both hold: after fixing j , choose $z \in \mathcal{O}_K \setminus \Lambda_j$. The first clause puts $H_j(z)$ in $\mathcal{O}_K$, while the second forbids it. The first clause also fails for the manual’s own principal example T_3 , whose half-integral branch formulas are integer-valued only on their assigned parity classes. The minimal condition needed to define $H : \mathcal{O}_K \rightarrow \mathcal{O}_K$ is equation (5); the iff condition equation (6) is the stronger branch-correctness property used in the Correspondence Principle. These two source clauses and their common integral domain are preserved rather than replaced by a field-wide predicate.

Definition 4.5 (proper and centered). A Hydra is

- (i) *proper* if $I - r_0 \in \text{End}_{\mathbb{F}}(K)$ is invertible;
- (ii) *centered* if $H_0(0) = 0$, equivalently $c_0 = 0$.

Properness gives a unique fixed point for the affine branch labelled 0, namely $(I - r_0)^{-1}c_0$, but neither makes that point zero nor makes a translation by it preserve the branch labelled 0. The residue-class permutation must also be tracked.

Proposition 4.6 (translation conjugacy and the zero-class branch). *Let $a \in \mathcal{O}_K$, let $\tau_a(x) = x + a$, and define $H^{(a)} = \tau_a^{-1} \circ H \circ \tau_a : \mathcal{O}_K \rightarrow \mathcal{O}_K$. Translation induces the permutation π_a of $\mathcal{A}_p$ determined uniquely by*

$$\Lambda_{\pi_a(j)} = \Lambda_j + a. \quad (7)$$

For $x \in \Lambda_j$, the branch of $H^{(a)}$ on Λ_j is

$$H_j^{(a)}(x) = \tau_a^{-1} H_{\pi_a(j)} \tau_a(x) = r_{\pi_a(j)} x + (r_{\pi_a(j)} a + c_{\pi_a(j)} - a). \quad (8)$$

Admissibility is preserved by this conjugacy, and exact-integrality is preserved when H is exact-integral. In the label-dependent terminology of the definition, $H^{(a)}$ is proper if and only if $I - r_{\pi_a(0)}$ is invertible. The branch of $H^{(a)}$ on the zero class Λ is centered if and only if

$$H_{\pi_a(0)}(a) = a. \quad (9)$$

In particular, if H is proper and $a_0 = (I - r_0)^{-1} c_0 \in \mathcal{O}_K$, then conjugation by τ_{a_0} is guaranteed to center the zero-class branch when $a_0 \in \Lambda$. If $a_0 \notin \Lambda$, the equation $H_0(a_0) = a_0$ alone gives no such conclusion; the exact additional condition is $H_{\pi_{a_0}(0)}(a_0) = a_0$.

Proof. Addition by a is a bijection of $\mathcal{O}_K$ carrying each coset Λ_j onto the coset $\Lambda_j + a$. Since the displayed enumeration contains every coset exactly once, equation (7) defines a permutation, with inverse π_{-a} . If $x \in \Lambda_j$, then $x + a \in \Lambda_{\pi_a(j)}$, so the piecewise definition of H selects $H_{\pi_a(j)}$. Applying τ_a^{-1} and expanding the affine map proves equation (8). If $x \in \Lambda_j$, admissibility of H puts $H_{\pi_a(j)}(x + a)$ in $\mathcal{O}_K$, so subtracting $a \in \mathcal{O}_K$ proves admissibility of $H^{(a)}$. If H is exact-integral, then, for $x \in \mathcal{O}_K$,

$$H_j^{(a)}(x) \in \mathcal{O}_K \iff H_{\pi_a(j)}(x + a) \in \mathcal{O}_K \iff x + a \in \Lambda_{\pi_a(j)} \iff x \in \Lambda_j,$$

which proves exact-integrality. At $j = 0$, centeredness means precisely $H_0^{(a)}(0) = 0$; by equation (8), this is equivalent to $r_{\pi_a(0)} a + c_{\pi_a(0)} - a = 0$, which is equation (9). The linear part of that same zero-class branch is $r_{\pi_a(0)}$, proving the properness criterion. Finally, $\pi_{a_0}(0) = 0$ exactly when $\Lambda + a_0 = \Lambda$, equivalently $a_0 \in \Lambda$; in that case the fixed-point equation for H_0 supplies the required equality. $\square$

4.2. Exact scalar prior-art boundary. The scalar integer specialization is not a new class of maps under a new name. Matthews's generalized Collatz maps already have exactly the same fixed-modulus residue-class presentation, while Kohl's residue-class-wise affine maps give the ambient mapping monoid [17, Sec. 2][18, Defs. 1.1 and 1.4]. The extra global-field data, non-scalar endomorphisms, centeredness, and properness in Siegel's formalism remain additional structure. In scalar coordinates, exact-integrality is a predicate of the displayed presentation, whereas Matthews attaches *relatively prime type* to the map at its least modulus. The following comparison proves the exact relation without identifying those two levels prematurely.

Proposition 4.7 (fixed-modulus scalar Hydra–Matthews equivalence). *Fix $D \in \mathbb{Z}_{\geq 2}$, put $K = \mathbb{F} = \mathbb{Q}$, $\mathcal{O}_K = \mathbb{Z}$, $\Lambda = D\mathbb{Z}$, and use the standard classes $\Lambda_i = i + D\mathbb{Z}$ for $0 \leq i < D$. Let*

$$H_i(x) = \alpha_i x + \beta_i, \quad \alpha_i \in \mathbb{Q}^\times, \quad \beta_i \in \mathbb{Q}.$$

Let $H : \mathbb{Z} \rightarrow \mathbb{Q}$ be the associated piecewise function

$$H(x) = H_i(x) \quad \text{for } x \in i + D\mathbb{Z}.$$

Then the following are equivalent for each i .

- (i) $H_i(i + D\mathbb{Z}) \subseteq \mathbb{Z}$.
- (ii) There are unique integers $m_i \neq 0$ and ρ_i such that

$$\rho_i \equiv im_i \pmod{D}, \quad H_i(x) = \frac{m_i x - \rho_i}{D} \quad (x \in \mathbb{Q}). \quad (10)$$

Consequently an admissible scalar Hydra with this fixed standard modulus is exactly a Matthews generalized Collatz presentation with the same displayed modulus, and its associated map has type $H : \mathbb{Z} \rightarrow \mathbb{Z}$. Write $d_{\text{Mat}}(H)$ for Matthews's least affine residue modulus, and put $g_i = \gcd(D, m_i)$. The full integral locus of the i th formal branch is

$$\{x \in \mathbb{Z} : H_i(x) \in \mathbb{Z}\} = i + \frac{D}{g_i}\mathbb{Z}.$$

Therefore

the scalar Hydra is exact-integral

$$\begin{aligned} \iff \gcd(D, m_i) = 1 \text{ for every } i \\ \iff \gcd(m_0 \cdots m_{D-1}, D) = 1. \end{aligned} \tag{11}$$

The last condition is the relatively-prime numerical condition for this displayed D -presentation. Matthews assigns the name relatively prime type using the least modulus $d_{\text{Mat}}(H)$ [17, Sec. 2]; consequently it is the same map-level condition only when $D = d_{\text{Mat}}(H)$.

The qualifier is necessary. The shortened $3x + 1$ map has least-modulus presentation $D = 2$ with multipliers $(1, 3)$, hence is of Matthews relatively prime type. Refining the same map to $D = 4$ gives multipliers $(2, 6, 2, 6)$ and remainder parameters $(0, -2, 0, -2)$ in equation (10). Each formal branch is then integral on a whole parity class, not on only its assigned class modulo 4; the refined $D = 4$ presentation is not exact-integral.

Proof. Assume (i). Since both i and $i + D$ belong to the assigned class,

$$m_i := D\alpha_i = H_i(i + D) - H_i(i) \in \mathbb{Z}.$$

The branch is an affine automorphism of $\mathbb{Q}$, so $m_i \neq 0$. Put

$$\rho_i := m_i i - DH_i(i) = -D\beta_i.$$

This is an integer and satisfies $\rho_i \equiv m_i i \pmod{D}$; expanding the definition gives equation (10). The coefficients of an affine function make $m_i = D\alpha_i$ and $\rho_i = -D\beta_i$ unique.

Conversely, if (ii) holds and $x = i + Dt$, then

$$H_i(x) = \frac{m_i i - \rho_i}{D} + m_i t \in \mathbb{Z},$$

so (i) holds. Applying the equivalence independently to all D classes identifies the two fixed-modulus presentations.

For $x \in \mathbb{Z}$, the congruence in equation (10) gives

$$\begin{aligned} H_i(x) \in \mathbb{Z} &\iff D \mid m_i x - \rho_i \\ &\iff D \mid m_i(x - i) \\ &\iff \frac{D}{g_i} \mid x - i. \end{aligned}$$

This proves the displayed integral locus. It equals the assigned class $i + D\mathbb{Z}$ exactly when $g_i = 1$. Requiring this for all i is equivalent to $\gcd(m_0 \cdots m_{D-1}, D) = 1$, which proves equation (11) and the source-term identification. $\square$

Proposition 4.8 (typed inclusion in the rcwa mapping monoid). *Inherit $D \in \mathbb{Z}_{\geq 2}$ and the branch data from theorem 4.7, assume its equivalent conditions hold for every $0 \leq i < D$, and define the piecewise map*

$$H : \mathbb{Z} \longrightarrow \mathbb{Z}, \quad H(x) = H_i(x) \quad \text{when } x \equiv i \pmod{D}.$$

Then H is a residue-class-wise affine (rcwa) mapping in Kohl's sense. If $d_{\text{Mat}}(H)$ is Matthews's least affine residue modulus and $\text{Mod}(H)$ is Kohl's least rcwa modulus, then

$$d_{\text{Mat}}(H) = \text{Mod}(H) \mid D. \tag{12}$$

If the displayed scalar Hydra is exact-integral, then

$$d_{\text{Mat}}(H) = \text{Mod}(H) = D. \quad (13)$$

More precisely,

$$\text{the displayed } D\text{-presentation is exact-integral} \iff \begin{cases} D = d_{\text{Mat}}(H), \\ H \text{ is of Matthews relatively prime type.} \end{cases} \quad (14)$$

Finally,

$$H \in \text{RCWA}(\mathbb{Z}) \iff H : \mathbb{Z} \rightarrow \mathbb{Z} \text{ is bijective,}$$

because $\text{RCWA}(\mathbb{Z})$ denotes Kohl's group of rcwa permutations, not the full rcwa mapping monoid.

Proof. On the class $i + D\mathbb{Z}$, formula equation (10) has Kohl's integer-coefficient form $(a_i x + b_i)/c_i$ with $(a_i, b_i, c_i) = (m_i, -\rho_i, D)$; hence D is an rcwa presentation modulus.

Matthews and Kohl minimize over the same positive integers in this scalar setting. Matthews asks that the restriction to each residue class be affine; Kohl writes the same rational-affine restriction as $(ax + b)/c$ with integer coefficients, which is possible after clearing denominators. Thus $d_{\text{Mat}}(H) = \text{Mod}(H)$.

For completeness, let $m = \text{Mod}(H)$ and $g = \gcd(m, D)$. Fix a class modulo g . Every class modulo m and every class modulo D lying above it have a nonempty infinite intersection by the Chinese remainder theorem. The affine formulas from the two presentations agree with H on that intersection, hence agree as rational affine functions. It follows that all formulas above the fixed class modulo g are identical; therefore g is also an rcwa presentation modulus. Minimality of m and $g \leq m$ force $g = m$, proving $m \mid D$. The last equivalence is precisely Kohl's definition of the group $\text{RCWA}(\mathbb{Z})$.

Now assume exact-integrality. If $m < D$, then $m \mid D$ and the two distinct D -classes $D\mathbb{Z}$ and $m + D\mathbb{Z}$ lie in the single class $m\mathbb{Z}$. Let A be the rational-affine formula of a modulus- m presentation on $m\mathbb{Z}$. It agrees with H_0 on the infinite set $D\mathbb{Z}$ and with H_m on the infinite set $m + D\mathbb{Z}$; equality on an infinite subset forces equality of rational-affine functions, so $H_0 = A = H_m$. Exact-integrality says that the integral locus of H_0 is $D\mathbb{Z}$ and that of H_m is $m + D\mathbb{Z}$. One function cannot have these two distinct integral loci, a contradiction. Hence $m = D$, proving equation (13). Together with equation (11), this proves the forward implication in equation (14): the displayed modulus is now the least modulus, and its multipliers satisfy Matthews's defining coprimality condition. Conversely, if $D = d_{\text{Mat}}(H)$ and H is of Matthews relatively prime type, its unique least- D multipliers satisfy $\gcd(m_0 \cdots m_{D-1}, D) = 1$; the earlier equivalence then proves that the displayed presentation is exact-integral. $\square$

Thus neither "Hydra" nor "rcwa group element" may be used as an unqualified synonym for the other. The shortened Collatz map itself is an rcwa mapping which is surjective but not injective [18, Rem. 1.2]; it therefore lies outside $\text{RCWA}(\mathbb{Z})$. Conversely, rcwa membership alone supplies no centeredness, properness, exact-integrality, branch-correctness, or global-field structure. The scalar equivalence between exact-integrality and Matthews's relatively-prime condition also requires that the displayed modulus be the least one, as proved above; it is not a consequence of rcwa membership alone.

4.3. The shortened $qx + 1$ Hydra. For odd q , equation (1) is a 2-Hydra over $K = \mathbb{F} = \mathbb{Q}$ with $\Lambda = 2\mathbb{Z}$ and branches

$$H_0(x) = \frac{x}{2}, \quad H_1(x) = \frac{qx + 1}{2}. \quad (15)$$

It is centered and proper because $c_0 = 0$ and $1 - r_0 = 1/2$. On integral inputs it has the exact branch-correctness property

$$H_0(x) \in \mathbb{Z} \iff x \in 2\mathbb{Z}, \quad H_1(x) \in \mathbb{Z} \iff x \in 1 + 2\mathbb{Z} \quad (x \in \mathbb{Z}).$$

The unshortened Collatz map is not exact-integral in this sense because its odd branch $3x + 1$ is integer-valued on both parity classes.

4.4. Words and order. Let Σ_p^* be the free monoid of finite words in $\mathcal{A}_p$, including the empty word $\emptyset$. Concatenation is denoted by $\mathbf{i} \wedge \mathbf{j}$. For $\mathbf{j} = (j_0, \dots, j_{m-1})$, set

$$H_{\mathbf{j}} = H_{j_0} \circ H_{j_1} \circ \dots \circ H_{j_{m-1}}, \quad H_{\emptyset} = \text{id}_K. \quad (16)$$

Thus

$$H_{\mathbf{i} \wedge \mathbf{j}} = H_{\mathbf{i}} \circ H_{\mathbf{j}}. \quad (17)$$

There are unique maps

$$M_H : \Sigma_p^* \rightarrow \text{End}_{\mathbb{F}}(K)^{\times}, \quad X_H : \Sigma_p^* \rightarrow K$$

such that

$$H_{\mathbf{j}}(x) = M_H(\mathbf{j})x + X_H(\mathbf{j}). \quad (18)$$

The second component is Siegel's *numen*; equivalently, $X_H(\mathbf{j}) = H_{\mathbf{j}}(0)$.

Proposition 4.9 (affine concatenation identities). *For all $\mathbf{i}, \mathbf{j} \in \Sigma_p^*$,*

$$M_H(\mathbf{i} \wedge \mathbf{j}) = M_H(\mathbf{i})M_H(\mathbf{j}), \quad (19)$$

$$X_H(\mathbf{i} \wedge \mathbf{j}) = M_H(\mathbf{i})X_H(\mathbf{j}) + X_H(\mathbf{i}) = H_{\mathbf{i}}(X_H(\mathbf{j})). \quad (20)$$

Proof. Substitute equation (18) twice into equation (17):

$$H_{\mathbf{i}}(H_{\mathbf{j}}(x)) = M_H(\mathbf{i})M_H(\mathbf{j})x + M_H(\mathbf{i})X_H(\mathbf{j}) + X_H(\mathbf{i}).$$

Uniqueness of the linear and translation parts gives both identities. $\square$

Corollary 4.10 (closed word formulas). *For $\mathbf{j} = (j_0, \dots, j_{m-1})$,*

$$M_H(\mathbf{j}) = r_{j_0}r_{j_1} \cdots r_{j_{m-1}}, \quad (21)$$

$$X_H(\mathbf{j}) = \sum_{a=0}^{m-1} (r_{j_0}r_{j_1} \cdots r_{j_{a-1}})c_{j_a}, \quad (22)$$

where an empty product is I and products mean composition in the displayed order.

Proof. Induct on m . Appending the last letter and applying equation (20) gives the new summand after the existing left prefix; equation (19) gives the product formula. $\square$

When the r_j do not commute, the product order in equations (21) and (22) is indispensable. Expressions such as $(1 - M_H(\mathbf{j}))^{-1}$ mean the inverse of an endomorphism of K , not scalar division.

4.5. Repeated words and affine fixed points. For a word $\mathbf{j}$, write $\mathbf{j}^{\wedge n}$ for its n -fold concatenation, and abbreviate $M = M_H(\mathbf{j})$ and $X = X_H(\mathbf{j})$.

Proposition 4.11 (geometric sum in the endomorphism algebra). *For $n \geq 1$,*

$$M_H(\mathbf{j}^{\wedge n}) = M^n, \quad (23)$$

$$X_H(\mathbf{j}^{\wedge n}) = (I + M + \dots + M^{n-1})X. \quad (24)$$

If $I - M$ is invertible, then

$$X_H(\mathbf{j}^{\wedge n}) = (I - M^n)(I - M)^{-1}X. \quad (25)$$

The affine map $H_{\mathbf{j}}$ has a unique fixed point precisely when $I - M$ is invertible, in which case

$$\text{Fix}(H_{\mathbf{j}}) = \{(I - M)^{-1}X\}. \quad (26)$$

Proof. The first two identities follow by induction from theorem 4.9. Because M commutes with every polynomial in M ,

$$(I - M)(I + M + \cdots + M^{n-1}) = I - M^n,$$

which gives equation (25). Finally, $H_{\mathbf{j}}(z) = z$ is exactly $(I - M)z = X$. $\square$

Remark 4.12. Properness requires $I - r_0$ to be invertible. It does not imply that $I - M_H(\mathbf{j})$ is invertible for every nonempty word. The latter is a separate hypothesis whenever equation (26) is used.

4.6. Correct words versus formal words. Every word defines an affine map on K , but it need not describe a genuine orbit segment of the piecewise map H . A chronological branch word $\varepsilon = (\varepsilon_0, \dots, \varepsilon_{m-1})$ is *correct* at $z \in \mathcal{O}_K$ if

$$H^{\circ k}(z) \in \Lambda_{\varepsilon_k} \quad (0 \leq k < m).$$

Then

$$H^{\circ m}(z) = H_{\varepsilon_{m-1}} \circ \cdots \circ H_{\varepsilon_0}(z) = H_{\text{rev}(\varepsilon)}(z). \quad (27)$$

Exact-integrality is designed to recover correctness from integrality of successive branch values. The passage from a formal fixed point of $H_{\mathbf{j}}$ to a periodic point of H therefore requires more than the affine equation alone; the needed arithmetic step is isolated in section 6.

5. THE NUMEN: FINITE DIGITS, INFINITE DIGITS, AND PLACE-DEPENDENT LIMITS

The numen is the translation cocycle of the affine word representation. Its extension to infinite digits is a second construction, and the topology in which that extension converges is part of the data. Keeping these two facts separate removes several ambiguities present when the word “continuation” is used without a codomain place.

5.1. Finite base- p words. For $\mathbf{j} = (j_0, \dots, j_{m-1}) \in \Sigma_p^*$ define

$$\text{Dig}_p(\mathbf{j}) = \sum_{a=0}^{m-1} j_a p^a. \quad (28)$$

The least significant digit occurs first. The map $\text{Dig}_p : \Sigma_p^* \rightarrow \mathbb{N}_0$ is surjective and is not injective: appending zeroes on the right does not change its value. These are the only identifications.

Lemma 5.1 (finite-word quotient). *If $\text{Dig}_p(\mathbf{i}) = \text{Dig}_p(\mathbf{j})$, then one of $\mathbf{i}, \mathbf{j}$ is obtained from the other by appending finitely many zeroes, after deleting zeroes already appended to both. Consequently $\Sigma_p^* / \sim \cong \mathbb{N}_0$, where $\mathbf{i} \sim \mathbf{j}$ means equality of digit sums.*

Proof. Every nonzero integer has a unique finite base- p expansion with nonzero last digit. The zero integer is represented by the empty word and by words consisting only of zeroes. The claim follows. $\square$

Proposition 5.2 (descent of the word numen). *If H is centered, then $X_H(\mathbf{j})$ depends only on $\text{Dig}_p(\mathbf{j})$. Hence there is a well-defined function*

$$X_H : \mathbb{N}_0 \longrightarrow K, \quad X_H(\text{Dig}_p(\mathbf{j})) = X_H(\mathbf{j}).$$

Proof. Centeredness gives $H_0(0) = 0$. Therefore

$$X_H(\mathbf{j} \wedge (0)) = H_{\mathbf{j}}(H_0(0)) = H_{\mathbf{j}}(0) = X_H(\mathbf{j}).$$

Iterate this identity and use theorem 5.1. $\square$

The multiplier requires a separate convention: unlike the centered numen, it does not descend through the finite-word quotient.

Definition 5.3 (canonical integer multiplier). For $n > 0$, let

$$\text{dig}_p(n) = (d_0(n), \dots, d_{L-1}(n))$$

be the unique shortest base- p word representing n , so that $d_{L-1}(n) \neq 0$, and put $\text{dig}_p(0) = \emptyset$. Define

$$M_H(n) := M_H(\text{dig}_p(n)), \quad M_H(0) = I. \quad (29)$$

Thus $M_H(n)$ always refers to the canonical word, not to an arbitrary word with digit sum n .

Indeed, the concatenation identity gives

$$M_H(\mathbf{j} \wedge (0)^{\wedge k}) = M_H(\mathbf{j})r_0^k.$$

Consequently equivalent representatives obtained by appending zeroes need not have the same multiplier. The word multiplier descends through all such representatives if and only if $r_0 = I$.

For a noncentered proper Hydra, the natural digit-recursive function still exists, but it is not the translation term $H_{\mathbf{j}}(0)$ alone. Its terminal value must be the fixed point of the zero branch.

Theorem 5.4 (digit-recursive characterization). *Let $(r_j, c_j)_{0 \leq j < p}$ be affine branches on K . Functions $f : \mathbb{N}_0 \rightarrow K$ satisfying*

$$f(pn + j) = r_j f(n) + c_j \quad (n \in \mathbb{N}_0, 0 \leq j < p) \quad (30)$$

are in bijection with solutions $a \in K$ of

$$(I - r_0)a = c_0. \quad (31)$$

The bijection sends f to $f(0) = a$. If H is proper, the solution is unique and $a = (I - r_0)^{-1}c_0$. If H is centered and proper, $a = 0$ and $f = X_H$.

Proof. Putting $n = j = 0$ in equation (30) gives equation (31). Conversely, choose a solution a . Set $f(0) = a$. Every positive n has a unique decomposition $n = pm + j$ with $0 \leq j < p$ and $m < n$, so equation (30) recursively defines one and only one value $f(n)$. The recursion automatically satisfies every displayed equation because the decomposition is unique.

If $\mathbf{d} = \text{dig}_p(n) = (d_0, \dots, d_{L-1})$, repeated substitution gives

$$f(n) = H_{d_0} \circ \dots \circ H_{d_{L-1}}(a) = M_H(\mathbf{d})a + X_H(\mathbf{d}). \quad (32)$$

For centered H , appending zeroes leaves the value unchanged and this is exactly the descended word numen of theorem 5.2. $\square$

Combining equation (22) with the canonical digit word gives the explicit finite formula

$$X_H(n) = \sum_{a=0}^{L-1} (r_{d_0} \cdots r_{d_{a-1}}) c_{d_a}, \quad \text{dig}_p(n) = (d_0, \dots, d_{L-1}), \quad (33)$$

for a centered Hydra, with the empty sum understood when $n = 0$. This formula, not an analogy with an orbit, is the primitive F-series of the later papers.

5.2. The base- p digit compactum. For any integer $p \geq 2$, put

$$\mathbb{Z}_p^{\text{dig}} = \varprojlim_N \mathbb{Z}/p^N \mathbb{Z}. \quad (34)$$

Every element has a unique digit expansion $\mathbf{z} = \sum_{a \geq 0} d_a(\mathbf{z})p^a$ with $d_a(\mathbf{z}) \in \mathcal{A}_p$. If p is prime this is the usual ring $\mathbb{Z}_p$. For composite p , equation (34) is still the compact base- p digit ring, but it is not a discrete valuation ring and should not be treated as a field's ring of integers.

Proposition 5.5 (digits, the inverse limit, and Haar probability). *Give $\mathcal{A}_p^{\mathbb{N}_0}$ the product topology, with each copy of $\mathcal{A}_p$ discrete. The map*

$$\Phi_p : \mathcal{A}_p^{\mathbb{N}_0} \longrightarrow \mathbb{Z}_p^{\text{dig}}, \quad (d_a)_{a \geq 0} \longmapsto \left(\sum_{a=0}^{N-1} d_a p^a \bmod p^N \right)_{N \geq 1}$$

is a homeomorphism. Under Φ_p , normalized Haar probability m_p on $\mathbb{Z}_p^{\text{dig}}$ is the product of the uniform probability measures on $\mathcal{A}_p$. Equivalently, for every $N \geq 1$ and $e_0, \dots, e_{N-1} \in \mathcal{A}_p$,

$$m_p\{\mathfrak{z} : d_0(\mathfrak{z}) = e_0, \dots, d_{N-1}(\mathfrak{z}) = e_{N-1}\} = p^{-N}. \quad (35)$$

Thus the coordinate digits are independent and each is uniform on $\mathcal{A}_p$.

Proof. The displayed residues are compatible under reduction, so Φ_p is well-defined. Conversely, take a compatible sequence $(x_N \bmod p^N)_N$. Its reduction modulo p determines a unique $d_0 \in \mathcal{A}_p$. After $d_0, \dots, d_{N-1}$ have been determined, compatibility makes

$$x_{N+1} - \sum_{a=0}^{N-1} d_a p^a$$

divisible by p^N , and its quotient modulo p determines the unique next digit d_N . This constructs the inverse of Φ_p . Both maps are continuous because each output coordinate or digit depends on only finitely many input coordinates; alternatively, the continuous bijection is a homeomorphism because the product is compact and the inverse limit is Hausdorff.

Fixing the first N digits specifies one coset of the open subgroup $p^N \mathbb{Z}_p^{\text{dig}}$. That subgroup has index p^N , so translation invariance and total mass one give every such cylinder mass p^{-N} . These are exactly the finite-dimensional cylinder probabilities of the uniform product measure, proving the measure identity and independence. $\square$

Write

$$[\mathfrak{z}]_{p^N} = \sum_{a=0}^{N-1} d_a(\mathfrak{z}) p^a, \quad (36)$$

$$\vartheta_p(\mathfrak{z}) = \frac{\mathfrak{z} - d_0(\mathfrak{z})}{p} = \sum_{a \geq 0} d_{a+1}(\mathfrak{z}) p^a. \quad (37)$$

The first is the canonical truncation and the second is the left digit shift. The digit and measure assertions used below are therefore consequences of theorem 5.5, including when p is composite.

5.3. The primitive F-series. Assume throughout this subsection that H is centered. Fix a place ℓ of K and view all r_j as continuous operators on the completion K_ℓ . Let

$$S_N(\mathfrak{z}) = X_H([\mathfrak{z}]_{p^N}) = \sum_{a=0}^{N-1} \left(\prod_{k=0}^{a-1} r_{d_k(\mathfrak{z})} \right) c_{d_a(\mathfrak{z})}. \quad (38)$$

Here and below the empty product is I . When $S_N(\mathfrak{z})$ converges in K_ℓ , define

$$X_{H,\ell}(\mathfrak{z}) = \lim_{N \rightarrow \infty} S_N(\mathfrak{z}). \quad (39)$$

This is a *rising extension*: its values are prescribed by limits along the canonical truncation sequence. It is not a topology-free interpolation.

Proposition 5.6 (pointwise convergence criterion). *Let $C = \max_j |c_j|_\ell$. Suppose that for a given $\mathfrak{z}$ there are $\eta > 0$ and $A \in \mathbb{R}$ such that*

$$\sum_{k=0}^{a-1} \log \|r_{d_k(\mathfrak{z})}\|_\ell \leq A - \eta a \quad (40)$$

for all sufficiently large a . Then the series in equation (38) converges absolutely in an archimedean completion and converges in a nonarchimedean completion. The same estimate makes the remainder term $(\prod_{k < N} r_{d_k})a_0$ tend to zero for every fixed $a_0 \in K_\ell$.

Proof. The norm of the a th summand is at most $C \exp(A - \eta a)$. This is summable in the archimedean case and tends to zero in the nonarchimedean case; a series in a complete nonarchimedean field converges exactly when its terms tend to zero. The identical product bound applies to the remainder. $\square$

Source issue 5.7 (the pointwise digit-density criterion). For a fixed digit sequence, write

$$\nu_{j,N} = \frac{1}{N} \#\{0 \leq k < N : d_k(\mathfrak{z}) = j\}, \quad \alpha_j = \log \|r_j\|_\ell.$$

The upper Lyapunov exponent is controlled by

$$\limsup_{N \rightarrow \infty} \sum_j \alpha_j \nu_{j,N} \leq \sum_{\alpha_j < 0} \alpha_j \liminf_{N \rightarrow \infty} \nu_{j,N} + \sum_{\alpha_j > 0} \alpha_j \limsup_{N \rightarrow \infty} \nu_{j,N}. \quad (41)$$

For a negative coefficient, an upper bound on $\alpha_j \nu_{j,N}$ requires a lower bound on the frequency, hence a $\liminf$; multiplication by $\alpha_j < 0$ reverses the inequality. For a positive coefficient, it requires an upper bound on the frequency, hence a $\limsup$. Terms with $\alpha_j = 0$ are immaterial. The pointwise criterion in the 2026 source reverses these choices, using a $\limsup$ for contractive branches and a $\liminf$ for expansive branches. Its displayed quantity is therefore a lower, not an upper, control and its negativity is not a sufficient convergence condition. We avoid that error here by assuming the direct eventual bound equation (40); negativity of the right side of equation (41) is one valid way to obtain that bound.

Theorem 5.8 (almost-everywhere extension). *Assume H is centered and proper and*

$$\prod_{j=0}^{p-1} \|r_j\|_\ell < 1. \quad (42)$$

Then equation (39) converges for Haar-almost every $\mathfrak{z} \in \mathbb{Z}_p^{\text{dig}}$ and defines a measurable K_ℓ -valued function on its conull domain. On that domain it obeys

$$X_{H,\ell}(p\mathfrak{z} + j) = r_j X_{H,\ell}(\mathfrak{z}) + c_j. \quad (43)$$

Proof. For Haar-almost every digit sequence, the strong law of large numbers gives

$$\frac{1}{N} \sum_{k=0}^{N-1} \log \|r_{d_k(\mathfrak{z})}\|_\ell \longrightarrow \frac{1}{p} \sum_{j=0}^{p-1} \log \|r_j\|_\ell < 0.$$

Thus equation (40) holds eventually, and theorem 5.6 applies. Each S_N is locally constant and therefore measurable; its pointwise limit is measurable. Finally, the digits of $p\mathfrak{z} + j$ are $j, d_0(\mathfrak{z}), \dots$. Separate the first summand in equation (38) and pass to the limit to obtain equation (43). $\square$

Theorem 5.9 (uniform contraction). *Suppose*

$$\rho = \max_{0 \leq j < p} \|r_j\|_\ell < 1.$$

Then the series in equation (38) converges uniformly on $\mathbb{Z}_p^{\text{dig}}$. Its sum is continuous and satisfies equation (43). More precisely, in the archimedean case,

$$\sup_{\mathfrak{z}} |X_{H,\ell}(\mathfrak{z}) - S_N(\mathfrak{z})|_\ell \leq \frac{C\rho^N}{1-\rho}, \quad (44)$$

while in the nonarchimedean case the right side may be replaced by $C\rho^N$.

Proof. The a th summand has norm at most $C\rho^a$, uniformly in $\mathfrak{z}$. The ordinary and ultrametric tail estimates give equation (44). Uniform limits of the locally constant functions S_N are continuous. The functional equation follows exactly as in theorem 5.8. $\square$

Source issue 5.10 (centeredness and uniqueness in the extension lemma). The 2026 extension lemma assumes properness but not centeredness, while its displayed F-series is the translation term $H_{\mathbf{j}}(0)$ and therefore uses terminal value 0. For a noncentered proper Hydra, the unique solution on $\mathbb{N}_0$ instead has terminal value $a = (I - r_0)^{-1}c_0$ and finite values $M_H(\mathbf{d})a + X_H(\mathbf{d})$ as in equation (32). The extra product term disappears under a contraction hypothesis, but it cannot be omitted formally. Likewise, a functional equation on all infinite digit strings need not determine an arbitrary function without a rising-limit, regularity, or boundedness condition. The uniqueness asserted here is uniqueness of the prescribed rising extension, and under uniform contraction uniqueness in the natural continuous bounded class.

5.4. The shortened $qx + 1$ numen. For the Hydra T_q , write $\chi_q = X_{T_q}$. From equations (15) and (30),

$$\chi_q(2n) = \frac{1}{2}\chi_q(n), \quad (45)$$

$$\chi_q(2n+1) = \frac{q\chi_q(n) + 1}{2}, \quad (46)$$

with $\chi_q(0) = 0$. If $\text{dig}_2(n) = (d_0, \dots, d_{L-1})$ and $s_a = d_0 + \dots + d_{a-1}$, then

$$M_q(n) = \frac{q^{d_0 + \dots + d_{L-1}}}{2^L}, \quad (47)$$

$$\chi_q(n) = \sum_{\substack{0 \leq a < L \\ d_a = 1}} \frac{q^{s_a}}{2^{a+1}}. \quad (48)$$

Here L is the length of the canonical shortest binary word; in particular, $M_q(0) = 1$. Appending zeroes would multiply the word multiplier by extra factors of $1/2$, so a padded representative must not be used in equation (47). For an odd prime q , the same series converges in $\mathbb{Q}_q$ for every $\mathfrak{z} \in \mathbb{Z}_2$: it is finite when the binary expansion has finitely many ones, and otherwise its nonzero terms have q -adic norms $q^{-s_a} \rightarrow 0$. Thus

$$\chi_q(\mathfrak{z}) = \sum_{\substack{a \geq 0 \\ d_a(\mathfrak{z}) = 1}} \frac{q^{s_a(\mathfrak{z})}}{2^{a+1}} \in \mathbb{Z}_q. \quad (49)$$

It is rising-continuous everywhere. It is 2-adic-to- q -adic continuous at every binary sequence with infinitely many ones, but not at the nonnegative integers: if $n \in \mathbb{N}_0$ has $k = \#_1(n)$ ones, a new 1 placed arbitrarily far beyond its last digit contributes a term of the fixed nonzero q -adic norm q^{-k} , independent of that remote position (it is a unit only when $n = 0$). This pointwise distinction is the prototype for the later theory of variable topologies of convergence.

6. PERIODIC CODES AND THE CORRESPONDENCE PRINCIPLE

The algebraic heart of the Correspondence Principle is a fixed-point identity for a repeated word. Its dynamical heart is branch correctness. Those are independent steps: the first takes place in an affine group, while the second identifies a formal composition with an iterate of a piecewise map.

6.1. Purely periodic digit strings. Let $\mathbf{j} = (j_0, \dots, j_{L-1})$ be a word of length $L \geq 1$, and put

$$n = \text{Dig}_p(\mathbf{j}) = \sum_{a=0}^{L-1} j_a p^a.$$

Its infinite repetition represents the base- p digit-ring element

$$B_p(\mathbf{j}) = B_{p,L}(n) = n(1 + p^L + p^{2L} + \cdots) = \frac{n}{1 - p^L}. \quad (50)$$

The length L is part of this expression. When $\mathbf{j}$ is the shortest word for $n > 0$, write $\lambda_p(n) = L$ and $B_p(n) = n/(1 - p^{\lambda_p(n)})$.

Theorem 6.1 (repeated-code fixed-point identity). *Let H be centered, let ℓ be a place of K , and set $M = M_H(\mathbf{j})$ and $X = X_H(\mathbf{j})$. Suppose $\|M\|_\ell < 1$. Then $I - M$ is invertible on K_ℓ , the rising numer converges at $B_p(\mathbf{j})$, and*

$$X_{H,\ell}(B_p(\mathbf{j})) = (I - M)^{-1}X. \quad (51)$$

This value is the unique fixed point of $H_{\mathbf{j}}$ in K_ℓ .

Proof. The mL -digit truncation of $B_p(\mathbf{j})$ is represented by $\mathbf{j}^{\wedge m}$. By theorem 4.11,

$$X_H(\mathbf{j}^{\wedge m}) = (I + M + \cdots + M^{m-1})X.$$

The operator series converges in the complete endomorphism algebra because $\|M\|_\ell < 1$, and its sum is $(I - M)^{-1}$. Truncations ending inside the next block have the same limit because the remaining finite prefix is multiplied by M^m . This proves the rising limit. The fixed-point assertion is equation (26). $\square$

Remark 6.2. In a scalar one-dimensional setting equation (51) is often written $X/(1 - M)$. For a general Hydra, M is an endomorphism and the typed formula is $(I - M)^{-1}X$.

Proposition 6.3 (rational points are the eventually periodic digit strings). *Under the diagonal embedding into the completions at the primes dividing p ,*

$$\mathbb{Q} \cap \mathbb{Z}_p^{\text{dig}} = \mathbb{Z}_{(p)} := \left\{ \frac{a}{b} \in \mathbb{Q} : \gcd(b, p) = 1 \text{ after reducing } a/b \right\}.$$

An element of $\mathbb{Z}_p^{\text{dig}}$ lies in this subring if and only if its base- p digit sequence is eventually periodic.

Proof. Write $p = \prod_{\ell|p} \ell^{e_\ell}$. The compatible Chinese-remainder isomorphisms give

$$\mathbb{Z}_p^{\text{dig}} = \varprojlim_N \mathbb{Z}/p^N \mathbb{Z} \xrightarrow{\sim} \prod_{\ell|p} \varprojlim_N \mathbb{Z}/\ell^{e_\ell N} \mathbb{Z} \xrightarrow{\sim} \prod_{\ell|p} \mathbb{Z}_\ell.$$

The subsequence $(e_\ell N)_{N \geq 1}$ is cofinal, which proves the second isomorphism. Under the resulting diagonal embedding of $\mathbb{Q}$ into $\prod_{\ell|p} \mathbb{Q}_\ell$, a reduced rational belongs to the displayed product of integer rings exactly when no prime $\ell \mid p$ divides its denominator. Hence $\mathbb{Q} \cap \mathbb{Z}_p^{\text{dig}} = \mathbb{Z}_{(p)}$ with the meaning stated above.

If a digit string has a prefix of length R with value u and thereafter repeats a block of length L with value $v = \sum_{h=0}^{L-1} e_h p^h$, then its value in the inverse limit is

$$u + p^R v(1 + p^L + p^{2L} + \cdots) = u + \frac{p^R v}{1 - p^L}.$$

Because $1 - p^L \equiv 1 \pmod{p}$, its denominator is coprime to p .

Conversely, let $x = a/b$ with $a, b \in \mathbb{Z}$ and $\gcd(b, p) = 1$. For every $N \geq 0$ there is a unique compatible residue $A_N \in \{0, \dots, p^N - 1\}$ satisfying $bA_N \equiv a \pmod{p^N}$; these residues define the digit expansion of x . Here $A_0 = 0$. Put

$$u_N = \frac{a - bA_N}{p^N} \in \mathbb{Z}.$$

If d_N is the next digit, then

$$u_{N+1} = \frac{u_N - bd_N}{p}, \quad d_N \in \mathcal{A}_p, \quad bd_N \equiv u_N \pmod{p}. \quad (52)$$

The congruence determines d_N uniquely from u_N because b is a unit modulo p . Iterating $|u_{N+1}| \leq (|u_N| + |b|(p-1))/p$ gives

$$|u_N| \leq p^{-N}|u_0| + |b|(1 - p^{-N}) \leq |u_0| + |b|.$$

Thus the integer sequence (u_N) takes values in a finite set, and the deterministic transition equation (52) must eventually repeat. Its associated digits are therefore eventually periodic. $\square$

Let H be centered, fix a place ℓ at which all branches are bounded, and suppose $\|M_H(\mathbf{j})\|_\ell < 1$. If $\mathbf{j}$ has a finite prefix $\mathbf{i}$ followed by the infinite repetition of $\mathbf{j}$, then

$$X_{H,\ell}(\mathbf{j}) = H_{\mathbf{i}}((I - M_H(\mathbf{j}))^{-1}X_H(\mathbf{j})). \quad (53)$$

This is an affine image of a formal periodic fixed point. It becomes a statement about the actual Hydra only after correctness is proved.

6.2. Classical 2-adic conjugacy and the exact periodic overlap. The parity-vector conjugacy predates the numen. Its digit order is chronological, whereas Siegel's displayed affine word acts from right to left. The difference is not cosmetic: it forces a word reversal in the precise locus where the two constructions have a common rational value.

Let q be any odd integer and extend T_q to $\mathbb{Z}_2$ using the parity branches in equation (15). Let

$$S = \vartheta_2 : \mathbb{Z}_2 \longrightarrow \mathbb{Z}_2, \quad S(z) = \frac{z - [z]_2}{2},$$

be the binary digit shift. For every prime ℓ , write $\iota_\ell : \mathbb{Q} \hookrightarrow \mathbb{Q}_\ell$ for the canonical field embedding.

Proposition 6.4 (the Bernstein–Lagarias conjugacy, with maps typed). *Define*

$$Q_q(x) = \sum_{i=0}^{\infty} [T_q^{\circ i}(x)]_2 2^i \quad (x \in \mathbb{Z}_2). \quad (54)$$

If $z \in \mathbb{Z}_2 \setminus \{0\}$, let either $I = \{1, \dots, r\}$ for some $r \geq 1$ or $I = \mathbb{Z}_{\geq 1}$, according as the set of one positions is finite or infinite, and write

$$z = \sum_{\ell \in I} 2^{d_\ell}, \quad d_1 \geq 0, \quad d_\ell < d_{\ell+1} \quad (\ell, \ell+1 \in I).$$

Define

$$\Phi_q(z) = - \sum_{\ell \in I} q^{-\ell} 2^{d_\ell}, \quad \Phi_q(0) = 0. \quad (55)$$

Then $Q_q, \Phi_q : \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ are mutually inverse 2-adic isometries, hence solenoidal homeomorphisms, and

$$Q_q \circ T_q = S \circ Q_q, \quad T_q = \Phi_q \circ S \circ \Phi_q^{-1}. \quad (56)$$

Thus Q_q is the chronological parity encoder and $\Phi_q = Q_q^{-1}$ is the conjugacy. This is the odd- $(a, b) = (q, 1)$ specialization of Bernstein–Lagarias, not a prime- q statement [20, Eqs. (1.3), (1.5), (1.6), (4.1), and (4.2)].

Proof. For odd q , H_0 maps $2\mathbb{Z}_2$ to $\mathbb{Z}_2$ and H_1 maps $1 + 2\mathbb{Z}_2$ to $\mathbb{Z}_2$. Thus the branch selected by the current parity is defined at every iterate, every digit in equation (54) lies in $\{0, 1\}$, and the displayed binary series converges. If $x \equiv y \pmod{2^N}$, induction through their common parity branches gives

$$T_q^{\circ i}(x) \equiv T_q^{\circ i}(y) \pmod{2^{N-i}} \quad (0 \leq i \leq N),$$

so the first N digits of $Q_q(x)$ and $Q_q(y)$ agree. Conversely, if those N digits agree and contain m ones, the same-branch affine calculation gives

$$T_q^{\circ N}(x) - T_q^{\circ N}(y) = \frac{q^m}{2^N}(x - y).$$

The left side lies in $\mathbb{Z}_2$ and q is a 2-adic unit, hence $x \equiv y \pmod{2^N}$. Therefore

$$x \equiv y \pmod{2^N} \iff Q_q(x) \equiv Q_q(y) \pmod{2^N}, \quad (57)$$

which proves that Q_q is an isometry onto its image.

The finite sum in equation (55) is immediate; in the infinite case the series converges because $|q^{-\ell}2^{d_\ell}|_2 = 2^{-d_\ell} \rightarrow 0$. Writing $z = \delta + 2w$ with $\delta \in \{0, 1\}$ and separating the first one when $\delta = 1$ gives the exact recursions

$$\Phi_q(2w) = 2\Phi_q(w), \quad \Phi_q(1 + 2w) = \frac{2\Phi_q(w) - 1}{q}. \quad (58)$$

The first value is even and the second is odd. Applying the corresponding branch of T_q therefore gives

$$T_q(\Phi_q(z)) = \Phi_q(Sz).$$

Iterating shows that the parity digits of $\Phi_q(z)$ are exactly the digits of z , so $Q_q\Phi_q = \text{id}_{\mathbb{Z}_2}$. Hence Q_q is surjective; together with equation (57), it is bijective and $\Phi_q = Q_q^{-1}$. Dropping the first parity digit in equation (54) proves $Q_qT_q = SQ_q$, and substitution of the inverse gives the second conjugacy formula. The congruence equivalence for every N proves that both inverse maps are solenoidal isometries and hence homeomorphisms. $\square$

Bernstein–Lagarias Appendix B is broader still. Given arbitrary solenoidal bijections $V_0, V_1 : \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$, it treats

$$U_{V_0, V_1}(x) = \begin{cases} V_0(x/2), & x \equiv 0 \pmod{2}, \\ V_1((x-1)/2), & x \equiv 1 \pmod{2}, \end{cases}$$

and proves that its chronological parity encoder is solenoidal bijective and $U = Q^{-1}SQ$, with a converse classification [20, App. B, Thms. B.1–B.2]. The shortened T_q has $V_0(y) = y$ and $V_1(y) = qy + (q+1)/2$; the latter is an affine isometry of $\mathbb{Z}_2$ for every odd q , with inverse $y \mapsto q^{-1}(y - (q+1)/2)$. This proves the exact inclusion in that class; it does not identify Appendix B's full class with general Hydras.

Theorem 6.5 (reversed purely periodic overlap). *Let q be odd, let $L \geq 1$, and let $\varepsilon = (\varepsilon_0, \dots, \varepsilon_{L-1})$ be a chronological binary block. If $m \geq 1$, write the positions of its m ones as*

$$0 \leq v_0 < \dots < v_{m-1} < L;$$

when $m = 0$, every position list and sum below is empty. Set

$$e = \sum_{i=0}^{L-1} \varepsilon_i 2^i, \quad \text{rev}_L(e) = \sum_{i=0}^{L-1} \varepsilon_i 2^{L-1-i}. \quad (59)$$

Put

$$R_{q,\varepsilon} := \frac{\sum_{k=0}^{m-1} q^{m-k-1} 2^{v_k}}{2^L - q^m} \in \mathbb{Q}. \quad (60)$$

Then

$$\Phi_q\left(\frac{e}{1-2^L}\right) = \iota_2(R_{q,\varepsilon}) \in \mathbb{Z}_2.$$

If, in addition, q is an odd prime, so that the edition's global $\chi_q : \mathbb{Z}_2 \rightarrow \mathbb{Z}_q$ is available, then

$$\chi_q\left(\frac{\text{rev}_L(e)}{1-2^L}\right) = \iota_q(R_{q,\varepsilon}) \in \mathbb{Z}_q. \quad (61)$$

Thus the two values have the same rational representative $R_{q,\varepsilon}$ after reversal. The reversal in equation (61) is mandatory.

Proof. If $m = 0$, every displayed value is zero, so assume first that $m \geq 1$. The one positions of the purely periodic 2-adic integer $e/(1-2^L)$ are

$$d_{rm+k+1} = v_k + rL \quad (r \geq 0, 0 \leq k < m).$$

Substitution into equation (55) and a 2-adically convergent geometric sum give

$$\begin{aligned} \Phi_q\left(\frac{e}{1-2^L}\right) &= - \sum_{r \geq 0} \sum_{k=0}^{m-1} q^{-(rm+k+1)} 2^{v_k+rL} \\ &= - \left(\sum_{k=0}^{m-1} q^{-(k+1)} 2^{v_k} \right) \sum_{r \geq 0} \left(\frac{2^L}{q^m} \right)^r \\ &= \iota_2 \left(\frac{\sum_{k=0}^{m-1} q^{m-k-1} 2^{v_k}}{2^L - q^m} \right), \end{aligned}$$

because $|2^L/q^m|_2 = 2^{-L} < 1$. This proves the asserted Φ_q evaluation with the rational representative defined in equation (60).

Now put $h_0(x) = x/2$ and $h_1(x) = (qx + 1)/2$. Induction on the chronological block gives

$$h_{\varepsilon_{L-1}} \circ \cdots \circ h_{\varepsilon_0}(x) = \frac{q^m x + \sum_{k=0}^{m-1} q^{m-k-1} 2^{v_k}}{2^L}. \quad (62)$$

To see the induction with every index fixed, for $0 \leq h \leq L$ put

$$m_h = \sum_{i=0}^{h-1} \varepsilon_i, \quad C_h = \sum_{\substack{0 \leq k < m \\ v_k < h}} q^{m_h-k-1} 2^{v_k}.$$

The claim after the first h chronological branches is

$$h_{\varepsilon_{h-1}} \circ \cdots \circ h_{\varepsilon_0}(x) = \frac{q^{m_h} x + C_h}{2^h}.$$

At $h = 0$ the left side denotes the empty composite $\text{id}_{\mathbb{Q}}$, so the claim is the identity. For $0 \leq h < L$, assume the claim at level h . If $\varepsilon_h = 0$, applying h_0 changes only the denominator from 2^h to 2^{h+1} , while $m_{h+1} = m_h$ and $C_{h+1} = C_h$. If $\varepsilon_h = 1$, then

$$h_1\left(\frac{q^{m_h} x + C_h}{2^h}\right) = \frac{q^{m_h+1} x + qC_h + 2^h}{2^{h+1}}.$$

Here $m_{h+1} = m_h + 1$; multiplication of C_h by q raises every old exponent by one, and the new odd position $v_{m_h} = h$ contributes the term $q^{0 \cdot 2^h}$. Thus the new numerator is exactly $q^{m_{h+1}} x + C_{h+1}$. At $h = L$ this is equation (62). Solving its affine fixed-point equation gives $R_{q,\varepsilon}$; its denominator is nonzero because an odd integer power cannot equal 2^L .

Under Siegel's convention, the displayed word $H_{j_0} \circ \cdots \circ H_{j_{L-1}}$ applies j_{L-1} first. It realizes the chronological block precisely when

$$(j_0, \dots, j_{L-1}) = (\varepsilon_{L-1}, \dots, \varepsilon_0),$$

whose little-endian code is exactly $\text{rev}_L(e)$. For prime q and $m \geq 1$, its linear part is $q^m/2^L$, of q -adic norm $q^{-m} < 1$. The repeated-code theorem therefore evaluates its rising numen at its unique fixed point, namely $\iota_q(R_{q,\varepsilon})$. The $m = 0$ case is again the centered zero word. This proves equation (61).

The reversal is genuinely necessary in the universal formula, not merely a different way to name the same argument. Take $q = 3$, $L = 2$, and $\varepsilon = (0, 1)$. Then $e = 2$ and $\text{rev}_2(e) = 1$, while the formulas already proved give

$$\Phi_3\left(\frac{2}{1-4}\right) = \iota_2(2), \quad \chi_3\left(\frac{1}{1-4}\right) = \iota_3(2), \quad \chi_3\left(\frac{2}{1-4}\right) = \iota_3(1).$$

Indeed, $h_1 \circ h_0$ is $x \mapsto (3x+2)/4$ with fixed point 2, whereas $h_0 \circ h_1$ is $x \mapsto (3x+1)/4$ with fixed point 1. Thus deleting rev_L makes the asserted identity false. $\square$

Source issue 6.6 (the unrestricted Böhm–Sontacchi Proposition 4 iterate identity is false). For $q = 3$, the right side of equation (62) is the formal branch-composition expression obtained by specializing Böhm–Sontacchi Proposition 1. Their printed Proposition 4 displays this expression as $u^n(x)$ while quantifying over every $x \in \mathbb{Z}$ and every run tuple $(n_0, \dots, n_m)$. That unrestricted left side is false. Take $x = 0$, $m = 1$, and $n_0 = n_1 = 1$. Then $n = 3$ and $v_0 = 1$, so the printed right side is $1/4$, whereas $u^3(0) = 0$. The same right side is correctly the formal value

$$f \circ g \circ f(0) = \frac{1}{4}, \quad f(x) = \frac{x}{2}, \quad g(x) = \frac{3x+1}{2}.$$

Consequently this edition uses the formal-composition identity and identifies it with $u^n(x)$ only after proving that the word is the actual parity itinerary, as in theorem 6.9.

The fixed point $R_{3,\varepsilon}$ of that correct formal composition is the candidate printed in Böhm–Sontacchi Proposition 5 [19, p. 263]. Their statement literally prints $v_m = n$; after the explicit edition renaming $n = L$, it prints $v_{i-1} < v_i$ for $1 \leq i < m$ and $v_m = L$, but does not visibly print $v_{m-1} < v_m$. For $m \geq 1$, the last inequality follows from the formal run construction, where $v_m - v_{m-1} = 1 + n_m > 0$; it is a reconstruction, not a silent correction of Proposition 5. The unnumbered paragraph immediately after its proof explicitly says the candidate may lie on a cycle of length L or a proper divisor of L [19, p. 264]. Thus the independently recovered historical formula is a return-time identity, not an exact-period classification.

Corollary 6.7 (global nonidentity). *For every odd prime q , the paired values with common rational representatives in theorem 6.5 do not identify the global maps Φ_q and χ_q . They have types*

$$\Phi_q : \mathbb{Z}_2 \rightarrow \mathbb{Z}_2, \quad \chi_q : \mathbb{Z}_2 \rightarrow \mathbb{Z}_q,$$

The two values have distinct rational representatives:

$$\Phi_q(1) = \iota_2\left(-\frac{1}{q}\right), \quad \chi_q(1) = \iota_q\left(\frac{1}{2}\right), \quad -\frac{1}{q} \neq \frac{1}{2} \quad \text{in } \mathbb{Q}. \quad (63)$$

The map Q_q must likewise retain its role as the inverse chronological parity encoder in equation (56); it cannot be substituted for Φ_q by a notation change.

Proof. Formula equation (55) at the single one position $d_1 = 0$ gives the first embedded value. The one-letter numen is $\chi_q(1) = H_1(0) = \iota_q(1/2)$. Equality of the rational representatives would force $q = -2$, contrary to oddness. The displayed domains, codomains, and inverse relation prove the remaining typing assertions. $\square$

Laarhoven–de Weger give a graph-theoretic presentation of the same classical conjugacy. Their symbol Φ is the chronological graph isomorphism satisfying $T = \Phi^{-1}\sigma\Phi$, so in the notation fixed here it plays the role of Q_3 , not Φ_3 . Their odd- $an + b$ analogue is exact, whereas their p -ary discussion uses “appropriately chosen” coefficients and does not cover arbitrary Hydras [21, Thms. 3.1 and 5.1 and Sec. 5.2].

6.3. Branch correctness for T_q . The shortened map admits a particularly clean correctness argument in $\mathbb{Q}_2$. Extend the two affine formulas in equation (15) to all of $\mathbb{Q}_2$.

Lemma 6.8 (one correct branch). *Let q be odd and $x \in \mathbb{Z}_2$. Exactly one of $H_0(x), H_1(x)$ belongs to $\mathbb{Z}_2$, namely $H_{[x]_2}(x)$. If a wrong branch is applied, the result has 2-adic valuation -1 . If $v_2(y) < 0$, then*

$$v_2(H_0(y)) = v_2(H_1(y)) = v_2(y) - 1.$$

Proof. If x is even, $x/2 \in \mathbb{Z}_2$ and $qx + 1$ is odd, so $(qx + 1)/2 \notin \mathbb{Z}_2$. If x is odd, $x/2 \notin \mathbb{Z}_2$ and $qx + 1$ is even, so $(qx + 1)/2 \in \mathbb{Z}_2$. This proves the first two assertions. If $v_2(y) < 0$, then $v_2(qy) = v_2(y) < 0 = v_2(1)$; the ultrametric inequality with unequal valuations gives $v_2(qy + 1) = v_2(y)$. Division by 2 lowers either valuation by one. $\square$

Proposition 6.9 (formal fixed points in $\mathbb{Z}_2$ are genuine). *Let $\mathbf{j}$ be a finite binary word. If $z \in \mathbb{Z}_2$ and $H_{\mathbf{j}}(z) = z$, then every branch in the right-to-left evaluation of $H_{\mathbf{j}}(z)$ is correct. Consequently*

$$T_q^{|\mathbf{j}|}(z) = z.$$

Proof. Start with $z \in \mathbb{Z}_2$ and evaluate the rightmost branch first. If any branch is wrong, theorem 6.8 says the intermediate value leaves $\mathbb{Z}_2$ with negative valuation; every later branch decreases that valuation further. The final value then cannot equal $z \in \mathbb{Z}_2$. Hence no branch was wrong, and the formal composition is the genuine iterate with the reversed chronological itinerary, as in equation (27). $\square$

6.4. The checked T_q periodic-point correspondence. Assume now that q is an odd prime, so that the numen $\chi_q : \mathbb{Z}_2 \rightarrow \mathbb{Z}_q$ is defined everywhere by equation (49).

Theorem 6.10 (Siegel’s periodic-word correspondence for T_q). *Let $n \geq 1$, let $L = \lambda_2(n)$, and let $\mathbf{j}$ be the shortest least-significant-digit-first binary word for n . Then*

$$\chi_q(B_2(n)) = \frac{\chi_q(n)}{1 - M_q(n)}, \quad M_q(n) = \frac{q^{\#_1(n)}}{2^L}. \quad (64)$$

The right side is a rational number. If it belongs to $\mathbb{Z}_2$, it is a periodic point of the 2-adic extension of T_q ; if it belongs to $\mathbb{Z}$, it is a periodic integer point of T_q .

Conversely, let x be a nonzero periodic integer point of exact period L . If $\mathbf{j}$ is the length- L word obtained by reversing one complete chronological itinerary of x and $n = \text{Dig}_p(\mathbf{j})$, then

$$x = \chi_q(B_{2,L}(n)), \quad B_{2,L}(n) = \frac{n}{1 - 2^L}. \quad (65)$$

At least one cyclic phase of the cycle has a reversed word whose final digit is 1; for that phase $L = \lambda_2(n)$ and $B_{2,L}(n) = B_2(n)$. Other phases may require the padded length- L code.

Proof. The word contains at least one 1, hence $|M_q(n)|_q = q^{-\#_1(n)} < 1$. Apply theorem 6.1; in the scalar case it gives exactly equation (64). If the fixed point lies in $\mathbb{Z}_2$, theorem 6.9 turns the formal fixed-point equation into a genuine periodicity equation.

Conversely, let $x \neq 0$ have period L and let $(\varepsilon_0, \dots, \varepsilon_{L-1})$ be its chronological parity itinerary. Put $\mathbf{j} = (\varepsilon_{L-1}, \dots, \varepsilon_0)$, so $H_{\mathbf{j}}(x) = T_q^L(x) = x$. The word contains a 1, since an all-zero cycle would satisfy $x/2^L = x$ and hence $x = 0$. The affine fixed point is unique because $M_q(\mathbf{j}) = q^{\#_1(\mathbf{j})}/2^L \neq 1$: an odd power of the odd integer q cannot equal a positive power of 2. Therefore $x = (1 - M_q(\mathbf{j}))^{-1}\chi_q(\mathbf{j})$,

which is $\chi_q(B_{2,L}(n))$ by theorem 6.1. Rotating the chronological starting point rotates the word. One can choose a rotation whose high digit is nonzero and thereby use the shortest-word abbreviation $B_2(n)$, but that rotation generally changes x to another point of its cycle. $\square$

Source issue 6.11 (shortest words do not encode every cycle phase). The stronger version in the 2007 source overclaims that every point in the numen image of a cycle is represented by $B_2(n)$ for the shortest positive word of n . For T_3 , the point 2 on the cycle $1 \leftrightarrow 2$ is encoded by the length-two word $(1, 0)$; its high zero is part of the repeated block, even though the ordinary integer represented by that word is 1. The synchronized code is $B_{2,2}(1) = 1/(1-4) = -1/3$, not $B_2(1) = 1/(1-2) = -1$. The padded formula equation (65) preserves the phase. The source also writes a truncation modulus 2^m for m repeated length- L blocks where the typed modulus is 2^{mL} ; taking the correct subsequence repairs the limit.

Source issue 6.12 (the centered zero orbit). The 2026 Correspondence Principle is displayed for every periodic $z \in \mathcal{O}_K$ but requires a preimage in $(\mathbb{Z}_p \setminus \mathbb{N}_0) \cap \mathbb{Q}$. A centered Hydra has the periodic point 0, encoded by the all-zero word, whose digit-ring element is $0 \in \mathbb{N}_0$. No general argument in the source produces a non-eventually-zero rational code with numen value 0. The earlier T_q theorem correctly treats nonzero periodic points. The general statement therefore needs either an explicit exception for 0, inclusion of the code 0, or an additional surjectivity assertion about the zero fibre.

6.5. The general 2026 Correspondence Principle. We now record the latest source theorem without upgrading its proof status. Let $H : \mathcal{O}_K \rightarrow \mathcal{O}_K$ be a proper, centered, exact-integral Λ -Hydra and $p = |\mathcal{O}_K/\Lambda|$. Normalize a finite-place additive valuation by $v(K^\times) = \mathbb{Z}$ and write

$$v(\Lambda) = \min\{v(\lambda) : 0 \neq \lambda \in \Lambda\}.$$

The source assumes:

- (H1) for each integer $n \geq 1$ there is a place ℓ_n of K such that $\|M_H(n)\|_{\ell_n} < 1$;
- (H2) there is a nontrivial nonarchimedean place v such that $\|r_j\|_v > 1$ for every branch and $v(\Lambda) \geq 1$.

Status record 6.13 (General CP (source)). Under (H1)–(H2), arXiv:2601.17030v3 asserts:

- (i) an integral point is periodic if and only if it is the value of X_H at a rational, non-eventually-zero base- p digit input;
- (ii) if K is a number field and an irrational digit input has numen value $x \in \mathcal{O}_K$, then $|H^{om}(x)|_w \rightarrow \infty$ as $m \rightarrow \infty$ for every nontrivial archimedean absolute value $|\cdot|_w$.

The source labels its proof “Sketch.” The repeated-word affine calculation needed for the forward direction is proved in full in theorem 6.1. The converse still depends on the source’s completion-level “correctness” argument, and the divergence clause depends on an identification of fibres not established by the displayed hypotheses. Accordingly, this edition records (i)–(ii) as source claims, not as reconstructed theorems.

Source issue 6.14 (defects inside the 2026 proof sketch). The sketch contains five points which matter mathematically.

- (a) It writes M_q in a theorem about a general H ; the typed symbol is M_H .
- (b) It writes a quotient $(1 - M_H^m)/(1 - M_H)$ although M_H is generally an endomorphism. The typed expression is $(I - M_H^m)(I - M_H)^{-1}$, and invertibility must be proved.
- (c) The proof of the divergence clause invokes a hypothesis “(iii)”, but the theorem lists only (i) and (ii).
- (d) From the periodic-point correspondence, existence of a rational code for a periodic value does not imply that every code in the same numen fibre is rational. Such an inference requires injectivity or a precise fibre theorem, neither of which is part of the displayed statement.

- (e) In dimension greater than one, $\|r_j\|_v > 1$ is only a statement about the *largest* expansion of r_j . It does not imply $|r_j x|_v > |x|_v$ for every nonzero x , which is the kind of branch-detection estimate used by the sketch. A sufficient typed replacement is a lower expansion bound $\inf_{x \neq 0} |r_j x|_v / |x|_v > 1$, equivalently $\|r_j^{-1}\|_v < 1$ for invertible r_j , together with the required translation/integrality control. In one scalar dimension the upper and lower factors coincide, but the general endomorphism statement cannot use the operator supremum alone.

These are not stylistic complaints; they delimit what has and has not been proved by the sketch.

6.6. Weak Collatz versus full Collatz. For T_3 , the *weak Collatz conjecture* in Siegel's contour paper is a periodic-orbit assertion: the only integer cycles are the known ones (with the sign and state-space convention stated in that paper). The full Collatz conjecture additionally requires every positive integer orbit to enter the positive $1 \leftrightarrow 2$ cycle. A characterization of periodic points, even a complete one, does not rule out divergent positive trajectories. The contour equivalence in section 8 is therefore kept under the name *weak*; it is not promoted to an equivalence with the full conjecture.

7. SET-SERIES, DREAMCATCHERS, AND SINGULARITY CONSERVATION

Siegel's 2019 paper encodes invariant sets by holomorphic generating series and then retains only their dominant radial boundary data. The resulting ℓ^2 object is called a *dreamcatcher*. This is a different analytic route from the numen: the numen codes branch compositions, whereas the permutation operator below acts on coefficient functions and detects sets invariant under inverse image.

7.1. Set-series and the historical Hydra class. For $V \subseteq \mathbb{N}_0$, define

$$\varsigma_V(w) = \sum_{v \in V} w^v, \quad |w| < 1, \quad (66)$$

$$\psi_V(z) = \varsigma_V(e^{2\pi i z}) = \sum_{v \in V} e^{2\pi i v z}, \quad \Im z > 0. \quad (67)$$

The coefficients are the indicator function $\mathbf{1}_{n \in V}$. The 2019 paper also uses exponential and Dirichlet set-series. Its displayed Dirichlet series $\sum_{v \in V} v^{-s}$ requires either $0 \notin V$ or an explicit treatment of the term $v = 0$.

The Hydra used in that paper is the following one-dimensional historical specialization. It is not the controlling 2026 definition.

Definition 7.1 (2019 ϱ -Hydra). Fix $\varrho \geq 2$. A 2019 ϱ -Hydra is a surjection $H : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ with

$$H(n) = \frac{a_j n + b_j}{d_j} \quad (n \equiv j \pmod{\varrho}), \quad (68)$$

where $a_j, d_j \geq 1$, $b_j \geq 0$, $\gcd(a_j, d_j) = 1$, and each assigned branch is integer-valued and nonnegative. Set

$$\mu_j = \frac{\varrho a_j}{d_j} \in \mathbb{N}. \quad (69)$$

It is *prime* when ϱ is prime and every a_j is 1 or a prime. It is *regulated* when $\mu_j = 1$ for at least one j .

The integrality of equation (69) follows by comparing $H(\varrho m + j)$ with $H(j)$, and coprimality then gives $d_j \mid \varrho$. Surjectivity is an extra hypothesis of the 2019 analytic framework; it is not part of the 2026 word formalism.

7.2. Permutation operators. Let $\ell^\infty(\mathbb{N}_0)$ carry its supremum norm. The coefficient pullback is the linear map

$$\mathbf{P}_H : \ell^\infty(\mathbb{N}_0) \longrightarrow \ell^\infty(\mathbb{N}_0), \quad (\mathbf{P}_H c)_n = c_{H(n)}.$$

Because the 2019 Hydra H is surjective,

$$\|\mathbf{P}_H c\|_\infty = \sup_{n \geq 0} |c_{H(n)}| = \sup_{m \geq 0} |c_m| = \|c\|_\infty;$$

thus $\mathbf{P}_H$ is an isometric injection (not necessarily a surjection). For $c \in \ell^\infty(\mathbb{N}_0)$, both

$$\mathcal{G}_{\mathbb{D}} c(w) = \sum_{n \geq 0} c_n w^n \quad (|w| < 1), \quad \mathcal{G}_{\mathbb{H}} c(z) = \sum_{n \geq 0} c_n e^{2\pi i n z} \quad (\Im z > 0)$$

converge normally on compact subsets of their displayed domains. The two analytic operators are therefore the conjugates of this specified coefficient map on the ranges of the injective generating maps:

$$\mathcal{Q}_H : \text{ran}(\mathcal{G}_{\mathbb{D}}) \longrightarrow \text{ran}(\mathcal{G}_{\mathbb{D}}), \quad \mathcal{Q}_H : \text{ran}(\mathcal{G}_{\mathbb{H}}) \longrightarrow \text{ran}(\mathcal{G}_{\mathbb{H}}).$$

$$\mathcal{Q}_H(\mathcal{G}_{\mathbb{D}} c) = \mathcal{G}_{\mathbb{D}}(\mathbf{P}_H c) = \sum_{n \geq 0} c_{H(n)} w^n, \quad (70)$$

$$\mathcal{Q}_H(\mathcal{G}_{\mathbb{H}} c) = \mathcal{G}_{\mathbb{H}}(\mathbf{P}_H c) = \sum_{n \geq 0} c_{H(n)} e^{2\pi i n z}. \quad (71)$$

For set-series this simply records inverse image:

$$\mathcal{Q}_H \psi_V = \psi_{H^{-1}(V)}, \quad \mathcal{Q}_H \psi_V = \psi_{H^{-1}(V)}. \quad (72)$$

Thus $H^{-1}(V) = V$ if and only if the corresponding set-series is a fixed point.

Let $\xi_m = e^{2\pi i/m}$. Splitting the input index modulo ϱ and using

$$H(\varrho n + j) = \mu_j n + H(j)$$

gives, for every $\psi \in \text{ran}(\mathcal{G}_{\mathbb{H}})$, the finite branch formula

$$(\mathcal{Q}_H \psi)(z) = \sum_{j=0}^{\varrho-1} \frac{1}{\mu_j} \sum_{k=0}^{\mu_j-1} \xi_{\mu_j}^{-kH(j)} e^{-2\pi i(b_j/a_j)z} \psi\left(\frac{\varrho z + k}{\mu_j}\right). \quad (73)$$

To verify the formula, expand the right side. The inner root-of-unity sum selects coefficients whose indices are congruent to $H(j)$ modulo μ_j ; the exponential prefactor converts the surviving exponent back to $\varrho n + j$.

Proposition 7.2 (fixed coefficient functions). *For a coefficient function $c : \mathbb{N}_0 \rightarrow \mathbb{C}$, define its coefficient pullback by*

$$(\mathbf{P}_H c)_n = c_{H(n)}.$$

Then $\mathbf{P}_H c = c$ if and only if

$$c_n = c_{H(n)} \quad (n \geq 0). \quad (74)$$

Equivalently, c is constant on every total orbit class of H .

Proof. By definition, $\mathbf{P}_H c = c$ is exactly equation (74). Iterating that identity makes c constant along forward orbits; if two points have a common forward image, their coefficients agree. Conversely, constancy on orbit classes implies equation (74). $\square$

Source issue 7.3 (“basis” in the Permutation Operator Theorem). The source says that the set-series of all orbit classes form a basis of the fixed-point space. Theorem 7.2 gives the exact coefficient-level statement. If infinitely many orbit classes have nonzero coefficients, the resulting decomposition is an infinite sum, not a finite Hamel-basis expansion. A topological basis claim requires a named function-space topology and a convergence proof.

7.3. Edgepoints and normalization. Let $\mathbb{H}_{+i} = \{z \in \mathbb{C} : \Im z > 0\}$ and $\psi \in \mathcal{A}(\mathbb{H}_{+i})$.

Definition 7.4 (edgepoint and virtual residue). An $x \in \mathbb{R}$ is an *edgepoint* of ψ when the finite limit

$$\mathrm{VRes}_x[\psi] = \lim_{y \downarrow 0} y\psi(x + iy)$$

exists. It is a *virtual pole* when that limit is nonzero. Write

$$P(\psi) = \{x \in \mathbb{R} : x \text{ is an edgepoint of } \psi\}.$$

For a Fourier set-series, the elementary estimate

$$|\psi_V(x + iy)| \leq \sum_{n \geq 0} e^{-2\pi ny} = \frac{1}{1 - e^{-2\pi y}}$$

implies

$$\lim_{y \downarrow 0} y^{1+\delta} \psi_V(x + iy) = 0 \quad (x \in \mathbb{R}, \delta > 0). \quad (75)$$

Thus the virtual residue records the largest possible radial growth order for a digital Fourier series.

Source issue 7.5 (ordinary versus virtual residues). If ψ has an ordinary simple pole at x with complex residue A , then $y\psi(x + iy) \rightarrow A/i = -iA$. The source's definition of a “classical virtual pole” subtracts $\mathrm{VRes}_x[\psi]/(z - x)$ and thereby omits this factor of i . This edition retains the term *virtual residue* for the radial normalization above but does not identify it numerically with the ordinary complex residue.

Proposition 7.6 (cross-section densities give the virtual residue). *Let $\alpha \geq 1$, $\beta \in \mathbb{Z}$, and assume that every limit*

$$d_k = \lim_{N \rightarrow \infty} \frac{1}{N} |V \cap (k + \alpha\mathbb{Z}) \cap [0, N]| \quad (0 \leq k < \alpha)$$

exists. Then β/α is an edgepoint of ψ_V and

$$\mathrm{VRes}_{\beta/\alpha}[\psi_V] = \frac{1}{2\pi} \sum_{k=0}^{\alpha-1} d_k \xi_\alpha^{\beta k}. \quad (76)$$

In particular, if V has natural density $d(V)$, then $\mathrm{VRes}_0[\psi_V] = d(V)/(2\pi)$.

Proof. Put $a_n = \mathbf{1}_{n \in V} \xi_\alpha^{\beta n}$. Splitting a Cesàro sum by residue class gives

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} a_n = \sum_{k=0}^{\alpha-1} d_k \xi_\alpha^{\beta k} =: L.$$

The elementary Abel implication for a bounded sequence now gives

$$(1 - r) \sum_{n \geq 0} a_n r^n \rightarrow L \quad (r \uparrow 1).$$

For completeness, if $A_N = \sum_{n < N} (a_n - L)$, then $A_N = o(N)$ and summation by parts gives $(1 - r) \sum_{n \geq 0} (a_n - L) r^n = (1 - r)^2 \sum_{N \geq 1} A_N r^{N-1} \rightarrow 0$; split the last sum at a fixed index after bounding $|A_N|/N$ on its tail. Finally set $r = e^{-2\pi y}$. Since $y/(1 - e^{-2\pi y}) \rightarrow 1/(2\pi)$,

$$y\psi_V(\beta/\alpha + iy) = \frac{y}{1 - r} (1 - r) \sum_{n \geq 0} a_n r^n \rightarrow \frac{L}{2\pi},$$

which proves the assertion. Only the Abel implication is used; no Tauberian converse is invoked. $\square$

Dreamcatchers therefore contain residue-class density data, not merely a list of analytic singularities.

7.4. The Square Sum Lemma. Equip subsets of $\mathbb{R}/\mathbb{Z}$ with counting measure. The result below is stated in Siegel's paper, which credits the proof idea to accepted MathOverflow answer 333801 by user *fedja* [16]. The argument below is instead this edition's finite-subset Bessel proof; it is not attributed to the answerer. It avoids the source proof's interchange of infinite sums before summability has been established.

Lemma 7.7 (Square Sum Lemma). *Let $V \subseteq \mathbb{N}_0$ and put $P' = P(\psi_V) \cap [0, 1)$. Then*

$$\sum_{t \in P'} |\text{VRes}_t[\psi_V]|^2 \leq \frac{1}{(2\pi)^2}. \quad (77)$$

In particular, the virtual-residue function belongs to $\ell^2(P')$ and has at most countable nonzero support.

Proof. Let $a_n = \mathbf{1}_{n \in V}$, let $r = e^{-2\pi y}$, and put on bounded sequences the semidefinite inner product

$$\langle u, v \rangle_y = (1 - r) \sum_{n \geq 0} u_n \overline{v_n} r^n.$$

For $t \in \mathbb{R}/\mathbb{Z}$, set $e_t(n) = e^{-2\pi i n t}$. Then

$$\begin{aligned} \langle e_s, e_t \rangle_y &= \frac{1 - r}{1 - r e^{2\pi i(t-s)}} \longrightarrow \mathbf{1}_{s=t}, \\ \langle a, e_t \rangle_y &= (1 - r) \psi_V(t + iy) \longrightarrow 2\pi \text{VRes}_t[\psi_V], \\ \|a\|_y^2 &= (1 - r) \sum_{n \geq 0} a_n r^n \leq 1. \end{aligned}$$

Fix a finite $F \subseteq P'$. The Gram matrix $G_y = (\langle e_s, e_t \rangle_y)_{s,t \in F}$ tends to the identity and is invertible for small y . The finite-dimensional projection inequality is

$$b_y^* G_y^{-1} b_y \leq \|a\|_y^2, \quad b_y = (\langle a, e_t \rangle_y)_{t \in F}.$$

Passing to the limit yields $(2\pi)^2 \sum_{t \in F} |\text{VRes}_t[\psi_V]|^2 \leq 1$. Taking the supremum over finite F gives equation (77). $\square$

7.5. Dreamcatchers and branch-invariant carriers.

Definition 7.8 (dreamcatcher over T). Let $\psi \in \mathcal{A}(\mathbb{H}_{+i})$, and let $T \subseteq P(\psi)$ be 1-periodic. The function ψ admits a dreamcatcher over T when

$$R_{\psi,T}(x) = \text{VRes}_x[\psi] \quad (x \in T) \quad (78)$$

belongs to $\ell^2(T/\mathbb{Z})$. Following the source, denote the class of such functions by $\mathcal{D}_1(T)$.

The carrier T is H -branch invariant when

$$\frac{\varrho T + k}{\mu_j} \subseteq T \quad (0 \leq j < \varrho, \ 0 \leq k < \mu_j). \quad (79)$$

This ensures that every virtual residue appearing after applying a branch in equation (73) is defined on the same carrier.

Lemma 7.9 (Virtual Residue Formula). *Let T be H -branch invariant and let $\psi \in \mathcal{D}_1(T) \cap \text{ran}(\mathcal{G}_{\mathbb{H}})$. At every $t \in T$, the function $\mathcal{Q}_H \psi$ has an edgepoint and*

$$\text{VRes}_t[\mathcal{Q}_H \psi] = \frac{1}{\varrho} \sum_{j=0}^{\varrho-1} e^{-2\pi i(b_j/a_j)t} \sum_{k=0}^{\mu_j-1} \xi_{\mu_j}^{-kH(j)} R_{\psi,T} \left(\frac{\varrho t + k}{\mu_j} \right). \quad (80)$$

Proof. Multiply equation (73) by y and substitute $z = t + iy$. The exponential prefactor converges to its value at t . For $A_{j,k}(z) = (\varrho z + k)/\mu_j$, the imaginary part of $A_{j,k}(t + iy)$ is $\varrho y/\mu_j$, hence

$$y\psi(A_{j,k}(t + iy)) = \frac{\mu_j}{\varrho} \left(\frac{\varrho y}{\mu_j} \right) \psi \left(A_{j,k}(t) + i \frac{\varrho y}{\mu_j} \right).$$

The factor $1/\mu_j$ in equation (73) cancels the μ_j here, leaving $1/\varrho$. The sum is finite, so its limit is the sum of the limits. $\square$

Definition 7.10 (dreamcatcher operator). For $R \in \ell^2(T/\mathbb{Z})$ let $\mathcal{E}_{H:T}R : T \rightarrow \mathbb{C}$ denote the raw finite expression

$$(\mathcal{E}_{H:T}R)(x) = \frac{1}{\varrho} \sum_{j=0}^{\varrho-1} e^{-2\pi i(b_j/a_j)x} \sum_{k=0}^{\mu_j-1} \xi_{\mu_j}^{-kH(j)} R \left(\frac{\varrho x + k}{\mu_j} \right). \quad (81)$$

Here $R(y)$ means the value on the class $y + \mathbb{Z}$; branch invariance ensures that every argument belongs to T . Define the maximal graph domain

$$\text{Dom}(\mathfrak{Q}_{H:T}) = \{R \in \ell^2(T/\mathbb{Z}) : \mathcal{E}_{H:T}R \text{ is 1-periodic on } T \text{ and } [\mathcal{E}_{H:T}R] \in \ell^2(T/\mathbb{Z})\}.$$

The dreamcatcher operator is the linear operator

$$\mathfrak{Q}_{H:T} : \text{Dom}(\mathfrak{Q}_{H:T}) \subseteq \ell^2(T/\mathbb{Z}) \longrightarrow \ell^2(T/\mathbb{Z}), \quad R \longmapsto [\mathcal{E}_{H:T}R].$$

Theorem 7.11 (Singularity Conservation Law). *Let T be H -branch invariant, let $\psi \in \mathcal{D}_1(T) \cap \text{ran}(\mathcal{G}_{\mathbb{H}})$ be fixed by $\mathcal{Q}_H$, and put $R = R_{\psi,T}$. Suppose $R \in \text{Dom}(\mathfrak{Q}_{H:T})$. Then*

$$\mathfrak{Q}_{H:T}R = R. \quad (82)$$

Proof. Since $\mathcal{Q}_H\psi = \psi$, the left side of equation (80) is $\text{VRes}_t[\psi] = R(t)$. Its right side is $\mathcal{E}_{H:T}R(t)$, whose descended class is $\mathfrak{Q}_{H:T}R$. Equality holds pointwise on T and hence in $\ell^2(T/\mathbb{Z})$. $\square$

Source issue 7.12 (undefined class and operator domain). The Virtual Residue Formula and SCL in the source assume $\psi \in \mathcal{D}_2(T)$, but only $\mathcal{D}_1(T)$ is defined. The intended symbol appears to be $\mathcal{D}_1(T)$. More seriously, the proof that equation (81) preserves $\ell^2(T/\mathbb{Z})$ treats the inclusion in equation (79) as saying that each affine branch map is a bijection of T . A bijection of $\mathbb{R}$ whose image sends T into T need not map T onto itself. The typed SCL above therefore uses the exact maximal graph domain and makes membership an explicit hypothesis for the particular dreamcatcher. It does not assert that the graph domain is all of $\ell^2(T/\mathbb{Z})$ or that the operator is bounded. Either conclusion needs two-sided branch invariance or a separate finite-multiplicity norm estimate.

A function ψ is *tame* in the source when every rational number is an edgepoint. In that case $T = \mathbb{Q}$ is available and, by theorem 7.7, the rational virtual residues of a Fourier set-series form an element of $\ell^2(\mathbb{Q}/\mathbb{Z})$.

7.6. The Finite Dreamcatcher Theorem. The later half of the 2019 paper studies $\ker(I - \mathfrak{Q}_H)$ purely on $\ell^2(\mathbb{Q}/\mathbb{Z})$. Its central statement is the following.

Status record 7.13 (Finite Dreamcatcher Theorem, source theorem). Let H be a prime, regulated 2019 Hydra. If

$$R \in \ker(I - \mathfrak{Q}_{H:\mathbb{Q}}) \subseteq \ell^2(\mathbb{Q}/\mathbb{Z})$$

and $\text{supp } R$ is finite modulo $\mathbb{Z}$, then

$$R(x) = R(0)\mathbf{1}_{x \in \mathbb{Z}} \quad (x \in \mathbb{Q}). \quad (83)$$

The source proof proceeds through the Off- ϱ Theorem, an auxiliary functional equation, two support propositions, and the Off- ϱ Infinity Theorem. This edition records the result with that proof status; it does not use the earlier flawed $\ell^2(T)$ -invariance argument as an independent proof of the finite-support theorem.

The result says that a finitely supported conserved singularity pattern has no nonintegral rational frequency. Combining it with the SCL gives the paper’s first weak rationality consequence.

Status record 7.14 (Weak Rationality Theorem 1, source consequence). If $V \subseteq \mathbb{N}_0$ is rational in the sense that its ordinary generating function is rational, and if $H^{-1}(V) = V$ for a prime regulated Hydra, then V is finite or cofinite in $\mathbb{N}_0$.

Indeed, rationality makes the virtual-pole support finite modulo $\mathbb{Z}$; invariance gives a permutation-operator fixed point; SCL and FDT force the only possible dominant rational boundary frequency to be 0. The remaining step uses the eventual periodicity of a 0–1 sequence with rational generating function.

Source issue 7.15 (results not promoted from the 2019 paper). Three later claims are kept out of the proved chain.

- (i) The proof of Weak Rationality Theorem 2 asserts that naturally measurable subsets of $\mathbb{N}_0$ form a σ -algebra and that natural density is countably additive. Both assertions are false in general, so that theorem is not canonized as proved.
- (ii) Theorem 12 assumes branch invariance of both the rational edgepoints and their rational complement. The source itself calls this double requirement the method’s “Achilles heel”; the edition retains the result only with that hypothesis if cited.
- (iii) The assertion that a limit of dreamcatchers is the dreamcatcher of the limiting function is explicitly Conjecture 1. A nearby corollary also uses an asymptotic fixed-point premise not present in its statement.

8. DIRICHLET SERIES, PERRON INVERSION, AND THE CONTOUR PROGRAMME

The paper designated Part I.5 moves the fixed-point numerator

$$(1 - M_q(n))x - \chi_q(n)$$

into the coefficients of a Dirichlet series. Perron inversion can then recover that numerator by a vertical contour integral. The coefficient extraction is sound; the advertised equivalence with every periodic point inherits the shortest-word defect isolated in equation (65).

8.1. The three Dirichlet series. Let q be an odd prime and $x \in \mathbb{C}$. With $M_q(n)$ defined using the shortest binary word of $n \geq 1$, put

$$\zeta_{M_q}(s) = \sum_{n=1}^{\infty} \frac{M_q(n)}{n^s}, \quad (84)$$

$$\zeta_{\chi_q}(s) = \sum_{n=1}^{\infty} \frac{\chi_q(n)}{n^s}, \quad (85)$$

$$\begin{aligned} F_q(s, x) &= \sum_{n=1}^{\infty} \frac{(1 - M_q(n))x - \chi_q(n)}{n^s} \\ &= (\zeta(s) - \zeta_{M_q}(s))x - \zeta_{\chi_q}(s). \end{aligned} \quad (86)$$

The source writes

$$\sigma_q = \log_2 \left(\frac{q+1}{2} \right) \quad (87)$$

for the common absolute-convergence threshold obtained from its summatory estimates. All contour identities below are first asserted on a line $\Re s = c > \sigma_q$.

8.2. Dyadic summation. The digit recursions imply exact block sums. Write

$$A_N = \sum_{0 \leq n < 2^N} \chi_q(n), \quad B_N = \sum_{0 \leq n < 2^N} M_q(n),$$

with $M_q(0) = 1$. Separating even and odd indices gives

$$A_{N+1} = \frac{q+1}{2} A_N + 2^{N-1}, \quad (88)$$

$$B_{N+1} = \frac{q+1}{2} B_N + \frac{1}{2}. \quad (89)$$

Solving these recurrences yields

$$A_N = \begin{cases} N2^N/4, & q = 3, \\ \frac{2^N}{q-3} \left[\left(\frac{q+1}{4} \right)^N - 1 \right], & q \neq 3, \end{cases} \quad (90)$$

$$B_N = \frac{q \left(\frac{q+1}{2} \right)^N - 1}{q-1}. \quad (91)$$

Proof. For equation (88), use

$$\chi_q(2m) + \chi_q(2m+1) = \frac{q+1}{2} \chi_q(m) + \frac{1}{2}$$

and sum over $0 \leq m < 2^N$. For equation (89), the usual $M_q(2m) = M_q(m)/2$ fails only at $m = 0$, because the shortest-word convention sets $M_q(0) = 1$. Hence the even sum is $1 + \frac{1}{2}(B_N - 1)$ and the odd sum is $qB_N/2$. The initial values are $A_0 = 0$ and $B_0 = 1$; elementary solution of the recurrences gives the two closed forms. $\square$

These identities explain the appearance of σ_q . A guaranteed absolute-convergence half-plane for a linear combination is the intersection of the component half-planes and therefore uses the maximum of their abscissae, not the source's printed "minimum." The exact abscissa of a particular linear combination can be smaller when leading terms cancel, so no equality of abscissae is inferred for arbitrary complex x .

8.3. Perron extraction of one coefficient. Let

$$a_n(x) = (1 - M_q(n))x - \chi_q(n).$$

For $0 < a < 1$, $n \geq 3$, and $c > \sigma_q$, the Perron difference kernel isolates $a_n(x)$:

Theorem 8.1 (coefficient contour identity). *Under the preceding hypotheses,*

$$a_n(x) = \text{PV} \left\{ \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{(n+a)^s - (n+a-1)^s}{s} F_q(s, x) ds \right\}. \quad (92)$$

Proof. For a Dirichlet series $F(s) = \sum_{m \geq 1} a_m m^{-s}$ absolutely convergent on $\Re s = c$, Perron's formula gives

$$\text{PV} \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s) X^s \frac{ds}{s} = \sum_{m < X} a_m$$

when X is not an integer. Apply this with $X = n + a$ and with $X = n + a - 1$. Since $0 < a < 1$, the two index sets differ only by $m = n$. Subtracting gives equation (92). $\square$

Consequently the integral vanishes exactly when

$$(1 - M_q(n))x = \chi_q(n). \quad (93)$$

This is a coefficient theorem. Its dynamical content is precisely whatever fixed-point correspondence is valid for the shortest word representing n .

Status record 8.2 (the printed contour “weak Collatz” theorem). The source states that periodicity is equivalent to the vanishing of equation (92) for some $n \geq 3$, with separate exclusions:

- (i) for $q = 3$, $x \in \mathbb{Z} \setminus \{-1, 0, 1\}$;
- (ii) for $q = 5$, $x \in \mathbb{Z} \setminus \{-1, 0\}$;
- (iii) for $q \geq 7$, apparently $x \in \mathbb{Z} \setminus \{0\}$, although the printed set is malformed.

Each part needs its own contour line $c > \sigma_3$, $c > \sigma_5$, or $c > \sigma_q$; the outer quantifiers in the printed theorem do not state this correctly.

Source issue 8.3 (the contour equivalence misses padded cycle words). The claimed biconditional is not valid for every listed periodic phase. For example, 2 is a periodic point of T_3 and is not among the source’s exclusions. If a shortest word fixed 2, branch correctness would force it to be a whole number of repetitions of the reversed itinerary (1, 0). Every such word has a high terminal zero and therefore is not shortest. Equivalently, $2 = \chi_3(B_{2,2}(1))$ uses the padded length-two block, while no shortest positive block yields equation (93) with $x = 2$. Thus theorem 8.1 is canonical, but the full periodicity equivalence is retained only as a source claim. A repaired phase-sensitive contour formulation must index the block length separately from its ordinary digit sum.

8.4. Corrected functional equations for f_q and F_q . Extend the coefficient sequence by $a_0(x) = 0$, and set

$$C_q(x) = \frac{(q-2)x + 1}{2}. \quad (94)$$

The digit recursions determine both the ordinary and Dirichlet generating functions without an analytic-continuation premise.

Proposition 8.4 (coefficient and ordinary generating-function recursions). *For $n \geq 1$ and $n \geq 0$, respectively,*

$$a_{2n}(x) = \frac{a_n(x) + x}{2}, \quad a_{2n+1}(x) = \frac{q}{2}a_n(x) - C_q(x). \quad (95)$$

Consequently, in the formal power-series ring $\mathbb{C}[[z]]$, and analytically for $|z| < 1$,

$$f_q(z, x) := \sum_{n=1}^{\infty} a_n(x)z^n = \frac{qz + 1}{2}f_q(z^2, x) + \frac{xz^2 - ((q-2)x + 1)z}{2(1 - z^2)}. \quad (96)$$

Proof. For $n \geq 1$, the shortest-word convention and equations (45) and (46) give

$$M_q(2n) = \frac{1}{2}M_q(n), \quad M_q(2n+1) = \frac{q}{2}M_q(n),$$

and the odd identities remain valid at $n = 0$ because $M_q(0) = 1$ and $\chi_q(0) = a_0(x) = 0$. Substitution in $a_n(x) = (1 - M_q(n))x - \chi_q(n)$ proves equation (95). Splitting the formal series into its positive even indices and its nonnegative odd indices gives

$$\begin{aligned} \sum_{n \geq 1} a_{2n}(x)z^{2n} &= \frac{1}{2}f_q(z^2, x) + \frac{xz^2}{2(1 - z^2)}, \\ \sum_{n \geq 0} a_{2n+1}(x)z^{2n+1} &= \frac{qz}{2}f_q(z^2, x) - \frac{C_q(x)z}{1 - z^2}. \end{aligned}$$

Their sum is equation (96). If the shortest word of n has length L , then $M_q(n) \leq (q/2)^L$ and $\chi_q(n) \leq \frac{1}{2} \sum_{j=0}^{L-1} (q/2)^j$. Thus $a_n(x)$ has polynomial growth in n , so the formal identity converges absolutely for $|z| < 1$. $\square$

Theorem 8.5 (corrected Dirichlet functional equation). *Let q be an odd prime, $x \in \mathbb{C}$, and $\Re s > \sigma_q$. Then every series below is absolutely convergent and*

$$F_q(s, x) = \frac{((q-1)x+1) - 2^s((q-2)x+1)}{2(2^s - 2^{\sigma_q})} \zeta(s) + \frac{q/2}{2^s - 2^{\sigma_q}} \sum_{m=1}^{\infty} \left(-\frac{1}{2}\right)^m \binom{s+m-1}{m} F_q(s+m, x). \quad (97)$$

The series over m converges locally uniformly in that half-plane.

Proof. Put $\alpha = (q+1)/2 = 2^{\sigma_q}$. Positivity of $M_q(n)$ and $\chi_q(n)$, together with equations (90) and (91), gives

$$\sum_{1 \leq n < 2^N} |a_n(x)| \leq |x|(2^N + B_N) + A_N.$$

For $q > 3$ the right side is $O_{q,x}(\alpha^N)$; for $q = 3$ it is $O_x(N2^N)$. Dyadic subdivision therefore proves absolute convergence of $F_q(s, x)$ for $\Re s > \sigma_q$.

Split the Dirichlet series by parity. The even recursion gives

$$\sum_{n \geq 1} \frac{a_{2n}(x)}{(2n)^s} = 2^{-s-1} (F_q(s, x) + x\zeta(s)).$$

For the odd part, the binomial series at $1/(2n)$ gives

$$\sum_{n \geq 1} \frac{a_n(x)}{(2n+1)^s} = 2^{-s} \sum_{m=0}^{\infty} \left(-\frac{1}{2}\right)^m \binom{s+m-1}{m} F_q(s+m, x). \quad (98)$$

To justify the interchange exactly, let K be a compact subset of $\Re s > \sigma_q$, put $u = \inf_{s \in K} \Re s$, and choose $R \geq \max(1, \sup_{s \in K} |s|)$. From the product formula,

$$\left| \binom{s+m-1}{m} \right| \leq R \prod_{j=1}^{m-1} \left(1 + \frac{R-1}{j+1}\right) \leq Re^{R-1} (m+1)^{R-1}.$$

Consequently the absolute value of the double series is bounded uniformly on K by

$$Re^{R-1} \left(\sum_{m \geq 0} 2^{-m} (m+1)^{R-1} \right) \left(\sum_{n \geq 1} |a_n(x)| n^{-u} \right) < \infty.$$

This proves both interchanges and local uniformity. Since

$$\sum_{n \geq 0} (2n+1)^{-s} = (1 - 2^{-s}) \zeta(s)$$

the odd recursion in equation (95) and equation (98) yield

$$F_q(s, x) = 2^{-s-1} (F_q(s, x) + x\zeta(s)) + \frac{q}{2} 2^{-s} \sum_{m=0}^{\infty} \left(-\frac{1}{2}\right)^m \binom{s+m-1}{m} F_q(s+m, x) - C_q(x)(1 - 2^{-s})\zeta(s).$$

Move the $m = 0$ and even copies of $F_q(s, x)$ to the left, multiply by 2^s , and use $x/2 + C_q(x) = ((q-1)x+1)/2$. This is equation (97). $\square$

Source issue 8.6 (the printed functional equation is false for $x \neq 0$). The source substitutes $z/(1-z)$ where the first even correction in $f_q(z^2, x)$ is $z^2/(1-z^2)$. Its resulting Dirichlet equation has zeta coefficient

$$-\frac{2^s((q-1)x+1)-1}{2(2^s-2^{\sigma_q})}.$$

As real $s \rightarrow +\infty$, the left side tends to $a_1(x) = -((q-2)x+1)/2$. The shifted series is bounded in this limit: the binomial identity and the estimate $F_q(s+m, x) = a_1(x) + O(2^{-s-m})$ give

$$\sum_{m \geq 1} \left(-\frac{1}{2}\right)^m \binom{s+m-1}{m} F_q(s+m, x) = a_1(x)((2/3)^s - 1) + O((2/3)^s).$$

The printed right side therefore tends to $-((q-1)x+1)/2$. Thus the printed identity fails for every $x \neq 0$; equation (97) is the corrected identity. No continuation or singularity claim is inherited from the false equation. A fresh continuation proof from the corrected equation is given below. The denominator lattice alone still does not establish an actual pole, because the numerator can vanish. The coefficient contour identity is unaffected: it uses only the original absolutely convergent Dirichlet series on $\Re s > \sigma_q$.

8.5. Meromorphic continuation from the corrected equation. Part I.5 uses its shift equation to continue F_q , and the public Web II exposition also reports a meromorphic continuation with a leftward half-lattice of singularities [7, 15]. Because the printed equation is false, the conclusion must be rebuilt from equation (97). Put

$$b_m(s) = \left(-\frac{1}{2}\right)^m \binom{s+m-1}{m}, \quad (99)$$

$$A_q(s, x) = \frac{((q-1)x+1) - 2^s((q-2)x+1)}{2}, \quad (100)$$

$$S_q(s, x) = \sum_{m=1}^{\infty} b_m(s) F_q(s+m, x). \quad (101)$$

Lemma 8.7 (normal convergence of the shifted tail). *For every compact $K \subset \mathbb{C}$, there are constants $C_K, A_K > 0$ such that*

$$|b_m(s)| \leq C_K 2^{-m} (m+1)^{A_K} \quad (s \in K, m \geq 1).$$

If m is large enough that $s+m$ lies in the original absolute-convergence half-plane for every $s \in K$, then

$$F_q(s+m, x) = a_1(x) + O_K(2^{-m}).$$

Hence the tail of equation (101) converges normally on K .

Proof. Take $R \geq \max(1, \sup_{s \in K} |s|)$. The product estimate used in the proof of theorem 8.5 gives

$$\left| \binom{s+m-1}{m} \right| \leq R e^{R-1} (m+1)^{R-1},$$

which proves the first assertion. Choose $r > \sigma_q$ and put $D_r = \sum_{n \geq 1} |a_n(x)| n^{-r} < \infty$. If $u = \inf_{s \in K} \Re s$ and $m+u \geq r$, then

$$\begin{aligned} |F_q(s+m, x) - a_1(x)| &\leq \sum_{n \geq 2} |a_n(x)| n^{-\Re s - m} \\ &\leq D_r 2^{r-u-m}. \end{aligned}$$

Multiplication by the first bound leaves a convergent exponential–polynomial majorant, uniformly on K . $\square$

Theorem 8.8 (global meromorphic continuation). *For every odd prime q and every $x \in \mathbb{C}$, the Dirichlet series $F_q(\cdot, x)$ has a unique meromorphic continuation to $\mathbb{C}$. If*

$$\Lambda_q = \left\{ \sigma_q + \frac{2\pi ik}{\log 2} : k \in \mathbb{Z} \right\},$$

then its poles are contained in the explicitly generated candidate set

$$\text{Poles } F_q(\cdot, x) \subseteq \mathcal{P}_q, \quad \mathcal{P}_q = \{\lambda - n : \lambda \in \Lambda_q, n \in \mathbb{N}_0\} \cup \{1 - n : n \in \mathbb{N}_0\}. \quad (102)$$

The inclusion need not be equality: cancellation can make a candidate singularity removable.

Proof. For $r \in \mathbb{N}_0$, set

$$H_r = \{s \in \mathbb{C} : \Re s > \sigma_q - r\}.$$

The defining series is holomorphic on H_0 . Suppose inductively that it has been continued meromorphically to H_r . If $s \in H_{r+1}$, then $s + m \in H_r$ for every $m \geq 1$. On a compact subset of H_{r+1} , only finitely many low-shift terms in equation (101) can meet poles of the existing continuation; by theorem 8.7, the remaining holomorphic tail converges normally. Thus $S_q(\cdot, x)$ is meromorphic on H_{r+1} , and

$$F_q(s, x) = \frac{A_q(s, x)\zeta(s) + (q/2)S_q(s, x)}{2^s - 2^{\sigma_q}} \quad (103)$$

defines a meromorphic extension to H_{r+1} . It agrees with the existing function on H_r , so the identity theorem gives uniqueness on the overlap. Since $\bigcup_{r \geq 0} H_r = \mathbb{C}$, induction proves the continuation.

At the first step, new candidates can occur only at the pole 1 of ζ and at the zero set Λ_q of the denominator. Each later shifted term replaces an existing candidate p by $p - m$, $m \geq 1$, while division introduces Λ_q again. Induction therefore gives the containing set in equation (102); it does not supply nonvanishing of the corresponding numerators. $\square$

8.6. Corrected principal parts for $q = 3$. Write $L = \log 2$ and use S_3 as in equation (101). Then

$$F_3(s, x) = \frac{\frac{1}{2}((2x+1) - 2^s(x+1))\zeta(s) + (3/2)S_3(s, x)}{2^s - 2}. \quad (104)$$

Proposition 8.9 (the corrected $q = 3$ principal parts). *Let $x \in \mathbb{C}$ and let γ be Euler's constant.*

(i) *At $s = 1$, F_3 has the genuine double-pole expansion*

$$F_3(s, x) = -\frac{1}{4L}(s-1)^{-2} + \left(\frac{3S_3(1, x)}{4L} - \frac{x}{2} - \frac{3}{8} - \frac{\gamma}{4L} \right) (s-1)^{-1} + O(1). \quad (105)$$

(ii) *If $s_k = 1 + 2\pi ik/L$ with $k \in \mathbb{Z} \setminus \{0\}$, then s_k is at most a simple pole and*

$$\text{Res}_{s=s_k} F_3(s, x) = \frac{3S_3(s_k, x) - \zeta(s_k)}{4L}. \quad (106)$$

It is removable precisely when $3S_3(s_k, x) = \zeta(s_k)$.

(iii) *The point $s = 0$ is a genuine simple pole and*

$$\text{Res}_{s=0} F_3(s, x) = -\frac{3}{16L}. \quad (107)$$

(iv) *Every $-k$, $k \in \mathbb{N}_0$, is at most a simple pole. With $R_k = \text{Res}_{s=-k} F_3(s, x)$, where $R_k = 0$ for a removable candidate,*

$$R_k = \frac{3}{16L(1-2^{k+1})(k+1)} + \frac{3}{2(1-2^{k+1})} \sum_{j=0}^{k-1} 2^j \binom{k}{j} R_j. \quad (108)$$

The sum is empty for $k = 0$. In particular, every R_k is independent of x ; the recurrence alone does not prove $R_k \neq 0$ for all k .

Proof. Near $s = 1$, put $t = s - 1$. Since S_3 is holomorphic there,

$$\begin{aligned}\frac{(2x+1) - 2^s(x+1)}{2} &= -\frac{1}{2} - (x+1)Lt + O(t^2), \\ 2^s - 2 &= 2Lt + L^2t^2 + O(t^3), \\ \zeta(s) &= t^{-1} + \gamma + O(t).\end{aligned}$$

Division gives equation (105); its nonzero $(s-1)^{-2}$ coefficient proves that the pole is genuinely double.

At s_k with $k \neq 0$, both S_3 and ζ are holomorphic, the denominator has derivative $2L$, and the coefficient of ζ in the numerator equals $-1/2$. This proves equation (106) and the stated removability criterion.

At $s = 0$, only the $m = 1$ term of S_3 can inherit the double pole at 1 to order one:

$$-\frac{s}{2}F_3(s+1, x) = \frac{1}{8L}s^{-1} + O(1).$$

All $m \geq 2$ terms are holomorphic there. Since $2^0 - 2 = -1$, equation (107) follows.

Finally, fix $k \geq 0$ and put $s = -k + t$. Inductively assume the candidates $0, -1, \dots, -(k-1)$ are at most simple. For $1 \leq m \leq k$,

$$\left(-\frac{1}{2}\right)^m \binom{-k+m-1}{m} = 2^{-m} \binom{k}{m};$$

these terms contribute $\sum_{j=0}^{k-1} 2^{-(k-j)} \binom{k}{j} R_j$ to the residue of S_3 . For $m = k+1$,

$$\left(-\frac{1}{2}\right)^{k+1} \binom{s+k}{k+1} = -\frac{t}{2^{k+1}(k+1)} + O(t^2),$$

and equation (105) shows that its contribution is $1/(2^{k+3}(k+1)L)$. Terms $m \geq k+2$ are holomorphic. Therefore

$$\text{Res}_{s=-k} S_3(s, x) = \frac{1}{2^{k+3}(k+1)L} + \sum_{j=0}^{k-1} 2^{-(k-j)} \binom{k}{j} R_j.$$

The zeta term in equation (104) is holomorphic at $-k$, and $2^{-k} - 2 = (1 - 2^{k+1})/2^k$. Multiplication by $(3/2)/(2^{-k} - 2)$ yields equation (108) and completes the induction. $\square$

Source issue 8.10 (the printed $q = 3$ pole claims and recurrence). The source's $q = 3$ residues inherit the false x -term and therefore contain spurious factors depending on $4x + 1$. Its headline assertion that every nonreal lattice point and every nonpositive integer is an actual pole does not prove noncancellation. Moreover, the recurrence displayed in the corollary is undefined at its stated initial index and conflicts with the formula reached in its proof. This edition retains the genuine double pole at 1, the genuine simple pole at 0, the exact Laurent coefficients, the candidate-pole containment, and the corrected recurrence equation (108); it does not turn unproved noncancellation into a pole classification.

9. (p, q) -ADIC FOURIER ANALYSIS AND WIENER-TAUBERIAN CYCLICITY

Siegel's phrase (p, q) -adic means that the domain topology and the value-field topology come from different primes. The cleanest theorem in this strand is a Wiener-Tauberian theorem for continuous functions on $\mathbb{Z}_p$ with values in a complete nonarchimedean field of residue characteristic $q \neq p$. We reconstruct that theorem after fixing the character and Haar normalizations.

9.1. The two-place setting. Let p be prime. Let K be a complete algebraically closed nonarchimedean field whose residue characteristic is a prime $q \neq p$. The strongest headline assumptions in the source also take K to be spherically incomplete, as is the standard completed algebraic closure $\mathbb{C}_q$. Since p is a unit in K , every p^N has absolute value 1 in K .

As an abstract group, the Pontryagin dual of the additive compact group $\mathbb{Z}_p$ is

$$\Gamma_p = \mathbb{Z}[1/p]/\mathbb{Z}. \quad (109)$$

Because $\text{char } K \neq p$ and K is algebraically closed, choose roots of unity

$$\zeta_0 = 1, \quad \text{ord}(\zeta_N) = p^N, \quad \zeta_{N+1}^p = \zeta_N \quad (N \geq 0). \quad (110)$$

Such a system can be constructed recursively. First choose a nontrivial root ζ_1 of $X^p - 1$; it exists because this polynomial has p distinct roots in K . After ζ_N is chosen with $N \geq 1$, take any root of $X^p - \zeta_N$ in K . Its order divides p^{N+1} and cannot divide p^N , so it has order exactly p^{N+1} .

For $t = a/p^N + \mathbb{Z} \in \Gamma_p$ and $z \in \mathbb{Z}_p$, let z_N be the image of z in $\mathbb{Z}/p^N\mathbb{Z}$, choose any integer lift $\tilde{z}_N$ of z_N , and set

$$\left\langle \frac{a}{p^N} + \mathbb{Z}, z \right\rangle := \zeta_N^{\tilde{a}z_N} \in K^\times. \quad (111)$$

Proposition 9.1 (the chosen K -valued duality). *Formula equation (111) is independent of every chosen representative and defines a pairing that is a group homomorphism in each variable,*

$$\langle \cdot, \cdot \rangle : \Gamma_p \times \mathbb{Z}_p \longrightarrow K^\times.$$

For each t , the map $\chi_t(z) = \langle t, z \rangle$ is locally constant, and

$$\Gamma_p \xrightarrow{\sim} \text{Hom}_{\text{lc}}((\mathbb{Z}_p, +), K^\times), \quad t \longmapsto \chi_t, \quad (112)$$

is an isomorphism of groups.

Proof. Changing a or $\tilde{z}_N$ by a multiple of p^N does not change equation (111), because ζ_N has order p^N . Suppose the same t is also represented by $b/p^M + \mathbb{Z}$. At a common level $L \geq M, N$,

$$ap^{L-N} \equiv bp^{L-M} \pmod{p^L}, \quad \zeta_L^{p^{L-N}} = \zeta_N, \quad \zeta_L^{p^{L-M}} = \zeta_M.$$

Reading both values at level L proves independence of the denominator and hence of the representative of t . At a common level, addition in either variable adds exponents modulo p^L ; this proves biadditivity. If $p^N t = 0$, then χ_t factors through $\mathbb{Z}_p/p^N\mathbb{Z}_p$, so it is locally constant.

If χ_t is trivial, evaluation at 1 in any presentation $t = a/p^N + \mathbb{Z}$ gives $\zeta_N^a = 1$, hence $p^N \mid a$ and $t = 0$; thus equation (112) is injective. Conversely, let $\chi : \mathbb{Z}_p \rightarrow K^\times$ be a locally constant homomorphism. Local constancy at 0 gives $p^N\mathbb{Z}_p \subseteq \ker \chi$ for some N , so χ factors through the cyclic group $\mathbb{Z}/p^N\mathbb{Z}$. Put $u = \chi(1)$. Then $u^{p^N} = 1$. The p^N distinct powers of ζ_N exhaust the roots of $X^{p^N} - 1$ in K , so $u = \zeta_N^a$ for a unique a modulo p^N . Therefore $\chi(z) = \zeta_N^{\tilde{a}z_N} = \chi_{a/p^N + \mathbb{Z}}(z)$, proving surjectivity and uniqueness. $\square$

The complex-looking notation $e^{2\pi i\{tz\}_p}$ in the source denotes the element equation (111); it is not a complex exponential living in a q -adic field. This compatible-root construction, rather than the source's characteristic-zero embedding notation, also works when K has positive characteristic different from p .

There is a unique normalized K -valued Haar integral, specified by

$$\int_{k+p^N\mathbb{Z}_p} dz = p^{-N}. \quad (113)$$

For continuous $f : \mathbb{Z}_p \rightarrow K$ it is the following limit in K :

$$\int_{\mathbb{Z}_p} f(z) dz = \lim_{N \rightarrow \infty} \frac{1}{p^N} \sum_{k=0}^{p^N-1} f(k). \quad (114)$$

Indeed, on functions constant modulo p^N the right side of equation (114) is $p^{-N} \sum_{k \bmod p^N} f(k)$; refinement from level N to $N+1$ repeats every value p times. Since $|p| = 1$ in K , these finite-level functionals have norm at most 1 and extend uniquely from the uniformly dense locally constant functions to $C(\mathbb{Z}_p, K)$. The extension is translation-invariant and sends 1 to 1. Conversely, any translation-invariant continuous functional sending 1 to 1 gives the same mass to all p^N residue classes, and their p^N disjoint indicators sum to 1; their common mass is therefore p^{-N} . Finally, the locally constant functions $z \mapsto f([z]_{p^N})$, where $[z]_{p^N} \in \{0, \dots, p^N - 1\}$ is the canonical residue representative, converge uniformly to f . Applying the continuous integral proves equation (114). This proves existence and uniqueness as well.

Put $\Gamma_p[p^N] = \{t \in \Gamma_p : p^N t = 0\}$; it consists of the p^N classes $a/p^N + \mathbb{Z}$, $0 \leq a < p^N$.

Proposition 9.2 (finite character orthogonality). *For $N \geq 0$ one has*

$$\sum_{t \in \Gamma_p[p^N]} \langle t, x \rangle = p^N \mathbf{1}_{x \in p^N \mathbb{Z}_p} \quad (x \in \mathbb{Z}_p), \quad (115)$$

$$\sum_{k \in \mathbb{Z}/p^N \mathbb{Z}} \langle t - u, k \rangle = p^N \mathbf{1}_{t=u} \quad (t, u \in \Gamma_p[p^N]). \quad (116)$$

In the second sum the residue class k is evaluated through any lift in $\mathbb{Z}_p$. Consequently,

$$\mathbf{1}_{z \equiv k \pmod{p^N}} = \frac{1}{p^N} \sum_{t \in \Gamma_p[p^N]} \langle t, z - k \rangle. \quad (117)$$

Proof. Writing x_N for any integer representing x modulo p^N , the first sum is

$$\sum_{a=0}^{p^N-1} (\zeta_N^{x_N})^a.$$

If $x \in p^N \mathbb{Z}_p$, every summand is 1. Otherwise $\zeta_N^{x_N} \neq 1$, while $(\zeta_N^{x_N})^{p^N} = 1$, so the finite geometric sum is 0. This proves equation (115). For the second identity write $t - u = b/p^N + \mathbb{Z}$ and apply the same geometric-sum argument to $\sum_{k=0}^{p^N-1} (\zeta_N^b)^k$. Division by p^N is valid because $\text{char } K \neq p$, and the first identity with $x = z - k$ gives equation (117). $\square$

9.2. Fourier transform and inversion. Let $C(\mathbb{Z}_p, K)$ carry the supremum norm. Let

$$c_0(\Gamma_p, K) = \{a : \Gamma_p \rightarrow K : \text{for every } \epsilon > 0, \#\{t : |a(t)| \geq \epsilon\} < \infty\}$$

with its supremum norm. Define

$$\widehat{f}(t) = \int_{\mathbb{Z}_p} f(z) \langle -t, z \rangle dz. \quad (118)$$

Theorem 9.3 (nonarchimedean Fourier isometry). *The Fourier transform extends to an isometric isomorphism*

$$\mathcal{F} : C(\mathbb{Z}_p, K) \xrightarrow{\sim} c_0(\Gamma_p, K). \quad (119)$$

For every $f \in C(\mathbb{Z}_p, K)$,

$$f(z) = \sum_{t \in \Gamma_p} \widehat{f}(t) \langle t, z \rangle, \quad (120)$$

where the series converges uniformly.

Proof. Suppose first that f is constant modulo p^N . For $t \in \Gamma_p[p^N]$, equation (114) gives

$$\widehat{f}(t) = \frac{1}{p^N} \sum_{k \bmod p^N} f(k) \langle -t, k \rangle. \quad (121)$$

If $t \notin \Gamma_p[p^N]$, choose $M > N$ with $t \in \Gamma_p[p^M]$. In the level- M Riemann sum, partitioning residues as $k = r + p^N h$ produces, for each $r \bmod p^N$, the factor

$$\sum_{h=0}^{p^{M-N}-1} \langle -t, p^N h \rangle.$$

This is a geometric sum with nontrivial ratio and ratio raised to p^{M-N} equal to 1, so it vanishes. Thus $\widehat{f}$ is supported on $\Gamma_p[p^N]$. Substitution of equation (121) and equation (115) now gives finite inversion. Conversely, equation (116) shows that every function supported on $\Gamma_p[p^N]$ is the transform of its finite inverse series.

All chosen characters, p^N , and p^{-N} have absolute value 1 in K . The ultrametric inequality applied to the forward and inverse finite transforms therefore gives

$$\sup_t |\widehat{f}(t)| \leq \sup_z |f(z)| \quad \text{and} \quad \sup_z |f(z)| \leq \sup_t |\widehat{f}(t)|.$$

Thus the finite transform is isometric. Locally constant functions are uniformly dense in $C(\mathbb{Z}_p, K)$, and finitely supported sequences are dense in $c_0(\Gamma_p, K)$; every finite character set is contained in $\Gamma_p[p^N]$ for some N . Completion therefore extends the compatible finite transforms to the isometric bijection equation (119). For $a \in c_0$, the inverse family $\sum_t a(t) \langle t, z \rangle$ converges uniformly, because the supremum norm of every tail is at most the largest omitted $|a(t)|$. This proves equation (120). $\square$

The source develops the same conclusion through van der Put expansions. For $n \geq 1$, the relevant basis function is the indicator of the ball $n + p^{\lambda_p(n)} \mathbb{Z}_p$; every continuous function has a unique uniformly convergent expansion in these indicators with coefficients tending to zero. That route supplies explicit formulas relating van der Put and Fourier coefficients, but the finite-transform completion above exposes the norm structure used by the Wiener theorem.

9.3. Convolution algebra. We use the following conventions throughout this chapter. For $a, b \in c_0(\Gamma_p, K)$,

$$(a * b)(t) = \sum_{s \in \Gamma_p} a(s) b(t - s). \quad (122)$$

For $f, g \in C(\mathbb{Z}_p, K)$ and $\mu \in C(\mathbb{Z}_p, K)'$,

$$(f * g)(z) = \int_{\mathbb{Z}_p} f(z - y) g(y) dy, \quad (123)$$

$$(f * \mu)(z) = (\mu * f)(z) = \mu(y \mapsto f(z - y)). \quad (124)$$

The last line is a definition of both displayed notations; it fixes the minus sign in the translate. Uniform continuity of f shows that $z \mapsto f(z - \cdot)$ is continuous into $C(\mathbb{Z}_p, K)$ in the supremum norm, so $f * \mu \in C(\mathbb{Z}_p, K)$ and $\|f * \mu\|_\infty \leq \|f\|_\infty \|\mu\|$. The Fourier–Stieltjes transform uses the same negative character as the function transform:

$$\widehat{\mu}(t) = \mu(z \mapsto \langle -t, z \rangle). \quad (125)$$

In particular $|\widehat{\mu}(t)| \leq \|\mu\|$, because every chosen character has supremum norm 1.

Proposition 9.4 (the three convolution identities). *The convolution in equation (122) is a well-defined continuous bilinear map $c_0 \times c_0 \rightarrow c_0$, with*

$$\|a * b\|_\infty \leq \|a\|_\infty \|b\|_\infty. \quad (126)$$

With the Fourier and Fourier–Stieltjes signs fixed by equations (118) and (125), one has

$$\widehat{fg} = \widehat{f} * \widehat{g}. \quad (127)$$

Moreover,

$$\widehat{f * g}(t) = \widehat{f}(t) \widehat{g}(t), \quad \widehat{f * \mu}(t) = \widehat{f}(t) \widehat{\mu}(t). \quad (128)$$

Thus equation (119) is an isometric Banach-algebra isomorphism from pointwise multiplication on $C(\mathbb{Z}_p, K)$ to convolution on $c_0(\Gamma_p, K)$.

Proof. For fixed t , the family $a(s)b(t-s)$ tends to zero off finite subsets: if $b \neq 0$ and $|a(s)b(t-s)| \geq \epsilon$, then $|a(s)| \geq \epsilon/\|b\|_\infty$, which occurs for only finitely many s . Completeness and the nonarchimedean criterion for summability therefore give equation (122). The ultrametric inequality gives equation (126). If b has finite support, $a * b$ is a finite sum of translates of a and hence belongs to c_0 . Approximating an arbitrary $b \in c_0$ by finitely supported b_n and using equation (126) proves $a * b \in c_0$ and continuity.

If f and g are constant modulo p^N , finite inversion and equation (116) give, for $t \in \Gamma_p[p^N]$,

$$\widehat{fg}(t) = \sum_{s \in \Gamma_p[p^N]} \widehat{f}(s)\widehat{g}(t-s).$$

Both sides vanish outside a sufficiently large finite character group, so this is equation (127) at finite level. Uniformly approximate f, g by locally constant functions. Pointwise multiplication, the Fourier isometry, and sequence convolution are continuous, so the identity passes to all $f, g \in C(\mathbb{Z}_p, K)$.

Fourier inversion gives, uniformly in $(z, y) \in \mathbb{Z}_p^2$,

$$f(z-y) = \sum_{s \in \Gamma_p} \widehat{f}(s)\langle s, z \rangle \langle -s, y \rangle.$$

Applying the continuous functional $g(y) dy$, respectively μ , term by term yields

$$(f * g)(z) = \sum_s \widehat{f}(s)\widehat{g}(s)\langle s, z \rangle, \quad (f * \mu)(z) = \sum_s \widehat{f}(s)\widehat{\mu}(s)\langle s, z \rangle.$$

The coefficient families lie in c_0 because $\widehat{f} \in c_0$ and $\widehat{g}, \widehat{\mu}$ are bounded. Uniqueness in Fourier inversion proves equation (128). $\square$

For $s \in \Gamma_p$, write

$$(\tau_s a)(t) = a(t+s).$$

Proposition 9.5 (finite translates and the closed principal ideal). *For every $a \in c_0(\Gamma_p, K)$,*

$$\tau_s a = a * \delta_{-s} \quad (s \in \Gamma_p), \tag{129}$$

$$\text{span}_K\{\tau_s a : s \in \Gamma_p\} = a * c_{00}(\Gamma_p, K), \tag{130}$$

$$\overline{\text{span}_K\{\tau_s a : s \in \Gamma_p\}} = \overline{a * c_0(\Gamma_p, K)}. \tag{131}$$

*The closures are taken in the supremum norm. Thus the finite translate span is generally not the algebraic principal ideal $a * c_0$; its closure is the closed principal ideal generated by a .*

Proof. For $t \in \Gamma_p$,

$$(a * \delta_{-s})(t) = \sum_{u \in \Gamma_p} a(u)\delta_{-s}(t-u) = a(t+s),$$

which proves equation (129). Every element of c_{00} is a finite linear combination of point masses, so bilinearity gives equation (130) in both directions.

Let $T_a : c_0(\Gamma_p, K) \rightarrow c_0(\Gamma_p, K)$ be $T_a(b) = a * b$. The convolution estimate gives $\|T_a(b)\|_\infty \leq \|a\|_\infty \|b\|_\infty$, hence T_a is continuous. For every $b \in c_0$, choose $b_N \in c_{00}$ with $\|b_N - b\|_\infty \rightarrow 0$. Then $T_a(b_N) \rightarrow T_a(b)$, so $a * c_0 \subseteq \overline{a * c_{00}}$. The reverse inclusion before taking closures follows from $c_{00} \subseteq c_0$. This proves equation (131). $\square$

Remark 9.6. The distinction cannot be removed even for the convolution unit. If $a = \delta_0$, then its finite translate span is $c_{00}(\Gamma_p, K)$, whereas $\delta_0 * c_0(\Gamma_p, K) = c_0(\Gamma_p, K)$. The group Γ_p is infinite, so these two spaces are unequal.

Theorem 9.7 (Wiener–Tauberian theorem for continuous (p, q) -adic functions). *For $\chi \in C(\mathbb{Z}_p, K)$ the following are equivalent.*

- (i) $1/\chi \in C(\mathbb{Z}_p, K)$;
- (ii) $\hat{\chi}$ is invertible in the convolution algebra $c_0(\Gamma_p, K)$;
- (iii) $\text{span}_K\{\tau_s \hat{\chi} : s \in \Gamma_p\}$ is dense in $c_0(\Gamma_p, K)$;
- (iv) $\chi(z) \neq 0$ for every $z \in \mathbb{Z}_p$.

Proof. The transform of 1 is the convolution unit δ_0 . If (i) holds, equation (127) shows that $\widehat{1/\chi}$ is a convolution inverse of $\hat{\chi}$. Conversely, if $b * \hat{\chi} = \delta_0$, surjectivity of equation (119) gives $b = \hat{g}$ for a unique $g \in C(\mathbb{Z}_p, K)$; injectivity and equation (127) then give $g\chi = 1$, so $g = 1/\chi$. This proves the equivalence of (i) and (ii).

If (ii) holds, then every $a \in c_0$ equals

$$a = \hat{\chi} * (\hat{\chi}^{-1} * a).$$

Choose $b_N \in c_{00}$ converging to $\hat{\chi}^{-1} * a$ in the supremum norm. Boundedness of convolution gives

$$\hat{\chi} * b_N \longrightarrow \hat{\chi} * (\hat{\chi}^{-1} * a) = a.$$

By equation (130), every $\hat{\chi} * b_N$ is a finite linear combination of translates of $\hat{\chi}$. This proves (iii).

Suppose (iii) holds but $\chi(z_0) = 0$. Fourier inversion makes evaluation at z_0 the nonzero continuous functional

$$L_{z_0}(a) = \sum_{t \in \Gamma_p} a(t) \langle t, z_0 \rangle \quad (a \in c_0).$$

The sum converges and $|L_{z_0}(a)| \leq \|a\|_\infty$ by the same nonarchimedean tail criterion used in Fourier inversion. For every s , a change of variables gives $L_{z_0}(\tau_s \hat{\chi}) = \langle -s, z_0 \rangle \chi(z_0) = 0$. Thus L_{z_0} vanishes on a dense subspace and hence on all of c_0 , contradicting $L_{z_0}(\delta_0) = 1$. Therefore (iii) implies (iv). Finally, if χ is continuous and nowhere zero on compact $\mathbb{Z}_p$, then $1/\chi$ is continuous (equivalently, $|\chi|$ is bounded away from zero), so (iv) implies (i). $\square$

9.4. Measures and the proposed measure theorem. A K -valued measure in the paper is a continuous functional $\mu \in C(\mathbb{Z}_p, K)'$. Its Fourier–Stieltjes transform is the precisely signed functional in equation (125); integral notation writes the same value as $\int_{\mathbb{Z}_p} \langle -t, z \rangle d\mu(z)$. Via equation (119) and the nonarchimedean duality $c'_0 = \ell^\infty$, this transform is naturally bounded, not necessarily vanishing at infinity.

Let

$$D_N(z) = p^N \mathbf{1}_{z \in p^N \mathbb{Z}_p}. \tag{132}$$

Then

$$(D_N * \mu)(z) = p^N \mu(z + p^N \mathbb{Z}_p) \tag{133}$$

with no sign ambiguity. Here $\mu(U) = \mu(\mathbf{1}_U)$ for a clopen set U . Indeed, the convention equation (124) gives

$$(D_N * \mu)(z) = \mu(y \longmapsto p^N \mathbf{1}_{z-y \in p^N \mathbb{Z}_p}) = p^N \mu(z + p^N \mathbb{Z}_p),$$

because $z - y \in p^N \mathbb{Z}_p$ if and only if $y \in z + p^N \mathbb{Z}_p$. The source calls the limit, when it exists, the pointwise Radon–Nikodym derivative of μ at z .

Status record 9.8 (measure Wiener–Tauberian claim). The paper asserts that the translates of $\hat{\mu}$ are dense enough to approximate every element of $c_0(\Gamma_p, K)$ if and only if every existing limit

$$\lim_{N \rightarrow \infty} (D_N * \mu)(z)$$

is nonzero. Since $\hat{\mu}$ need not itself lie in c_0 , the precise typed density condition is

$$c_0(\Gamma_p, K) \subseteq \overline{\text{span}_K\{\tau_s \hat{\mu} : s \in \Gamma_p\}}^{\ell^\infty}. \tag{134}$$

The proof establishes the forward implication from equation (134) to nonvanishing of every existing local density limit.

Source issue 9.9 (the converse for measures is not proved). The purported second half assumes both a zero local-density limit and the density of translates, then derives a contradiction. This is the contrapositive of the already proved implication; it is not the converse “no zero limit implies density.” The proposed condition is also vacuous for a measure for which no limit in equation (133) exists. A valid converse requires an annihilator/closed-ideal argument together with an existence or regularity hypothesis. The measure equivalence is therefore not promoted to a theorem in this edition.

9.5. The characteristic distribution of a numen. Suppose $X_{H,\ell} : \mathbb{Z}_p \rightarrow K_\ell$ is measurable and satisfies equation (43). Push normalized real Haar probability forward by $X_{H,\ell}$ and call the resulting probability measure $\nu_{H,\ell}$. For an additive character γ_t of K_ℓ , define

$$\hat{\nu}_{H,\ell}(t) = \int_{\mathbb{Z}_p} \gamma_t(-X_{H,\ell}(z)) dz. \quad (135)$$

Partitioning $\mathbb{Z}_p$ into $j + p\mathbb{Z}_p$ and using the numen recursion gives

$$\hat{\nu}_{H,\ell}(t) = \frac{1}{p} \sum_{j=0}^{p-1} \gamma_t(-c_j) \hat{\nu}_{H,\ell}(r_j^* t), \quad (136)$$

where r_j^* is pullback on the character group: $\gamma_{r_j^* t}(x) = \gamma_t(r_j x)$. When r_j is scalar multiplication, this is the source’s shorter notation $r_j t$.

Equivalently, the pushforward measure satisfies

$$\nu_{H,\ell} = \frac{1}{p} \sum_{j=0}^{p-1} (H_j)_* \nu_{H,\ell}. \quad (137)$$

Existence follows from the numen and its functional equation. Uniqueness of a probability measure solving equation (137) is a separate IFS theorem and requires, for example, that all H_j be strict contractions on a complete metric space. The latest manual’s unconditional uniqueness wording is not used without such a hypothesis.

10. RISING-CONTINUITY, FRAMES, AND PRODUCTS OF F-SERIES

The 2022 dissertation, the 2023 paper on variable topologies, and the 2025 paper on products all study series whose target topology depends on the input digit sequence. The stable core is: canonical truncation limits, point-selected completions (frames), digit-multiplicative F-series, and finite-level Fourier recursions. Several Banach-algebra and product-of- distributions conclusions in the sources require repairs recorded below.

10.1. Rising-continuity. Let p be prime, let K be a complete nonarchimedean field, and let $f : \mathbb{Z}_p \rightarrow K$.

Definition 10.1 (rising-continuity). The function f is *rising-continuous* when

$$f(z) = \lim_{N \rightarrow \infty} f([z]_{p^N}) \quad (z \in \mathbb{Z}_p). \quad (138)$$

Write $\tilde{C}(\mathbb{Z}_p, K)$ for the class of such functions.

For an arbitrary function $f : \mathbb{Z}_p \rightarrow K$, define the source’s level differences by

$$(\nabla_{p^0} f)(z) = f(0), \quad (139)$$

$$(\nabla_{p^N} f)(z) = f([z]_{p^N}) - f([z]_{p^{N-1}}), \quad N \geq 1, \quad (140)$$

and its extended del quantity by

$$\|f\|_{\nabla} = \sup_{N \geq 0} \sup_{z \in \mathbb{Z}_p} |(\nabla_{p^N} f)(z)|_K \in [0, \infty]. \quad (141)$$

Proposition 10.2 (the exact domain of the del norm). *If $f \in \tilde{C}(\mathbb{Z}_p, K)$, then*

$$\|f\|_{\nabla} = \sup_{z \in \mathbb{Z}_p} |f(z)|_K \quad (142)$$

as an equality in $[0, \infty]$. Consequently

$$\tilde{C}_b(\mathbb{Z}_p, K) := \{f \in \tilde{C}(\mathbb{Z}_p, K) : \|f\|_{\nabla} < \infty\}$$

is precisely the bounded rising-continuous class, and it is a Banach space under $\|\cdot\|_{\nabla}$.

Proof. For every N and z , the ultrametric inequality bounds $|(\nabla_{p^N} f)(z)|_K$ by $\sup_w |f(w)|_K$, including $N = 0$. Conversely, the finite telescoping identity

$$f([z]_{p^N}) = \sum_{h=0}^N (\nabla_{p^h} f)(z)$$

and the ultrametric inequality give $|f([z]_{p^N})|_K \leq \|f\|_{\nabla}$. Rising-continuity and continuity of the absolute value let $N \rightarrow \infty$, proving the reverse inequality in equation (142). Thus the two finite norms agree. Bounded K -valued functions are complete under the supremum norm. If (f_m) is a uniformly Cauchy sequence of bounded rising-continuous functions with uniform limit f , then for fixed z the ultrametric estimate

$$|f(z) - f([z]_{p^N})|_K \leq \max\{|f(z) - f_m(z)|_K, |f_m(z) - f_m([z]_{p^N})|_K, |f_m([z]_{p^N}) - f([z]_{p^N})|_K\}$$

first becomes small by choosing m and then N . Hence f is rising-continuous, so the displayed subspace is closed and complete. $\square$

Every continuous function is rising-continuous because $[z]_{p^N} \rightarrow z$ in $\mathbb{Z}_p$. The converse is false: the condition tests one distinguished convergent sequence at each point, not every convergent sequence. For every odd prime q , the shortened numen $\chi_q : \mathbb{Z}_2 \rightarrow \mathbb{Z}_q$ is the principal example (the preceding definition with $p = 2$ and $K = \mathbb{Q}_q$): it is rising-continuous everywhere but discontinuous at each nonnegative integer.

Define the truncation of f at level N by

$$f_N(z) = f([z]_{p^N}) = \sum_{k=0}^{p^N-1} f(k) \mathbf{1}_{z \equiv k \pmod{p^N}}. \quad (143)$$

Proposition 10.3 (pointwise versus uniform truncation). *For an arbitrary $f : \mathbb{Z}_p \rightarrow K$:*

- (i) $f_N \rightarrow f$ pointwise if and only if f is rising-continuous;
- (ii) $f_N \rightarrow f$ uniformly if and only if f is continuous.

Proof. Part (i) is the definition. Every f_N is locally constant and hence continuous, so a uniform limit is continuous. Conversely, a continuous function on compact $\mathbb{Z}_p$ is uniformly continuous. Since $|z - [z]_{p^N}|_p \leq p^{-N}$ uniformly in z , uniform continuity gives $\sup_z |f(z) - f_N(z)| \rightarrow 0$. $\square$

10.2. Van der Put paths. Set $\lambda_p(0) = 0$ and

$$\lambda_p(n) = 1 + \lfloor \log_p n \rfloor \quad (n \geq 1).$$

For $n \geq 1$, let n^- be the integer obtained by deleting the highest nonzero base- p digit of n . Define

$$v_0(z) = 1, \quad (144)$$

$$v_n(z) = \mathbf{1}_{z \equiv n \pmod{p^{\lambda_p(n)}}}, \quad n \geq 1, \quad (145)$$

$$c_0(f) = f(0), \quad (146)$$

$$c_n(f) = f(n) - f(n^-), \quad n \geq 1. \quad (147)$$

Theorem 10.4 (rising van der Put representation). *A function $f : \mathbb{Z}_p \rightarrow K$ is rising-continuous if and only if*

$$f(z) = \sum_{n=0}^{\infty} c_n(f) v_n(z) \quad (148)$$

converges at every z . Equivalently, along every truncation path,

$$c_{[z]_{p^N}}(f) \text{ ind } \lambda_p([z]_{p^N}) = N \longrightarrow 0. \quad (149)$$

The expansion is unique.

Proof. At a fixed z , exactly one possible new van der Put ball appears when the N th digit is nonzero: its index is $[z]_{p^N}$ and its coefficient is the difference between two successive effective truncation values. Therefore the partial sum through level N telescopes to $f([z]_{p^N})$. In a complete nonarchimedean field, the path series converges exactly when its terms tend to zero, giving equation (149); its sum is $f(z)$ exactly when equation (138) holds. Evaluation successively on the nested balls recovers every coefficient, proving uniqueness. $\square$

Corollary 10.5 (unique rising continuation). *Let $g : \mathbb{N}_0 \rightarrow K$. A rising-continuous extension to $\mathbb{Z}_p$ exists if and only if $g([z]_{p^N})$ converges in K for every $z \in \mathbb{Z}_p$. When it exists it is unique and equals*

$$\tilde{g}(z) = \lim_{N \rightarrow \infty} g([z]_{p^N}).$$

Proof. If a rising-continuous extension f exists, then $g([z]_{p^N}) = f([z]_{p^N}) \rightarrow f(z)$ for every z , so the displayed limit is necessary and determines f uniquely. Conversely, suppose every displayed limit exists and use it to define $\tilde{g}$. If $m \in \mathbb{N}_0$, then $[m]_{p^N} = m$ for all sufficiently large N , hence $\tilde{g}(m) = g(m)$. Therefore

$$\tilde{g}([z]_{p^N}) = g([z]_{p^N}) \longrightarrow \tilde{g}(z),$$

which is exactly rising-continuity. Thus $\tilde{g}$ is the required extension, and the necessity argument already proved uniqueness. $\square$

This is the precise uniqueness condition missing from several functional-equation statements in the 2026 manual: uniqueness on $\mathbb{N}_0$ does not by itself imply uniqueness on all infinite digit strings, but rising-continuity does.

10.3. Frames: a topology selected at each point. Let F be a global field and let $\text{Comp}(F)$ denote a chosen collection of complete valued extensions of F .

Definition 10.6 (frame). An F -frame on a set X is a map

$$\mathcal{F} : X \rightarrow \text{Comp}(F).$$

A function f is compatible with $\mathcal{F}$ when $f(x) \in \mathcal{F}(x)$ for every x . A sequence f_N converges to f in the frame when

$$f_N(x) \longrightarrow f(x) \quad \text{in } \mathcal{F}(x) \quad (x \in X). \quad (150)$$

Thus a frame is not one exotic topology on a disjoint union. It is a rule for which ordinary completion is to judge convergence at each input. Pointwise sums and products of compatible functions are meaningful because both operands at a fixed point lie in the same selected field.

The standard two-place example in the body of the 2023 paper uses complex convergence on $\mathbb{N}_0$ and q -adic convergence on $\mathbb{Z}_p \setminus \mathbb{N}_0$. Its abstract describes the assignments in the opposite order; the body definition controls this edition.

Source issue 10.7 (an unrestricted frame function need not have a sup norm). One source proposition declares all functions from compact X to a complete field to be a Banach space under the supremum norm. Compactness bounds continuous functions, not arbitrary functions. The statement becomes valid after restricting to bounded functions (and proving completeness pointwise plus uniformly), or to continuous functions in a fixed target. The same issue reappears in the dissertation's proposed Banach algebra of all rising-continuous functions: rising-continuity alone does not imply boundedness or finiteness of the displayed “del” norm.

10.4. Digit-multiplicative F-series. For a digit prefix of fixed length N , use the padded count

$$\#_{p;j}^{(N)}(z) = \#\{0 \leq h < N : d_h(z) = j\}. \quad (151)$$

This count includes high zero digits. The ordinary digit count of the integer $[z]_{p^N}$ silently discards them and is not interchangeable with equation (151) when a zero-digit multiplier occurs.

Fix a complete valued field E containing nonzero parameters $d, q_0, \dots, q_{p-1}$. For $z \in \mathbb{Z}_p$, put

$$u_N(z) = d^{-N} \prod_{j=0}^{p-1} q_j^{\#_{p;j}^{(N)}(z)}, \quad D_E = \left\{ z \in \mathbb{Z}_p : \sum_{N \geq 0} u_N(z) \text{ converges in } E \right\}.$$

Define the E -valued F-series on its exact convergence domain by

$$S_{d;\mathbf{q}}^E : D_E \longrightarrow E, \quad S_{d;\mathbf{q}}^E(z) = \sum_{N=0}^{\infty} u_N(z). \quad (152)$$

For $0 \leq k < p$, the digit identity

$$\#_{p;j}^{(N+1)}(pz + k) = \#_{p;j}^{(N)}(z) + \mathbf{1}_{j=k}$$

gives the termwise tail identity

$$u_{N+1}(pz + k) = \frac{q_k}{d} u_N(z).$$

Since $q_k/d \neq 0$, it follows without an unstated convergence condition that $pz + k \in D_E$ if and only if $z \in D_E$, and on this common domain

$$S_{d;\mathbf{q}}^E(pz + k) = 1 + \frac{q_k}{d} S_{d;\mathbf{q}}^E(z). \quad (153)$$

A frame may select a different completion at different inputs, but an identity comparing z with $pz + k$ is typed only after both values are placed in one specified field E (or after a specified identification of their target fields).

The primitive numen series equation (38) is the affine version: its N th term contains a digit-dependent translation c_{d_N} and the ordered product of all earlier linear multipliers. Fix a complete valued field E and scalars $a_0, \dots, a_{p-1}, b_0, \dots, b_{p-1} \in E$. In this scalar commuting case put

$$D_X = \left\{ z \in \mathbb{Z}_p : \sum_{N=0}^{\infty} b_{d_N(z)} \prod_{h=0}^{N-1} a_{d_h(z)} \text{ converges in } E \right\}$$

and define

$$X : D_X \longrightarrow E, \quad X(z) = \sum_{N=0}^{\infty} b_{d_N(z)} \prod_{j=0}^{p-1} a_j^{\#_{p;j}^{(N)}(z)}. \quad (154)$$

For $0 \leq k < p$ and $z \in D_X$ one has $pz + k \in D_X$ and

$$X(pz + k) = a_k X(z) + b_k. \quad (155)$$

Conversely, if $a_k \neq 0$, then $pz + k \in D_X$ implies $z \in D_X$. Indeed, the zeroth summand at $pz + k$ is b_k , and its summand of index $N + 1$ is a_k times the summand of index N at z . Thus convergence in the forward direction is automatic even when $a_k = 0$, while division by a_k in the converse is valid exactly under the displayed nonvanishing hypothesis.

For the Collatz numen, the 2023 paper writes the useful frame identity

$$\chi_3(z) = -\frac{1}{2} + \frac{1}{4} \sum_{N=0}^{\infty} \frac{3^{\#_{2:1}^{(N)}(z)}}{2^N}. \quad (156)$$

On nonnegative integers the right side is read archimedeanly; on $\mathbb{Z}_2 \setminus \mathbb{N}_0$ it is read 3-adically. This differing-codomain identity does not follow from theorem 10.5, whose domain and codomain are fixed. It follows directly from one finite identity. Indeed, write

$$A_N(z) = \frac{3^{\#_{2:1}^{(N)}(z)}}{2^N}, \quad S_N(z) = \sum_{h=0}^{N-1} A_h(z),$$

and let $d_h(z) \in \{0, 1\}$ be the h th digit. The N -digit numen partial sum is

$$\chi_{3,N}(z) = \frac{1}{2} \sum_{h=0}^{N-1} d_h(z) A_h(z).$$

Since $A_{h+1} = (\frac{1}{2} + d_h)A_h$ and $A_0 = 1$, telescoping gives the exact finite relation

$$\chi_{3,N}(z) = \frac{1}{4} S_N(z) - \frac{1}{2} + \frac{1}{2} A_N(z). \quad (157)$$

If $z \in \mathbb{N}_0$, its one-count is eventually constant and $A_N(z) \rightarrow 0$ in $\mathbb{R}$. If $z \in \mathbb{Z}_2 \setminus \mathbb{N}_0$, its binary expansion has infinitely many one digits, so $|A_N(z)|_3 = 3^{-\#_{2:1}^{(N)}(z)} \rightarrow 0$. In the first case $\chi_{3,N}(z)$ is eventually the finite numen value; in the second it converges in $\mathbb{Q}_3$ by the primitive numen series. Passing to the limit in equation (157) proves equation (156) separately in the stated completion at every input.

Source issue 10.8 (pointwise convergence claim in the 2023 paper). One necessity proof chooses a general nonnatural input and then replaces it by $z = -1$. That proves a global obstruction, not the asserted pointwise necessity for the original digit sequence. Placewise convergence should be checked from the actual prefix products, as in theorem 5.6, rather than from that substitution.

10.5. Products and the exact finite recursion. Let R be a commutative ring with identity, let $a_{j,k}, b_{j,k} \in R$, and let $X_1, \dots, X_d : \mathbb{Z}_p \rightarrow R$ be degree-one digit-recursive functions satisfying

$$X_j(pz + k) = a_{j,k} X_j(z) + b_{j,k} \quad (1 \leq j \leq d, 0 \leq k < p). \quad (158)$$

For a multi-index $\mathbf{n} = (n_1, \dots, n_d) \in \mathbb{N}_0^d$, put

$$X_{\mathbf{n}} = \prod_{j=1}^d X_j^{n_j}.$$

If $\mathbf{m} \leq \mathbf{n}$ coordinatewise, define

$$\binom{\mathbf{n}}{\mathbf{m}} = \prod_j \binom{n_j}{m_j}, \quad (159)$$

$$a_{\mathbf{m},k} = \prod_j a_{j,k}^{m_j}, \quad (160)$$

$$b_{\mathbf{n}-\mathbf{m},k} = \prod_j b_{j,k}^{n_j-m_j}, \quad (161)$$

$$r_{\mathbf{m},\mathbf{n},k} = \binom{\mathbf{n}}{\mathbf{m}} a_{\mathbf{m},k} b_{\mathbf{n}-\mathbf{m},k}. \quad (162)$$

Proposition 10.9 (multinomial branch recursion). *For every k ,*

$$X_{\mathbf{n}}(pz + k) = \sum_{\mathbf{m} \leq \mathbf{n}} r_{\mathbf{m},\mathbf{n},k} X_{\mathbf{m}}(z). \quad (163)$$

Proof. Raise each equation in equation (158) to the power n_j , expand by the binomial theorem, multiply over j , and collect the term with multi-index $\mathbf{m}$. $\square$

This finite algebraic identity is independent of every limiting assumption in the 2025 paper.

10.6. Finite Fourier recursion and formal solutions. Fix through the end of theorem 10.11 a complete nonarchimedean field K of residue characteristic $q \neq p$, a compatible system of p -power roots of unity, and the normalized K -valued Haar integral of section 9. Assume the branch parameters and the functions under transformation are K -valued. A point-varying frame by itself is not a single coefficient field in which Fourier sums can be added.

Let

$$\alpha_{\mathbf{m},\mathbf{n}}(t) = \frac{1}{p} \sum_{k=0}^{p-1} r_{\mathbf{m},\mathbf{n},k} \langle -t, k \rangle, \quad \alpha_{\mathbf{n}} = \alpha_{\mathbf{n},\mathbf{n}}. \quad (164)$$

Fix $\mathbf{n}$ and assume explicitly that $X_{\mathbf{m}} \in C(\mathbb{Z}_p, K)$ for every $\mathbf{m} \leq \mathbf{n}$. Then all Fourier transforms below are the normalized Haar integrals of equation (118). The substitution $x = pz + k : \mathbb{Z}_p \xrightarrow{\sim} k + p\mathbb{Z}_p$ gives

$$\int_{k+p\mathbb{Z}_p} f(x) dx = \frac{1}{p} \int_{\mathbb{Z}_p} f(pz + k) dz$$

for continuous f . Applying this identity to $f(x) = X_{\mathbf{n}}(x) \langle -t, x \rangle$, using equation (163), and interchanging only finite sums gives

$$\hat{X}_{\mathbf{n}}(t) = \sum_{\mathbf{m} \leq \mathbf{n}} \alpha_{\mathbf{m},\mathbf{n}}(t) \hat{X}_{\mathbf{m}}(pt). \quad (165)$$

At finite truncation level the identity is exact before any limiting integrability claim, but it has a necessary one-level shift. Put

$$X_{j,N}(z) = X_j([z]_{p^N}), \quad X_{\mathbf{n},N}(z) = \prod_{j=1}^d X_{j,N}(z)^{n_j}.$$

Then

$$\hat{X}_{\mathbf{n},N+1}(t) = \sum_{\mathbf{m} \leq \mathbf{n}} \alpha_{\mathbf{m},\mathbf{n}}(t) \hat{X}_{\mathbf{m},N}(pt). \quad (166)$$

Indeed, $[pz + k]_{p^{N+1}} = p[z]_{p^N} + k$. More generally, for $f_N(z) = f([z]_{p^N})$,

$$\hat{f}_N(t) = \frac{\mathbf{1}_{p^N t=0}}{p^N} \sum_{r=0}^{p^N-1} f(r) \langle -t, r \rangle. \quad (167)$$

At $t = 0$, separate the diagonal term in equation (165):

$$(1 - \alpha_{\mathbf{n}}(0))\hat{X}_{\mathbf{n}}(0) = \sum_{\mathbf{m} < \mathbf{n}} \alpha_{\mathbf{m},\mathbf{n}}(0)\hat{X}_{\mathbf{m}}(0). \quad (168)$$

Proposition 10.10 (formal compatibility classification). *Assume the lower-degree transforms are fixed.*

- (i) *If $\alpha_{\mathbf{n}}(0) \neq 1$, the formal recursion has a unique value at 0, and hence a unique value at every $t \in \Gamma_p$ by repeated use of equation (165) until $p^N t = 0$.*
- (ii) *If $\alpha_{\mathbf{n}}(0) = 1$ and the right side of equation (168) is nonzero, no formal solution exists.*
- (iii) *If $\alpha_{\mathbf{n}}(0) = 1$ and that right side is zero, the value $\hat{X}_{\mathbf{n}}(0)$ is free and the solutions form a one-dimensional affine family.*

Proof. Equation (168) gives the three alternatives at 0. Every $t \in \Gamma_p$ has p -power order, so $p^N t = 0$ for some N ; running equation (165) backward from that terminal value determines all other values. $\square$

The paper calls the locus $\alpha_{\mathbf{n}}(0) = 1$ a breakdown locus. Theorem 10.10 shows why breakdown does not automatically mean nonexistence: the compatibility sum decides between no solution and an affine family.

10.7. M-functions and limits. For scalar branch multipliers $r_0, \dots, r_{p-1} \in K^\times$, the associated M-function is the prefix cocycle

$$M_N(z) = \prod_{h=0}^{N-1} r_{d_h(z)} = r_0^N \prod_{j=1}^{p-1} \left(\frac{r_j}{r_0} \right)^{\#_{p^j}^{(N)}(z)}. \quad (169)$$

It satisfies

$$M_{N+L}(z) = M_N(z)M_L(\vartheta_p^{\circ N} z).$$

These cocycles are the proposed majorants for passing from equation (167) to a framewise distributional limit.

Before stating the exact limit consequence, set

$$\begin{aligned} \mathcal{S}(\mathbb{Z}_p, K) &= \{\phi : \mathbb{Z}_p \rightarrow K : \phi \text{ is locally constant modulo } p^N \text{ for some } N\}, \\ \mathcal{S}(\mathbb{Z}_p, K)' &= \text{Hom}_K(\mathcal{S}(\mathbb{Z}_p, K), K). \end{aligned}$$

Proposition 10.11 (the exact consequence of transform convergence). *Fix $\mathbf{n}$. Suppose that for every $\mathbf{m} \leq \mathbf{n}$ and every $t \in \Gamma_p$ the limit*

$$A_{\mathbf{m}}(t) = \lim_{N \rightarrow \infty} \hat{X}_{\mathbf{m},N}(t)$$

exists in the fixed complete field K . Then

$$A_{\mathbf{n}}(t) = \sum_{\mathbf{m} \leq \mathbf{n}} \alpha_{\mathbf{m},\mathbf{n}}(t) A_{\mathbf{m}}(pt) \quad (t \in \Gamma_p). \quad (170)$$

Moreover, every function $A : \Gamma_p \rightarrow K$ defines a unique algebraic distribution $T_A \in \mathcal{S}(\mathbb{Z}_p, K)'$ by

$$\langle T_A, \phi \rangle = \sum_{t \in \Gamma_p} \hat{\phi}(t) A(-t), \quad \phi \in \mathcal{S}(\mathbb{Z}_p, K), \quad (171)$$

where the sum is finite. With the Fourier sign of equation (118), this convention gives $\langle T_A, \langle -t, \cdot \rangle \rangle = A(t)$. In particular, if

$$\langle \mu_{\mathbf{m},N}, \phi \rangle = \int_{\mathbb{Z}_p} X_{\mathbf{m},N}(z) \phi(z) dz,$$

then $\mu_{\mathbf{m},N} \rightarrow T_{A_{\mathbf{m}}}$ pointwise on $\mathcal{S}(\mathbb{Z}_p, K)$ for every $\mathbf{m} \leq \mathbf{n}$.

Proof. Take $N \rightarrow \infty$ in equation (166). Its sum over $\mathbf{m} \leq \mathbf{n}$ is finite, so convergence of the displayed terms is all that is required and yields equation (170). Every locally constant test function has a unique finite character expansion $\phi(z) = \sum_t \hat{\phi}(t) \langle t, z \rangle$ by finite Fourier inversion. Hence equation (171) is a well-defined K -linear functional, and evaluating it on the character $\langle -t, \cdot \rangle$ proves the final identity and uniqueness. Finally, finite Fourier inversion for ϕ gives

$$\langle \mu_{\mathbf{m}, N}, \phi \rangle = \sum_t \hat{\phi}(t) \hat{X}_{\mathbf{m}, N}(-t),$$

where the sum is finite. Termwise passage to the limit is therefore exact and yields equation (171) with $A = A_{\mathbf{m}}$. $\square$

Status record 10.12 (conditional analytic theorem in the 2025 paper). The paper explicitly introduces a Main Limit Assumption before asserting that the formal Fourier data are realized by frame-quasi-integrable products. Theorem 10.11 isolates exactly what follows if all of the required finite-transform limits exist in one fixed completion.

The later Main Limit Lemma proposes sufficient conditions in terms of strong frame summability of M_N , uniform control of shifted X , and $(\mathcal{F}, M)$ -type decay. The printed proof does not establish those sufficient conditions, so the ensuing base case, induction, power theorem, and main product corollary remain conditional.

10.8. Test functions, distributions, and what the product means. More generally, for a commutative coefficient ring R with identity, let

$$\mathcal{S}(\mathbb{Z}_p, R) = \{\phi : \mathbb{Z}_p \rightarrow R : \phi \text{ is locally constant modulo } p^N \text{ for some } N\}, \quad (172)$$

$$\mathcal{S}(\mathbb{Z}_p, R)' = \text{Hom}_R(\mathcal{S}(\mathbb{Z}_p, R), R). \quad (173)$$

The second module is the algebraic space of (p, R) -adic distributions. If μ is a distribution and ϕ a test function, multiplication by a test function is unambiguous:

$$\langle \phi \mu, \psi \rangle = \langle \mu, \phi \psi \rangle \quad (\psi \in \mathcal{S}(\mathbb{Z}_p, R)).$$

Here the distribution occupies the first slot and the test function the second. This convention is used throughout the subsection. Products of two arbitrary distributions are not defined by this rule.

Source issue 10.13 (the product algebra is transported, not intrinsic). The 2025 paper defines, for its designated family,

$$(X^m(z) dz)(X^n(z) dz) := X^{m+n}(z) dz$$

and defines the corresponding “convolution powers” to be the recursively constructed formal Fourier data. The exact formal object is the polynomial algebra $R[T]$, with T^n labelled by the symbol $[X^n(z) dz]$ and with $T^m T^n = T^{m+n}$. This construction exists without realizing any symbol as a distribution.

Suppose separately that distributions $\mu_n \in \mathcal{S}(\mathbb{Z}_p, R)'$ have actually been constructed for the named symbols, and define the R -linear realization map

$$\Phi : R[T] \longrightarrow \mathcal{S}(\mathbb{Z}_p, R)', \quad \Phi \left(\sum_n a_n T^n \right) = \sum_n a_n \mu_n.$$

The rule

$$\mu_m \odot \mu_n := \mu_{m+n} \quad (174)$$

descends to a well-defined multiplication on $\text{im } \Phi$ if and only if $\ker \Phi$ is an ideal of $R[T]$. Indeed, well-defined descent is equivalent to $\Phi(f) = \Phi(f') \Rightarrow \Phi(fg) = \Phi(f'g)$ for every g , which is equivalent to closure of $\ker \Phi$ under multiplication by every polynomial. When this holds, the realized algebra is canonically $R[T]/\ker \Phi$; when Φ is injective it is the polynomial algebra itself. If $\ker \Phi$ is not an ideal, the declaration equation (174) depends on the chosen polynomial representative and is not a product on the image.

The paper also warns that ordinary convolution of its Fourier data generally diverges. The formal product above is therefore not a proof that arbitrary distributions possess pointwise products, nor that standard Fourier convolution realizes the declared multiplication.

10.9. Failure and repair ledger for the product paper. The following defects affect theorem status rather than notation alone.

- (1) Haar probability of a residue class modulo p^N is printed in two places as $1/p$; it is p^{-N} . The corresponding finite character expansion needs the same correction.
- (2) In the archimedean Wiener algebra, absolutely summable Fourier coefficients must carry the ℓ^1 norm $\sum_t |\hat{f}(t)|$, not the coefficient supremum norm printed in the paper.
- (3) The Main Limit Lemma repeatedly infers that a positive sequence tending to zero is monotone on a tail, and that a tail sum is O of its first term. Neither follows from convergence or from a root bound. The base case also uses the false estimate $Na_N = O(a_N)$.
- (4) The localization $R[(1 - a_0)^{-1}]$ can be quotiented by an ideal I only after requiring I to be disjoint from every power of $1 - a_0$; the weaker printed condition $1 - a_0 \notin I$ does not suffice. A fraction field of the quotient additionally requires a domain, for example a prime ideal. Multivariable polynomial localizations are not Dedekind domains in dimension greater than one.
- (5) A final corollary says a breakdown index gives no solution for any initial data, contradicting the compatible affine-family case already proved in the paper and restated in theorem 10.10.

A repair programme must first correct Haar and Wiener normalization, impose domain-safe localization hypotheses, replace the false monotonicity steps by a genuine monotone majorant or regular-variation estimate, and prove rising continuity of the proposed limits. Only then can the analytic product and inversion conclusions be promoted from conditional claims.

11. SYNTHESIS: TERMINOLOGY, DEPENDENCIES, AND EXACT THEOREM STATUS

This section is the control panel for the edition. It does not add another layer of analogy to the preceding mathematics. It identifies the same objects in Siegel's terminology and in standard mathematical language, records the hypotheses needed to pass from one object to the next, and marks the precise point at which each proved chain stops. A status assigned below belongs to a particular statement in a particular witness; it is not a judgment on an entire paper or on the programme as a whole.

11.1. Authorship and editorial boundary.

Status record 11.1 (Authorship boundary). The mathematical corpus edited in this volume is the work of Maxwell C. Siegel. In particular, the source constructions and the names *Hydra map*, *numen*, *dreamcatcher*, (p, q) -*adic analysis*, *rising-continuity*, and *F-series* remain attributed to Siegel. The standard-language column in the crosswalk below explains those objects; it does not rename them and does not transfer their authorship.

Typed recastings with explicit source crosswalks, corrected quantifiers, padded-word conventions, replacement hypotheses, counterexamples to printed assertions, and proofs explicitly supplied in this edition are editorial work. They are described as reconstructions or repairs, not retroactively attributed to Siegel as source text. Conversely, a correction supplied by this edition does not change authorship of the underlying corpus.

11.2. Terminology crosswalk. The third column gives a standard description of the typed source object. The fourth records an identification which must *not* be made.

Siegel term	Typed object in this edition	Standard-language description	Exactness boundary
Hydra map	A residue-class assignment of affine branches $H_j = (r_j, c_j)$ on $\mathcal{O}_K$, admissible on the assigned classes	A piecewise-affine arithmetic dynamical system with a finite digit alphabet	The 2019 surjective ϱ -Hydra is a historical specialization; it is not identical to the 2026 global-field definition.
Scalar integer Hydra	The specialization with $D \in \mathbb{Z}_{\geq 2}$, $K = \mathbb{F} = \mathbb{Q}$, $\mathcal{O}_K = \mathbb{Z}$, $\Lambda = D\mathbb{Z}$, and rational-affine branches integral on their assigned classes	Exactly a fixed-modulus Matthews generalized Collatz presentation, hence an rcwa mapping in Kohl's mapping monoid; see theorems 4.7 and 4.8	The displayed modulus need not be minimal, an rcwa mapping need not be bijective, and rcwa membership supplies none of the Hydra-specific centeredness, properness, or branch-recognition predicates. Exact-integrality of a displayed D -presentation is equivalent to Matthews relatively prime type only together with $D = d_{\text{Mat}}(H)$.
Affine lattice morphism	$\text{Aff}_{\mathbb{F}}(K) = \text{GL}_{\mathbb{F}}(K) \ltimes K$, acting by $x \mapsto rx + c$	An affine $\mathbb{F}$ -linear automorphism of the affine space underlying K	The source phrase does not imply that such a map preserves $\mathcal{O}_K$, Λ , or any lattice; each lattice condition is stated separately.
Exact-integral Hydra	$H_j(x) \in \mathcal{O}_K$ exactly when $x \in \Lambda_j$, for $x \in \mathcal{O}_K$	A branch recognition or branch-correctness condition	Mere admissibility only says that the selected branch maps its selected class into $\mathcal{O}_K$; the stronger field-wide predicate on all $x \in K$ is not substituted.
Hydra word	$H_{\mathbf{j}} = H_{j_0} \circ \cdots \circ H_{j_{L-1}}$	A word in an affine semigroup, or a symbolic cylinder word	The rightmost map acts first; chronological itineraries must therefore be reversed under this convention.
M_H	The linear part of $H_{\mathbf{j}}$	A multiplicative linear cocycle or monoid representation	In the global-field theory it is generally an endomorphism, not a scalar.
Numen X_H	The translation part $H_{\mathbf{j}}(x) = M_H(\mathbf{j})x + X_H(\mathbf{j})$	An affine translation cocycle, equivalently $H_{\mathbf{j}}(0)$ for every affine word	Centeredness is what makes this word cocycle descend through appended zero digits. For a noncentered Hydra the digit-recursive solution also contains the terminal zero-branch fixed point.

Siegel term	Typed object in this edition	Standard-language description	Exactness boundary
Primitive F-series	The truncation expansion of X_H along the digits of z	A digit-recursive affine series	Its infinite value has no meaning until a codomain completion and a convergence condition are specified.
Correspondence Principle	Repeated symbolic code $\mapsto$ affine fixed point, followed by branch correctness	A periodic symbolic-coding theorem for a piecewise-affine map	A formal fixed point of a branch composition is not automatically a periodic point of the piecewise map.
Classical parity conjugacy	$Q_q : \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ is the chronological parity encoder and $\Phi_q = Q_q^{-1}$ conjugates the binary shift to T_q	The Bernstein–Lagarias solenoidal conjugacy, with the Böhm–Sontacchi return formula at $q = 3$	For odd prime q , neither classical map is the cross-adic numen $\chi_q : \mathbb{Z}_2 \rightarrow \mathbb{Z}_q$. Their purely periodic values have the same rational representative under separate canonical embeddings only after reversing the chronological word; see theorems 6.5 and 6.7.
Set-series	$\varsigma_V(w) = \sum_{v \in V} w^v$ and $\psi_V(z) = \sum_{v \in V} e^{2\pi i v z}$	Ordinary and Fourier generating functions of an indicator sequence	The Dirichlet set-series needs a separate convention when $0 \in V$.
Permutation operator	$P_H : \ell^\infty(\mathbb{N}_0) \rightarrow \ell^\infty(\mathbb{N}_0)$, $(P_H c)_n = c_{H(n)}$, together with its conjugates on the named generating-map ranges	An isometric coefficient pull-back when the 2019 Hydra H is surjective	It is not the transfer operator which sums over inverse branches, and the isometric injection P_H need not be surjective.
Edgepoint and virtual residue	$\text{VRes}_x \psi = \lim_{y \downarrow 0} y \psi(x + iy)$ when finite	A radial first-order boundary coefficient	It differs by a factor from the ordinary complex residue of a simple pole.
Dreamcatcher	The trace $R_{\psi, T}(t) = \text{VRes}_t[\psi]$ in $\ell^2(T/\mathbb{Z})$	A square-summable boundary-singularity profile on a specified periodic carrier $T \subseteq P(\psi)$	It is not the original holomorphic function and need not determine that function.
Singularity Conservation Law	The fixed-point identity $\mathfrak{Q}_{H:T} R_{\psi, T} = R_{\psi, T}$ on the maximal graph domain defined by the raw finite expression $\mathcal{E}_{H:T}$	A typed boundary-value intertwining identity	One-sided branch invariance makes the raw expression defined; it does not prove that $\mathfrak{Q}_{H:T}$ is an everywhere-defined bounded operator on $\ell^2(T/\mathbb{Z})$.
(p, q) -adic analysis	Functions on $\mathbb{Z}_p$ with values in a complete field of residue characteristic $q \neq p$	Harmonic analysis with different domain and codomain places	It is not analysis over a single object carrying simultaneously the p - and q -adic topologies.

Siegel term	Typed object in this edition	Standard-language description	Exactness boundary
Rising-continuity	$f(z) = \lim_N f([z]_{p^N})$ on the bounded subspace $\tilde{C}_b(\mathbb{Z}_p, K)$ the del norm equals the supremum norm	Sequential continuity along the one canonical truncation path at each point, with an exact Banach subspace after boundedness is imposed	It is weaker than topological continuity and does not itself imply boundedness.
Frame	A point-indexed choice $z \mapsto \mathcal{F}(z)$ of a complete valued extension in which to read a limit	A field of pointwise convergence topologies, or a bundle of chosen completions	It is not one normed target space unless additional uniform structure is imposed.
M-function	$M_N(z) = \prod_{h < N} r_{d_h}(z)$	A multiplicative digit cocycle or prefix product	Decay of $M_N(z)$ does not by itself make its norm sequence monotone.
Quasi-integrability	In one fixed complete field K , pointwise limits $A_{\mathbf{m}}(t) = \lim_N \hat{X}_{\mathbf{m}, N}(t)$; each resulting function $A : \Gamma_p \rightarrow K$ acts on locally constant tests by the finite sum equation (171)	An algebraic distributional realization of specified limiting Fourier data	A merely formal solution of the Fourier recursion does not establish the required finite-transform limits.
Breakdown locus	The resonance equation $\alpha_{\mathbf{n}}(0) = 1$	A zero-mode resonance locus for the formal Fourier recurrence	It does not mean automatic nonexistence: the lower-term compatibility sum decides between no solution and an affine family.
Algebra of distributions	The declared rule $(X^m dz)(X^n dz) := X^{m+n} dz$ on designated symbols, realized by an R -linear map $\Phi : R[T] \rightarrow \mathcal{S}(\mathbb{Z}_p, R)'$	A polynomial algebra transported from powers of the named F-series	Multiplication descends to $\text{im } \Phi$ exactly when $\ker \Phi$ is an ideal; even then it is not an intrinsic product of arbitrary distributions or ordinary convergent Fourier convolution.

11.3. The typed dependency and result map. There are four principal chains. Each arrow below has the hypotheses shown in its row; no arrow is supplied merely by similarity of notation.

Chain	Input and indispensable hypothesis	Output, and the next permitted step
Number-field state space	A number field K , its full Minkowski map $\iota_K : K \rightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$, and an integral basis of $\mathcal{O}_K$	$\iota_K(\mathcal{O}_K)$ is a full lattice and hence locally finite; the preperiodic-or-escaping dichotomy then applies to self-maps of this full image. A single archimedean projection is not substituted; see theorems 3.3 and 3.4.
Digit compactum	An integer $p \geq 2$, the discrete alphabet $\mathcal{A}_p$, and the compatible residue system $\varprojlim_N \mathbb{Z}/p^N \mathbb{Z}$	The digit map is a homeomorphism $\mathcal{A}_p^{\mathbb{N}_0} \simeq \mathbb{Z}_p^{\text{dig}}$, and Haar probability is the uniform product law with cylinder mass p^{-N} ; see theorem 5.5.

Chain	Input and indispensable hypothesis	Output, and the next permitted step
Affine foundation	Admissible branches $H_j = (r_j, c_j)$ and the affine semidirect-product law	The piecewise map H and the exact decomposition $H_{\mathbf{j}}(x) = M_H(\mathbf{j})x + X_H(\mathbf{j})$; see theorem 4.9.
Scalar prior-art boundary	$K = \mathbb{F} = \mathbb{Q}$, $\mathcal{O}_K = \mathbb{Z}$, standard classes modulo a displayed $D \in \mathbb{Z}_{\geq 2}$, and rational-affine branches integral on their assigned classes	Unique integers $m_i \neq 0$ and $\rho_i \equiv im_i \pmod{D}$ give $H_i(x) = (m_i x - \rho_i)/D$; the resulting map is rcwa and $d_{\text{Mat}}(H) = \text{Mod}(H) \mid D$. Exact-integrality is equivalent to $\gcd(m_0 \cdots m_{D-1}, D) = 1$ for the displayed presentation; it forces both least moduli to equal D and hence makes the map Matthews relatively-prime. Conversely, Matthews relatively prime type gives this exact-integral presentation only when $D = d_{\text{Mat}}(H)$. Membership in Kohl's group $\text{RCWA}(\mathbb{Z})$ requires a separate global bijectivity proof; see theorems 4.7 and 4.8.
Affine recentering	An integral translation τ_a and the induced quotient permutation $\Lambda_{\pi_a(j)} = \Lambda_j + a$	The conjugated branch on Λ_j is $\tau_a^{-1} H_{\pi_a(j)} \tau_a$; its zero-class branch is centered exactly when $H_{\pi_a(0)}(a) = a$. A fixed point of H_0 suffices without another branch equation only when $a \in \Lambda$; see theorem 4.6. Properness after translation is likewise the exact condition that $I - r_{\pi_a(0)}$ be invertible.
Finite numen	A specified terminal value a satisfying $(I - r_0)a = c_0$; for a centered proper Hydra this forces $a = 0$	A unique digit-recursive function on $\mathbb{N}_0$ with $f(0) = a$ and $f(pn + j) = r_j f(n) + c_j$; see theorem 5.4.
Infinite numen	A named completion K_ℓ and point-wise prefix-product decay, almost-everywhere geometric-mean contraction, or uniform contraction	A rising, measurable, or continuous extension, respectively, satisfying the same digit equations; see theorems 5.6, 5.8 and 5.9.
Rational digit input	The canonical CRT identification $\mathbb{Z}_p^{\text{dig}} \simeq \prod_{\ell \mid p} \mathbb{Z}_\ell$ and the diagonal copy of $\mathbb{Q}$	The intersection is exactly $\mathbb{Z}_{(p)} = \{a/b : \gcd(b, p) = 1\}$, and these are exactly the eventually periodic base- p digit strings; see theorem 6.3.
Periodic code	A repeated word $\mathbf{j}$ and invertibility of $I - M_H(\mathbf{j})$ (obtained, for example, from $\ M_H(\mathbf{j})\ _\ell < 1$)	The unique <i>formal</i> fixed point $(I - M_H(\mathbf{j}))^{-1} X_H(\mathbf{j})$; see theorem 6.1.
Classical conjugacy comparison	An odd integer q , the chronological parity encoder Q_q , and a length- L parity block with code e and reversed code $\text{rev}_L(e)$	The typed isometric conjugacy $Q_q T_q = S Q_q$ for every odd q and the common rational representative $R_{q, \epsilon}$ of equation (60) under the separate embeddings; when q is prime, its cross-adic numen evaluation is equation (61). The latter equality is neither a global map identity nor a composite- q extension of χ_q .
Genuine periodicity	The preceding formal fixed point lies in the integral state space, and branch correctness is proved	A periodic point of the piecewise map. This implication is proved for T_q by theorem 6.9; the general 2026 implication remains at source-sketch status.

Chain	Input and indispensable hypothesis	Output, and the next permitted step
Dreamcatcher input	A function $\psi \in \mathcal{D}_1(T) \cap \text{ran}(\mathcal{G}_{\mathbb{H}})$, a branch-invariant carrier T , and the co-efficient pullback $\mathcal{Q}_H$ on its declared range	The Virtual Residue Formula equation (80); cross-section densities provide an independent exact Abelian criterion for its radial limits by theorem 7.6.
Dreamcatcher conservation	The preceding data and a verified $R_{\psi,T}$ -membership in the maximal graph domain of $\mathfrak{Q}_{H:T}$	The fixed-point identity $\mathfrak{Q}_{H:T}R_{\psi,T} = R_{\psi,T}$ in theorem 7.11; no all-space boundedness is inferred.
Finite rational obstruction	Prime and regulated 2019 Hydra plus finite support modulo $\mathbb{Z}$	The Finite Dreamcatcher conclusion $R = R(0)\mathbf{1}_{\mathbb{Z}}$; rational invariant sets are then finite or cofinite.
Contour coefficient	The exact coefficients $a_n(x) = (1 - M_q(n))x - \chi_q(n)$ and a line $c > \sigma_q$ of absolute convergence	Perron inversion recovers one coefficient by equation (92); the integral vanishes exactly when that particular word numerator vanishes.
Dirichlet recursion	The coefficient recursions equation (95) and $\Re s > \sigma_q$	The corrected ordinary and Dirichlet generating-function identities equations (96) and (97).
Dirichlet continuation	The corrected shift equation and the normal-tail estimate theorem 8.7	Unique meromorphic continuation to $\mathbb{C}$, the candidate-pole containment equation (102), and the corrected $q = 3$ principal parts. No actual pole is inferred merely from a denominator zero.
Contour dynamics	A synchronized block length and the T_q branch- correctness theorem	A phase-sensitive periodicity statement using $B_{2,L}(n)$; shortest-word data alone do not encode every phase.
Function Fourier theory	$q \neq p$, a specified tower $\zeta_{N+1}^p = \zeta_N$ of primitive roots, normalized K -valued Haar integration, and a complete algebraically closed nonarchimedean target	The explicit pairing $\Gamma_p \times \mathbb{Z}_p \rightarrow K^\times$, classification of locally constant K -valued characters, both finite orthogonality identities, the isometry $C(\mathbb{Z}_p, K) \simeq c_0(\Gamma_p, K)$, and the precisely signed product-convolution identities; see theorems 9.1 to 9.4.
Function cyclicity	The preceding Banach-algebra isomorphism	The four-way Wiener–Tauberian equivalence in theorem 9.7. Its dense finite translate span is $\widehat{\chi} * c_{00}(\Gamma_p, K)$; only after closure does it equal the closed principal ideal $\overline{\widehat{\chi} * c_0(\Gamma_p, K)}$.
Measure cyclicity	A bounded Fourier–Stieltjes transform and existing local density limits	Density of translates implies nonvanishing of each existing limit. The reverse arrow is not proved.
Rising Banach domain	A complete nonarchimedean target K and the bounded rising-continuous class $\widetilde{C}_b(\mathbb{Z}_p, K)$	The del norm equals the supremum norm and makes this exact subspace Banach; unrestricted rising-continuity still permits infinite del quantity; see theorem 10.2.
Product algebra, finite level	Degree-one affine digit recursions in a common commutative coefficient ring; for analytic transforms, one fixed nonarchimedean field K and continuous K -valued products	The multinomial branch identity equation (163), the continuous Fourier recursion, the universally exact truncation recursion, and the zero-mode compatibility classification theorem 10.10.

Chain	Input and indispensable hypothesis	Output, and the next permitted step
Product distributions	For every $\mathbf{m} \leq \mathbf{n}$ and $t \in \Gamma_p$, convergence of $\widehat{X}_{\mathbf{m},N}(t)$ in one fixed complete field K	The limiting functions satisfy the exact Fourier recursion and define unique algebraic distributions on $\mathcal{S}(\mathbb{Z}_p, K)$; the truncation distributions converge pointwise on that test space by theorem 10.11. The printed Main Limit sufficient conditions do not establish this hypothesis.

In compact form, the verified central spine is

$$\{H_j = (r_j, c_j)\} \longrightarrow (M_H, X_H) \longrightarrow X_H|_{\mathbb{N}_0} \longrightarrow X_{H,\ell} \longrightarrow (I - M_H(\mathbf{j}))^{-1} X_H(\mathbf{j}),$$

where the fourth arrow requires a place and convergence, and the last value becomes a periodic point only after branch correctness. The three analytic branches are

$$\psi_V \longrightarrow \text{VRes } \psi_V \longrightarrow \mathfrak{Q}_{H:T} R, \quad (M_q, \chi_q) \longrightarrow F_q \longrightarrow a_n(x),$$

and

$$C(\mathbb{Z}_p, K) \xrightarrow{\mathcal{F}} c_0(\Gamma_p, K), \quad (X_j)_j \longrightarrow X_{\mathbf{n},N} \longrightarrow \widehat{X}_{\mathbf{n},N}.$$

Within these three displayed analytic branches, the unresolved steps are the everywhere-defined boundedness of the dreamcatcher operator, the shortest-word contour biconditional, the converse measure cyclicity criterion, and source-sufficient hypotheses for the general product limit. The exact maximal graph domain and the exact consequence of pointwise finite-transform convergence are proved. The separate source-status table below records additional unproved or false claims outside those displayed branches.

11.4. Result-status table. The five codes below are mutually exclusive as editorial dispositions. “Proof checked” means that the source proof and its named dependencies were read against the typed statement used here; it does not mean computer formal verification. When this edition gives a complete proof or changes a hypothesis and proves the changed statement, the stronger code **R** is used instead.

- R:** reconstructed and proved in this edition;
S: source theorem with its proof checked against the typed hypotheses;
C: conditional, conjectural, or formal statement not promoted to an unconditional theorem;
U: superseded witness retained for mathematical history;
W: withdrawn witness or statement invalid as printed.

Code	Result or witness	Source	Exact disposition
R	Preperiodic-or-escaping dichotomy	2601.17030v3	Reconstructed and proved in theorem 3.4 with the necessary locally finite hypothesis replacing bare uniform discreteness.
R	Full Minkowski lattice foundation	Standard number-field foundation supplied by this edition	The full map, its domain and codomain, the integral-basis matrix, nonzero discriminant, and local finiteness are proved in theorem 3.3; no one-embedding substitute is used.
R	Digit compactum and Haar product law	2412.02902v1 and 2601.17030v3, rebuilt	The explicit map from digit sequences to the inverse limit, its inverse, homeomorphism property, and every cylinder mass are proved in theorem 5.5, including composite p .

Code	Result or witness	Source	Exact disposition
R	Rational points and eventually periodic digits	Edition-supplied arithmetic foundation	The compatible CRT isomorphism, diagonal intersection $\mathbb{Q} \cap \mathbb{Z}_p^{\text{dig}} = \mathbb{Z}_{(p)}$, and both directions of eventual periodicity are proved in theorem 6.3.
R	Affine word calculus and repeated-word fixed point	2601.17030v3, reorganized	Proved in theorems 4.9 and 4.11; endomorphism order and invertibility are explicit.
R	Exact scalar prior-art identification	Matthews’s generalized Collatz form and Kohl’s rcwa definition	The fixed-displayed-modulus equivalence, the typed inclusion in the rcwa mapping monoid, divisibility of the displayed modulus by the minimal modulus $d_{\text{Mat}}(H) = \text{Mod}(H)$, the exact equivalence between scalar exact-integrality and the conjunction of Matthews relatively-prime type with a displayed least modulus, equality of the minimal modulus with the displayed modulus under exact-integrality, and the separate bijectivity condition for $\text{RCWA}(\mathbb{Z})$ are proved in theorems 4.7 and 4.8.
R	Translation conjugacy and zero-class centering	2601.17030v3, corrected	The induced residue-class permutation, every conjugated branch, preservation of admissibility and exact-integrality, and the necessary-and-sufficient centering equation are proved in theorem 4.6.
R	Digit recursion and the finite numen	2007.15936v3, 2412.02902v1, 2601.17030v3	Reconstructed and proved in theorem 5.4, including the noncentered terminal-value alternative.
R	Pointwise, almost-everywhere, and uniform rising extension	2412.02902v1, 2601.17030v3	Proved under the displayed prefix-product, geometric-mean, and uniform contraction hypotheses in theorems 5.6, 5.8 and 5.9.
R	Phase-sensitive T_q correspondence	2007.15936v3	The affine identity, 2-adic branch correctness, and padded length- L converse are proved in theorem 6.10.
R	Classical conjugacy and reversed periodic overlap	Böhm–Sontacchi (1978), Bernstein–Lagarias (1996), and Laarhoven–de Weger (2012)	The parity map and inverse conjugacy are typed and proved for every odd q in theorem 6.4. The two adic values are proved to be the images of one rational representative under their separate canonical embeddings. The mandatory reversal, the $q = 3$ specialization to the Böhm–Sontacchi Proposition 5 candidate, its period-dividing boundary, and global nonidentity with χ_q are proved in theorems 6.5 and 6.7.
R	Square Sum Lemma, Virtual Residue Formula, and typed SCL	1909.09733v3	The square-sum proof-idea attribution is bound to accepted MathOverflow answer 333801 by user <i>fedja</i> [16]; this edition’s finite-subset Bessel proof remains editorial and is not attributed to the answerer. VRF/SCL are proved with the operator-domain premise stated.

Code	Result or witness	Source	Exact disposition
R	Exact coefficient pullback	1909.09733v3, built	re- The map $P_H : \ell^\infty \rightarrow \ell^\infty$ is proved to be an isometric injection from surjectivity, and the two generating-series operators are defined only as its conjugates on their declared ranges.
R	Cross-section densities and virtual residues	1909.09733v3, built	re- The residue-class Cesàro limits, summation-by-parts Abel passage, and exact factor $1/(2\pi)$ are proved in theorem 7.6; no Tauberian converse is used.
R	Dyadic sums and coefficient contour identity	2111.07883v2	Recalculated and proved in equations (90) and (91) and theorem 8.1.
R	Nonarchimedean Fourier isometry and function Wiener-Tauberian theorem	2208.11082v2	Reconstructed from finite Fourier duality. The compatible-root pairing, character classification, both finite orthogonality formulas, the isometry, the typed convolution morphisms, and the cyclicity theorem are proved in theorems 9.1 to 9.4 and 9.7.
R	Rising van der Put theorem and unique rising continuation	2412.02902v1	Proved by the exact telescoping path argument in theorems 10.4 and 10.5.
R	Del norm on the bounded rising class	2412.02902v1, corrected	cor- The level operators are defined for every N , the extended del quantity is proved equal to the supremum quantity on rising-continuous functions, and $\tilde{C}_b(\mathbb{Z}_p, K)$ is proved complete in theorem 10.2.
R	Product branch recursion and formal compatibility trichotomy	2507.13358v1	Proved at the finite/formal level in theorems 10.9 and 10.10; no limiting distribution is inferred.
R	Fixed-target F-series domains and the χ_3 frame identity	2307.00436v1 and 2412.02902v1, rebuilt	and Padded digit counts give the exact convergence domains D_E and D_X , every forward/converse domain implication, and the scalar and affine branch equations in one complete field. The two-place χ_3 identity follows instead from the finite telescoping formula equation (157); no disputed pointwise-necessity argument is used.
R	Pointwise finite-transform limits and algebraic distributions	2507.13358v1, constructed	re- Under the exact fixed- K pointwise convergence hypothesis, the limiting Fourier recursion, unique functional on locally constant tests, and pointwise convergence of the truncation distributions are proved in theorem 10.11.
S	Finite Dreamcatcher Theorem	1909.09733v3	Source theorem; the Off- ϱ theorem, auxiliary functional equation, support lemmas, Off- ϱ Infinity theorem, and final dependency chain were checked. It is retained with the source-proof status recorded in theorem 7.13.

Code	Result or witness	Source	Exact disposition
S	Weak Rationality Theorem 1	1909.09733v3	Source consequence; its dependency on rational generating functions, SCL, finite virtual-pole support, and the Finite Dreamcatcher Theorem was checked. This disposition does not extend to Weak Rationality Theorem 2.
R	Corrected coefficient, ordinary generating-function, and Dirichlet functional equations for F_q	2111.07883v2, corrected	The printed ordinary generating-function derivation has a wrong first even term. The recursions and both corrected identities are proved directly in theorems 8.4 and 8.5 on the proved initial absolute-convergence half-plane $\Re s > \sigma_q$.
R	Global continuation and corrected $q = 3$ principal parts	2111.07883v2 and Web II, rebuilt	Normal convergence and strip induction prove theorem 8.8; direct Laurent calculation proves theorem 8.9. The pole set is contained in the stated candidate set; equality is not asserted.
C	General global-field Correspondence Principle and divergence clause	Corre- 2601.17030v3	Conditional/source formulation only. The source labels the proof a sketch, and the zero fibre, correctness, and irrational-fibre steps are not established by its displayed hypotheses.
C	Theorem 12 rational-edgepoint conclusion	1909.09733v3	May be cited only with branch invariance of both rational edgepoints and rational non-edgepoints; the source itself identifies this as the method's Achilles heel.
C	Identification of a limiting dreamcatcher with the virtual residue of the limiting function	1909.09733v3	Explicitly Conjecture 1, not a theorem.
W	Printed F_q functional equation and its dependent singularity chain	2111.07883v2	The printed identity fails for every $x \neq 0$, as certified by the $s \rightarrow +\infty$ limit in Chapter 7. Its residue formulas contain spurious x -terms, and its actual-pole assertions lack noncancellation proofs. The continuation theorem and every retained principal part are instead re-derived from the corrected equation.
C	Self-similar numen measure uniqueness	2601.17030v3	Existence of the pushforward and its functional equation are typed; uniqueness requires a separate contraction or IFS hypothesis.
C	Main Limit, product distributions, power theorem, and inversion	2507.13358v1	Formal or conditional. Exact pointwise convergence of every required finite transform in one fixed K would produce the limiting recursion and algebraic distributions by theorem 10.11; the printed sufficient-condition proof and induction do not establish that convergence.

Code	Result or witness	Source	Exact disposition
C	Declared uct/convolution of named distributions	prod- 2507.13358v1 algebra	The symbolic polynomial algebra exists formally. After the designated distributions exist, its multiplication descends to their realized image if and only if the realization kernel is an ideal. It is not ordinary distribution multiplication or convergent convolution.
U	Parity-vector Fourier cessor	prede- 2002.12762v1	Preserved as a historical stage but superseded by the numen formulations; v2 is the withdrawal notice declaring the work obsolete.
U	Original 2111.07883v1	2111.07883v1	Materially different title and subject; superseded for the logical work by the current contour paper v2, not silently merged with it.
U	Earlier general Hydra conven- tions	pre-2601 corpus	The 2026 manual v3 controls the general name and word formalism. The specialized 2019 ϱ -Hydra remains the correct historical object for the dreamcatcher theorems.
U	Earlier witnesses of the 2026 manual	2601.17030v1–v2	v3 is the controlling witness and includes the correction to Proposition 7.6; earlier text is retained only for version history.
W	Number-ring functional- equation paper	2111.07882 v1/v2	v2 withdraws the work for an error in the quotient-ring section. It is not used to certify a general number-ring theorem.
W	Weak Rationality Theorem 2 proof	1909.09733v3	Invalid as printed: naturally measurable sets do not form a σ -algebra and natural density is not a countably additive measure on such a class.
W	Shortest-word contour bicon- ditional for every listed cycle phase	2111.07883v2, rely- ing on 2007.15936	Invalid as printed; the point 2 of the T_3 cycle already requires a padded block. The coefficient identity survives.
W	Measure Wiener–Tauberian equivalence	2208.11082v2	The proof gives density $\Rightarrow$ no existing zero local density and then repeats its contrapositive; the converse is not proved and can be vacuous.
W	Banach space of all compati- ble frame functions	2307.00436v1	Invalid without boundedness or continuity: compactness of the domain does not bound an arbitrary function.
W	Banach algebra of all rising- continuous functions	2412.02902v1	Invalid as defined: rising-continuity does not imply boundedness or finite del norm. This edition instead introduces and proves the exact Banach space $\widehat{C}_b(\mathbb{Z}_p, K)$ in theorem 10.2; the unrestricted source claim remains invalid.
W	Main Limit proof and result- ing unconditional product the- orems	2507.13358v1	Invalid as a proof: decay is repeatedly treated as eventual monotonicity; root bounds are used to claim a tail is the order of its first term; and $Na_N = O(a_N)$ is asserted.

Code	Result or witness	Source	Exact disposition
W	Simultaneous universal and exact integrality clauses	2601.17030v3	Inconsistent for $p \geq 2$ and incompatible with T_3 . Both source clauses quantify over $\mathcal{O}_K$. Equations (5) and (6) preserves that domain and separates the two mathematically usable predicates.
W	Pointwise digit-density convergence criterion	2601.17030v3	Invalid as printed: the source uses a limsup for frequencies of contractive branches and a liminf for expansive branches, reversing the inequalities needed for an upper Lyapunov bound. The corrected bound is equation (41).
W	Multidimensional repeated-block identity and resulting Correspondence Principle	2412.02902v1	Invalid as printed when coordinate digit lengths differ. Repetition of one synchronized vector block of common length N gives coordinate $n_i/(1 - p^N)$, not $n_i/(1 - p^{\lambda_p(n_i)})$. The multidimensional correspondence built on the latter identity is not promoted.

11.5. Source-by-source repair ledger. This ledger gives the minimum repair actually used by the edition. An entry of “not promoted” means that no repaired theorem is smuggled into a later argument.

Witness	Source defect or unstable point	Editorial treatment in this edition
1909.09733v3	The paper alternates between the defined class $\mathcal{D}_1(T)$ and an undefined $\mathcal{D}_2(T)$; its operator proof treats one-sided branch inclusion as bijectivity; its virtual residue is not the ordinary complex residue; the Permutation Operator Theorem says “basis” without a topology.	Declare $\mathcal{D}_1(T)$ as the edition’s explicit source-level choice and intersect it with the declared generating-operator range wherever $\mathcal{Q}_H$ is used; define the maximal graph domain and state its membership premise in SCL; retain the radial normalization; prove the exact co-efficient pullback and Abelian cross-section residue formula; and replace the Hamel-basis wording by the coefficient-orbit statement. Exclude Weak Rationality Theorem 2 and retain the limit identification as Conjecture 1.
2007.15936v3	The shortest binary word loses a high zero that belongs to a periodic block; one repeated-block truncation is written modulo 2^m rather than 2^{mL} . Its general integer-domain Hydra definition also requires the selected branch value to be a <i>nonnegative</i> integer iff the input has the assigned residue, which fails for negative inputs (including the negative fixed point of T_3).	Keep the block length in $B_{2,L}(n) = n/(1 - 2^L)$, reverse the chronological itinerary explicitly, and prove correctness in $\mathbb{Z}_2$. The shortest notation $B_2(n)$ is used only for a phase whose high block digit is nonzero. On domain $\mathbb{Z}$, branch recognition means <i>integer</i> iff assigned residue; nonnegativity belongs only to a separately restricted state space such as $\mathbb{N}_0$.

Witness	Source defect or unstable point	Editorial treatment in this edition
Böhm–Sontacchi (1978)	Printed Proposition 4 identifies a formal affine branch composition with $u^n(x)$ while quantifying independently over every $x \in \mathbb{Z}$ and every run tuple. This is false: for $x = 0$, $m = 1$, and $n_0 = n_1 = 1$, its right side is $1/4$ whereas $u^3(0) = 0$. The same right side is correctly $f \circ g \circ f(0) = 1/4$.	Derive the formal-composition identity from their Proposition 1 and identify it with an iterate only after the branch word is proved to be the actual parity itinerary. Recover the Proposition 5 fixed-return candidate independently from the edition’s branch-correctness theorem; retain the unnumbered paragraph after Proposition 5 on p. 264 as the exact period-dividing locator; and mark the last position inequality as a reconstruction from the formal run data, not as text printed in Proposition 5.
2111.07883v2	The contour theorem inherits the shortest-word phase defect; its common $c > \sigma_q$ quantifier is not correctly specialized to the $q = 3, 5$ parts; the $q \geq 7$ excluded set is malformed; one abscissa paragraph says “minimum”; the derivation of $f_q(z^2, x)$ uses $z/(1-z)$ in place of $z^2/(1-z^2)$; and the resulting false Dirichlet identity underlies the later singularity discussion.	Prove the Perron coefficient identity on $c > \sigma_q$ and state each specialization with its own line. Re-derive the coefficient, ordinary generating-function, and Dirichlet recursions; prove the corrected identity first on $\Re s > \sigma_q$; prove normal convergence and continue it strip by strip; compute the corrected $q = 3$ principal parts; and treat the remaining lattice as candidates unless noncancellation is separately proved.
2208.11082v2	The body restatement omits some headline hypotheses; a radius in the differentiation discussion tends in the wrong direction; and the measure WTT proves only one implication. Its notation also asks for an embedding $\overline{\mathbb{Q}} \hookrightarrow K$, which is unavailable in positive characteristic even though the stated field class allows that case.	Carry the full nonarchimedean setting and residue-characteristic restriction; construct only the required compatible p -power roots directly in K ; prove representative independence, character classification, and finite orthogonality; fix $(f * \mu)(z) = \mu(y \mapsto f(z - y))$ so the local-density coset is exact; use balls shrinking to a point; reconstruct the function WTT from finite Fourier duality; retain only density $\Rightarrow$ nonzero existing local densities for measures.
2307.00436v1	The abstract reverses the complex and q -adic parts of the standard frame; one $X_{p,q}$ display swaps p and q ; a convergence proof replaces the chosen input by -1 ; and all compatible functions are declared Banach under a possibly infinite sup norm.	The body frame definition and the original $S_{p,q}$ summand control; define the exact fixed-target domains D_E and D_X and test convergence on the actual digit path; prove the branch equations only on those domains; and derive the two-place χ_3 identity from its finite telescoping formula. Normed claims are restricted to a bounded or continuous class.

Witness	Source defect or unstable point	Editorial treatment in this edition
2412.02902v1	The definition of rising-continuity is pointwise, but a footnote asserts all such functions are bounded and the later del-norm Banach algebra uses that assertion. In the multidimensional block construction, coordinates of unequal digit lengths are repeated with their individual denominators even though the vector word has one synchronized common length.	Retain the pointwise definition and the valid truncation/van der Put theory. The prefix-free example $f(p^k) = q^{-k}$, with $f = 0$ elsewhere, is rising-continuous and unbounded in $\mathbb{Q}_q$. Define every del operator, prove that the del and supremum quantities agree on the rising-continuous class, and prove completeness of the exact bounded subspace $\tilde{C}_b(\mathbb{Z}_p, K)$. For a common block length N , use $n_i/(1 - p^N)$ in every coordinate; the printed multidimensional Correspondence Principle is not retained without a new proof from that corrected synchronization.
2507.13358v1	Haar mass and the finite character factor are printed as $1/p$ instead of p^{-N} ; the archimedean Wiener norm is printed as a coefficient supremum instead of ℓ^1 ; one finite recursion double-counts the diagonal; the Main Limit proof uses false monotonicity and tail estimates; localization and fraction-field hypotheses are insufficient; the paper's own warning says ordinary convolution usually diverges.	Correct the finite normalizations and use $\mathbf{m} < \mathbf{n}$ after separating the diagonal. Retain the multinomial and formal compatibility algebra. Prove the exact fixed- K consequence of pointwise finite-transform convergence, including the algebraic distribution and pointwise test-space limit. Require ideals disjoint from the localization set and a domain before taking a fraction field. Classify the source's limiting product/inversion results as conditional, and make the named transported multiplication descend to a realized image only under the exact kernel-ideal condition.
2601.17030v3	The Hydra definition simultaneously imposes universal branch integrality and an incompatible iff branch condition; properness is used as though it implied centeredness; the source's "affine lattice morphisms" are typed as all affine $\mathbb{F}$ -linear automorphisms and need not preserve a lattice; scalar division notation is used for an endomorphism; bare uniform discreteness is used as local finiteness; the pointwise density criterion reverses liminf and limsup; and the Correspondence proof is a sketch with an unlisted hypothesis, an unresolved zero orbit, and an unsupported fibre inference. Its preliminary ultrametric proposition says equality holds iff the summands have unequal norms; unequal norms are sufficient, not necessary.	Separate admissibility from exact-integrality; call $\text{Aff}_{\mathbb{F}}(K)$ the affine automorphism group and state lattice preservation separately; track the quotient permutation under translation and use $H_{\pi_a(0)}(a) = a$ as the exact centering condition; retain the zero-branch fixed point in the noncentered recursion; use $(I - M)^{-1}$; impose local finiteness for the escape dichotomy; use the corrected Lyapunov upper bound; and prove only the affine repeated-word calculation. The false biconditional for ultrametric equality is not used. Record the general Correspondence Principle and divergence statement at source-sketch status; do not infer self-similar-measure uniqueness without contraction.

The two withdrawn logical works require no mathematical repair inside this volume. The witness arXiv:2002.12762v1 is retained only as a superseded predecessor and its v2 record is the withdrawal notice. The witness arXiv:2111.07882 is retained only as a historical record of the number-ring programme; its quotient-ring theorem is not imported through another paper.

11.6. Exact coverage and explicit nonclaims. The edition provides exact mathematical coverage of the following objects.

- (1) The full Minkowski embedding of a number-field integer ring as a locally finite lattice and the resulting exact preperiodic-or-escaping dichotomy.
- (2) Residue-class affine dynamics, affine-automorphism typing, the translation-induced coset permutation and exact centering condition, the distinction between admissibility and branch recognition, affine word composition, and the ordered cocycles M_H and X_H ; together with the exact scalar fixed-modulus equivalence to Matthews's form and the typed inclusion into Kohl's rcwa mapping monoid.
- (3) The base- p digit compactum as an explicit inverse limit and product probability space, including composite p ; its compatible CRT decomposition; and the exact equivalence between diagonal rational points and eventually periodic digit strings.
- (4) The finite digit recursion for the numen, its rising extension in a specified completion, and pointwise, almost-everywhere, and uniform convergence criteria with their hypotheses displayed.
- (5) The repeated-code fixed-point identity, the complete branch-correctness argument for shortened T_q , and the phase-sensitive padded-word periodic correspondence; the complete classical parity encoder/conjugacy typing; the valid formal-composition identity behind Böhm–Sontacchi's printed Proposition 4; the independently recovered Proposition 5 return candidate; and the reversed purely periodic common rational representative under the separate Bernstein–Lagarias and numen embeddings.
- (6) Set-series, the exact isometric coefficient pullback and its declared generating-map ranges, edgepoints, the Abelian cross-section density formula, the Square Sum Lemma, the Virtual Residue Formula, the maximal graph-domain Singularity Conservation Law, and the source-status Finite Dreamcatcher Theorem.
- (7) The Dirichlet series ζ_{M_q} , ζ_{χ_q} , and F_q ; exact dyadic block sums; Perron extraction of one coefficient; the exact coefficient recursions; and the corrected ordinary and Dirichlet functional equations in $\Re s > \sigma_q$; unique meromorphic continuation to $\mathbb{C}$; a rigorous candidate-pole set; and corrected $q = 3$ principal parts and negative-integer residue recurrence.
- (8) The explicit compatible-root K -valued character pairing on $\mathbb{Z}_p$, classification and finite orthogonality, normalized nonarchimedean Haar and Fourier analysis, the isometric Fourier transform, precisely signed function/sequence/function–measure convolution, and the function Wiener–Tauberian equivalence, together with the one proved direction of the measure claim.
- (9) Rising-continuity, canonical truncations, the exact del operators and bounded Banach subspace, van der Put path expansions, point-selected frames, fixed-target digit-multiplicative and affine F-series, and the finite proof of the two-place χ_3 identity.
- (10) Multinomial product recursions over a common coefficient ring, exact continuous and finite Fourier recursions, the formal compatibility trichotomy at a breakdown index, the exact distributional consequence of fixed-field finite-transform convergence, and the kernel-ideal condition for a transported polynomial product to descend.

It makes none of the following claims.

- (N1) It does not prove the full Collatz conjecture, the weak Collatz conjecture, or a classification of all integer cycles.
- (N2) It does not claim that the scalar integer Hydra is a new class distinct from Matthews's generalized Collatz presentation or Kohl's rcwa mapping monoid. It does not identify an rcwa mapping with an rcwa permutation, equate a displayed Hydra modulus with the minimal rcwa modulus in general or without the scalar exact-integrality hypothesis, or infer Hydra-specific predicates from rcwa membership.

- (N3) It does not identify Q_q , Φ_q , and χ_q as global maps, suppress the mandatory chronological-word reversal, promote a return under T_q^L to exact period L , or use Bernstein–Lagarias’s composite-odd- q conjugacy to enlarge the prime- q codomain theorem for χ_q . It does not use Böhm–Sontacchi’s false unrestricted Proposition 4 identity to replace a formal branch word by u^L without first proving that the word is the actual parity itinerary.
- (N4) It does not claim that every phase of a cycle is represented by the shortest positive digit word, or that a periodic-point theorem rules out divergent positive trajectories.
- (N5) It does not prove the full 2026 global-field Correspondence Principle, its divergence clause, general injectivity of a numen, or a theorem identifying every point of a numen fibre.
- (N6) It does not claim that the 2019 natural-density argument is a measure-theoretic proof, that rational edgepoint carriers are automatically branch invariant, or that a limit of dreamcatchers is automatically the dreamcatcher of the limiting holomorphic function.
- (N7) It does not retain any continuation, residue, or pole conclusion merely because it was derived from the source’s false printed F_q equation. The newly proved global continuation does not imply that every candidate lattice point is an actual pole, does not establish a complete pole classification or the source’s asserted hyperexponential growth, and does not justify shifting the Perron contour left of the absolute-convergence half-plane.
- (N8) It does not prove the converse Wiener–Tauberian theorem for arbitrary measures, nor uniqueness of the self-similar numen measure without an independent regularity or contraction hypothesis.
- (N9) It does not treat every rising-continuous or frame-compatible function as bounded, and it does not assert the dissertation’s Banach-algebra theorem on the unrestricted class.
- (N10) It does not define a general product of distributions, identify the formal convolution powers with ordinarily convergent convolution, or promote the 2025 Main Limit, product, measure, and inversion conclusions beyond their conditional status. The edition proves what follows from actual pointwise finite-transform convergence; it does not claim the source’s defective sufficient-condition argument establishes that premise.
- (N11) It does not promote the dissertation’s multidimensional Correspondence Principle: its repeated vector block uses coordinatewise digit lengths where one common synchronized block length is required.
- (N12) It does not revive the withdrawn quotient-ring paper or merge materially different arXiv versions into one theorem.

These exclusions are part of the mathematical result. They identify the exact frontier of the reconstructed corpus: the affine and finite-level structures are strong and explicit; several infinite-level analytic passages remain conditional; and the celebrated dynamical conjecture is a target of the programme, not a conclusion of this edition.

12. CANONICAL SOURCE REGISTER

This register fixes the bibliographic identity, version policy, and evidential role of every source used to define Siegel’s canonical research corpus. A “lineage” below is an arXiv identifier, not a claim that every version under that identifier is the same mathematical text. Likewise, an author-site “last modified” date is provenance metadata, not a rule of mathematical precedence.

12.1. Authority order. Precedence is decided claim by claim, in the following order.

- (1) An explicit authorial correction or forward convention controls the object it names. Thus 2601.17030v3 controls the definitions of Hydra maps and numens: it warns that earlier definitions, including the dissertation’s, may be inconsistent and declares its definitions the standard going forward. For frames, F-series, and rising-continuity, 2307.00436v1 explicitly takes precedence where it conflicts with the dissertation.

- (2) Subject to (1), the latest substantive, non-withdrawn arXiv version controls the theorem stated by that logical work. A metadata-only withdrawal revision contains no replacement mathematics.
- (3) The dissertation supplies the largest continuous exposition and all material not replaced by a later specific source. Its three public witnesses are collated rather than silently identified.
- (4) A journal-layout PDF is a same-work witness. It is not presumed to be textually identical to the arXiv source, and typography alone does not make it a later mathematical correction.
- (5) An author-site post controls its own expository wording and any material unique to that post. It does not supersede an arXiv theorem merely because the page reports a later modification date.
- (6) A materially distinct earlier version remains part of the critical corpus. A withdrawn work remains part of the historical corpus, but it cannot certify a current theorem.

This order fixes source selection; it does not certify truth. A controlling witness may still contain a typographical error, an omitted hypothesis, or a proof sketch, each of which is recorded without alteration.

12.2. The ten arXiv lineages. Submission dates below are UTC dates in the archived arXiv metadata. The similarity figures are derived TeX-line routing metrics, not altered source witnesses. The reproducible procedure is the following. For the first TeX file in each version profile, `scripts/profile_version_deltas.py` reads UTF-8 text with replacement of decoding errors, splits it into lines, deletes the suffix beginning at an unescaped percent sign, replaces each nonempty whitespace run by one space, trims the result, and discards empty lines. If the resulting line sequences have lengths N_1, N_2 and the matching blocks returned by Python's `difflib.SequenceMatcher` with `autojunk=False` contain M lines, the reported value is

$$\frac{2M}{N_1 + N_2}.$$

The inputs are `public_records/maxwell_siegel_arxiv_2026-08-16.json` and `dossiers/_structural/LATEX_PROFILE_INDEX.jsonl`; exact compared paths, source SHA-256 hashes, line counts, edit counts, and full-precision values are in `dossiers/_structural/VERSION_DELTAS.jsonl`, whose SHA-256 is 40520bb04b54100bbc3a640f752c9fba0fbba4378545b3ad8b01dbb459d22647. This derived diagnostic only orders collation work. It neither changes a source nor proves semantic equivalence, semantic difference, or permission to merge witnesses.

Lineage	Work	Witness history	Canonical treatment
1909.09733	<i>Conservation of Singularities in Functional Equations Associated to Collatz-Type Dynamical Systems; or, Dreamcatchers for Hydra Maps</i>	v1, 20 Sep. 2019; v2, 4 Oct. 2019; v3, 19 Oct. 2019. Similarities: v1–v2, 0.6041; v2–v3, 0.8736.	Use v3; retain v1 and v2 as substantial same-title witnesses. This is the earlier unit-disk/profinite dreamcatcher branch, not a chapter-version of the dissertation's central (p, q) -adic branch.
2002.12762	<i>Fourier Analysis of the Parity-Vector Parameterization of the Generalized Collatz $px + 1$ maps</i>	v1, 24 Feb. 2020, substantive paper; v2, 14 Dec. 2023, metadata-only withdrawal.	Withdrawn as obsolete and under the wrong subject classification. Preserve v1 only as a parity/Fourier precursor and historical witness.

Lineage	Work	Witness history	Canonical treatment
2007.15936	<i>The Collatz Conjecture & Non-Archimedean Spectral Theory: Syracuse Random Variables Part I—Arithmetic Dynamical Systems and Non-Archimedean Value Distribution Theory</i>	v1, 19 Jul. 2020, titled <i>Syracuse Random Variables and the Periodic Points of Collatz-type maps</i> ; v2, 28 Mar. 2023; v3, 24 Apr. 2023. Similarities: v1–v2, 0.0248; v2–v3, 0.9790.	Use v3 for the shortened- $qx + 1$ numen and Correspondence Principle. Treat v1 as a materially distinct predecessor, not a minor draft. Collate the journal-layout PDF and Web installments I–II.
2111.07882	<i>Functional Equations Associated to Collatz-Type Maps on Integer Rings of Algebraic Number Fields</i>	v1, 10 Nov. 2021, substantive paper; v2, 16 Apr. 2022, metadata-only withdrawal.	Withdrawn for an error in Section 3 concerning a quotient of a ring of algebraic integers modulo an ideal. Historical only; do not use it to certify the number-field extension.
2111.07883	<i>The Collatz Conjecture & Non-Archimedean Spectral Theory: A p-Adic Characterization Part I.5—How To Write The (Weak) Collatz Conjecture As A Contour Integral</i>	v1, 10 Nov. 2021, titled <i>of the Periodic Points of a Class of Collatz-Type Maps on the Integers</i> ; v2, 15 Dec. 2023. Similarity 0.0551.	Use v2 for the Dirichlet/Mellin–Perron branch. Preserve v1 separately as a substantially different foundational witness to the Correspondence Principle.
2208.11082	<i>The Collatz Conjecture & Non-Archimedean Spectral Theory: Part II—(p, q)-Adic Fourier Analysis and Wiener’s Tauberian Theorem</i>	v1, 18 Aug. 2022, titled <i>Wiener Tauberian Theorems and the Value Distributions of (p, q)-adic Functions</i> ; v2, 20 Jun. 2025. Similarity 0.2870.	Use v2; preserve v1 as a materially distinct shorter witness. Collate Web installment III.
2307.00436	<i>Infinite Series Whose Topology of Convergence Varies From Point to Point</i>	v1, 1 Jul. 2023.	Use v1 for frames, F-series, rising-continuity, and variable topology. It expressly takes precedence over conflicting dissertation terminology.
2412.02902	<i>(p, q)-adic Analysis and the Collatz Conjecture</i>	v1, 3 Dec. 2024; arXiv witness of the 2022 University of Southern California dissertation.	Use the TeX as the machine-readable backbone, collated with the August and corrected October 2022 site PDFs. Later specific sources override it only within their declared scope.
2507.13358	<i>Algebras of p-Adic Distributions Induced by Pointwise Products of F-Series</i>	v1, 1 Jul. 2025.	Use v1. The March 2025 Kansas talk is a precursor and expository witness, not a replacement source.
2601.17030	<i>The Hydra Map and Numen Formalisms for Collatz-Type Problems</i>	v1, 19 Jan. 2026; v2, 29 Jan. 2026; v3, 12 Feb. 2026. Similarities: v1–v2, 0.9691; v2–v3, 0.9935.	Use v3 as the present definition authority and global-field manual. The v3 metadata states that minor errors in Proposition 7.6 were rectified.

The changed titles and direct collation of 2007.15936v1, 2111.07883v1, and 2208.11082v1 require separate witness records. The routing values above only prioritize that collation: neither a shared arXiv identifier nor a line metric is a semantic merge instruction.

12.3. The Web installments and the paper-series numbering. The author site’s four-part numbering is not the paper-series numbering. To prevent false citations, this edition always prefixes the site objects with *Web installment*.

Site label	Exact title	Page dates	Relation and status
Web I	<i>The Collatz Conjecture & Non-Archimedean Spectral Theory–Part I–The Numen of a Hydra Map</i>	Published 2022-08-09 21:45:22 UTC; page reports modification 2024-07-11 21:24:00 UTC.	Expository witness for the construction half of paper Part I, arXiv:2007.15936v3. It is not an additional arXiv Part I.
Web II	<i>The Collatz Conjecture & Non-Archimedean Spectral Theory–Part II–The Correspondence Principle</i>	Published 2022-08-10 20:55:28 UTC; page reports modification 2022-10-11 19:55:07 UTC.	Expository witness for the Correspondence-Principle half of <i>paper Part I</i> . Its final Dirichlet/Perron aside is a precursor to paper Part I.5, arXiv:2111.07883v2. It is not paper Part II.
Web III	<i>The Collatz Conjecture & Non-Archimedean Spectral Theory–Part III–(p, q)-adic Fourier Analysis & Tauberian Spectral Theory</i>	Published 2022-08-12 22:51:30 UTC; archived page reports modification 2025-12-26 22:12:20 UTC.	Evolving site exposition overlapping <i>paper Part II</i> , arXiv:2208.11082v2. Unique material may be used after theorem-level collation; its modification date alone confers no authority.
Web IV	<i>The Collatz Conjecture & Non-Archimedean Spectral Theory–Part IV–A Fourier Analysis of the Numen of the Shortened $qx+1$ map</i>	Published 2022-08-15 19:14:38 UTC; archived page reports modification 2025-12-26 22:13:00 UTC.	Site-only numen/Fourier application, chiefly parallel to the dissertation’s one-dimensional Fourier material. There is no corresponding arXiv paper Part IV in this corpus.

The archived Web I and Web II objects are current snapshots acquired on 23 August 2026. Word-Press’s public endpoint did not expose their revision histories without authentication. Consequently their published and modified timestamps do not identify the contents of every intervening revision.

12.4. Dissertation and supplemental author-site witnesses. The dissertation has three distinct public witnesses:

- (a) the August 2022 author-site PDF, 476 pages;
- (b) the October 2022 author-site PDF explicitly labelled corrected, 478 pages; and
- (c) arXiv:2412.02902v1, submitted 3 December 2024, 467 pages, with an extractable TeX source.

Their normalized texts and pagination differ. The arXiv TeX is therefore an editable base, not evidence that the explicitly corrected October PDF is a byte-for-byte or proposition-for-proposition predecessor. Any difference used in the edition must be collated at the relevant passage.

The remaining author-site witnesses have the following roles.

Part I journal-layout PDF.:

Same-work layout witness for 2007.15936; use for publication pagination and typographic variants only after comparison with v3.

Infinite-series journal-layout PDF.:

Same-work layout witness for 2307.00436; it does not silently replace the available TeX.

Arithmetic Properties of F -series talk notes.:

March 2025 Emporia State University talk; supplementary precursor to 2507.13358.

Episode 1 script.:

June 2024 companion exposition for the dissertation and the (p, q) -adic programme; useful orientation, not a primary theorem source.

 p -adic L -functions seminar and Fourier-series notes.:

Siegel-authored background with high contextual relevance. These are supplementary notes, not parts of the Collatz research lineage.

12.5. Corpus boundary. The core edition includes the ten arXiv lineages, all materially distinct versions identified above, the four Web installments, the three dissertation witnesses, the two same-work layout witnesses, and the research-specific talk and companion script. The p -adic L -function seminar and Fourier-series notes are Siegel-authored apparatus, not core research works. Authorship and core status are separate questions. Siegel's generic course notes in introductory real analysis, linear algebra, vector calculus, complex variables, partial and ordinary differential equations, and probability remain Siegel-authored works, but they are outside the Collatz, Hydra, numen, dreamcatcher, and F -series research corpus.

External supporting literature is not part of Siegel's corpus, but it is used with exact citations to establish prior art, standard theorems, and comparison maps.

12.6. External prior-art comparison ledger. The following sources are outside Siegel's corpus. They are present because an exact critical edition must distinguish Siegel's added structure from an older identical special case, and must distinguish a shared value formula from an identity of maps. Each row records the strongest comparison proved in this edition and the boundary beyond which the citation may not be used.

Object	Primary source	Exact proved overlap	Nonidentity boundary
Scalar integer Hydra at displayed modulus $D \in \mathbb{Z}_{\geq 2}$	Matthews, Sec. 2 [17]	Admissibility is equivalent to unique integers $m_i \neq 0$ and $\rho_i \equiv im_i \pmod{D}$ with $H_i(x) = (m_i x - \rho_i)/D$; see theorem 4.7. Exact-integrality is equivalent to the displayed-presentation condition $\gcd(m_0 \cdots m_{D-1}, D) = 1$.	Matthews attaches relatively prime type to the least-modulus presentation. Thus exact-integrality is equivalent to Matthews relatively prime type only together with $D = d_{\text{Mat}}(H)$. A nonminimal refinement may destroy exact-integrality. Neither comparison identifies Siegel's non-scalar or global-field structure or adds centeredness or properness.

Object	Primary source	Exact proved overlap	Nonidentity boundary
Residue-class-wise affine mapping	Kohl, Defs. 1.1 and 1.4 and Rem. 1.2 [18]	Every scalar integer Hydra in the preceding row is rcwa, and $d_{\text{Mat}}(H) = \text{Mod}(H) \mid D$; under exact-integrality both least moduli equal D ; see theorem 4.8.	The group $\text{RCWA}(\mathbb{Z})$ contains only bijective rcwa maps. The shortened Collatz map is rcwa but is not injective; a Hydra is therefore not automatically an rcwa group element.
Formal affine word and $q = 3$ return candidate	Böhm–Sontacchi, Prop. 1; printed Props. 4–5; unnumbered paragraph after Prop. 5 on p. 264 [19]	Proposition 1 specializes to the valid formal composition in equation (62); at $q = 3$, its fixed point is the rational $R_{3,\varepsilon}$ printed as the Proposition 5 candidate after the explicit renaming $n = L$.	Printed Proposition 4’s unrestricted identification with $u^L(x)$ is false; it may be made only for a matching itinerary. The unnumbered p. 264 paragraph allows exact period properly dividing L . The final strict position inequality is reconstructed from the formal run data, not inserted into printed Proposition 5.
2-adic parity conjugacy	Bernstein–Lagarias, Eqs. (1.3)–(1.6), (4.1)–(4.2), and App. B [20]	The chronological parity encoder Q_q , inverse conjugacy Φ_q , and every odd- q classical specialization are exact. For odd prime q , the Φ_q value is $\iota_2(R_{q,\varepsilon})$ and the reversed χ_q value is $\iota_q(R_{q,\varepsilon})$ as in theorem 6.5.	Q_q and Φ_q retain their inverse roles in $\mathbb{Z}_2$; for odd prime q , neither is the cross-adic map $\chi_q : \mathbb{Z}_2 \rightarrow \mathbb{Z}_q$. Composite odd q in the classical conjugacy does not enlarge the edition’s prime- q numeren theorem.
De Bruijn graph presentation	Laarhoven–de Weger, Thms. 3.1 and 5.1 and Sec. 5.2 [21]	Their graph isomorphism is the chronological parity map denoted Q_3 here; their odd- $an+b$ analogue is an exact classical bridge.	Their paper’s symbol Φ has the orientation of this edition’s Q_3 , not Bernstein–Lagarias’s Φ_3 ; its p -ary discussion uses appropriately chosen coefficients and is not an arbitrary- Hydra theorem.

12.7. Source-issue ledger.

Code	Source	Canonicalization issue
SR-01	Corpus index	A shared logical-work identifier records bibliographic lineage, not semantic sameness. The reproducible routing values 0.0248, 0.0551, and 0.2870 do not establish that distinction and do not authorize a merge; the separately archived versioned sources and direct collation control witness status.
SR-02	1909.09733	A common title is not an identity proof. Versions v1 and v2 remain separately archived in the apparatus even though v3 is the base; the value 0.6041 is only a reproducible collation-routing diagnostic.

Code	Source	Canonicalization issue
SR-03	2002.12762	v2 is a metadata-only withdrawal declaring the paper obsolete and under the wrong subject classification. It supplies no corrected paper; v1 is historical only.
SR-04	2111.07882	v2 withdraws the work for an error in Section 3 about a quotient ring. The archived comment contains the contradictory phrase “was not incorrectly described”; the withdrawal and existence of an error are clear, but the intended corrected assertion must not be guessed.
SR-05	2601.17030	The paper expressly replaces potentially inconsistent earlier Hydra definitions. v3 also reports corrections to minor errors in Proposition 7.6. Earlier Hydra terminology is historical unless re-derived in the v3 formalism.
SR-06	2307.00436	The paper expressly takes precedence over the dissertation where their frames/F-series terminology conflicts. It is more general in some respects and less comprehensive in others; only the conflicting definitions are superseded.
SR-07	Dissertation	The August 2022, corrected October 2022, and December 2024 arXiv witnesses are textually distinct. “Later upload” and “corrected” are recorded attributes, not a license to discard the other witnesses.
SR-08	Web numbering	Web II is part of the web sequence but belongs chiefly to arXiv <i>paper Part I</i> ; Web III overlaps arXiv <i>paper Part II</i> . Bare labels such as “Part II” are therefore prohibited in source-critical references.
SR-09	Web I	The current snapshot calls <i>numen</i> Greek whereas paper Part I correctly identifies it as Latin; its displayed q -adic decay formula omits the subscript q on the norm; and its uniqueness proof says equality on $\mathbb{Z}_q$ where the input domain must be $\mathbb{Z}_2$. Paper Part I v3 controls these points. Hash-bound locator: <code>author_site/wordpress_2026-08-23/part_i_numen_hydra_map_2024-07-11.txt</code> , SHA-256 <code>6dfcf3ee7c4056978deae669094e08f41acbfbdf7f1410b2cafb3d58293f682ae</code> , lines 143, 271–275, and 327–329.
SR-10	Web II	Claim 5’s statement omits the conclusion $\chi_q(z/2) = 0$, although its proof supplies it. The page calls its terminal periodic-point formulation Version 3, while paper Part I v3 canonically labels its corresponding published formulation Version 4. Its divergent result says “belongs to a divergent trajectory”; paper Part I v3 states directly that the value is a divergent point. Hash-bound locator: <code>author_site/wordpress_2026-08-23/part_ii_correspondence_principle_2022-10-11.txt</code> , SHA-256 <code>b5d37b71f4778109fbd853f621b0562c6aeca1487141968d2cde3887c8ba46c6</code> , lines 283–287, 353–355, and 359–373.
SR-11	Web I–IV	Publication and modification timestamps are provenance only. No page supersedes a formal source solely because its modification date is later, and the public snapshots do not furnish a complete revision history.
SR-12	Web III–IV	These are active, evolving expositions. Statements unique to them require theorem-level checking before incorporation; overlap with paper Part II or the dissertation is not presumed identity.
SR-13	Layout PDFs	The Part I and infinite-series layouts are distinct files and must be collated for wording and theorem numbering. They are not replacement TeX sources merely because they resemble journal typography.

Code	Source	Canonicalization issue
SR-14	2111.07883v2; Web II	In the calculation of $f_q(z^2, x)$ immediately before the printed ordinary generating-function equation, the first term is written as $z/(1-z)$ instead of $z^2/(1-z^2)$. The resulting Dirichlet functional equation is false for $x \neq 0$, and its $q = 3$ residue formulas contain spurious x -terms. Part I.5 and the public Web II nevertheless identify the continuation programme. Chapter 7 recovers it by deriving the corrected equation, proving normal convergence and global strip-by-strip continuation, and recalculating the principal parts. Denominator zeros remain candidate poles unless noncancellation is proved.
SR-15	Böhm–Sontacchi (1978)	Printed Proposition 4 is false under its stated unrestricted quantifiers. With $x = 0$, $m = 1$, and $n_0 = n_1 = 1$, it gives the right side $1/4$, while $u^3(0) = 0$; the right side is instead the formal composition $f \circ g \circ f(0) = 1/4$. The edition therefore derives the formal affine identity from Proposition 1, invokes u^L only after proving actual-itinerary correctness, and recovers the Proposition 5 fixed-return candidate independently. The possibility of exact period properly dividing L is located in the unnumbered paragraph after Proposition 5 on p. 264, not attributed generically to Propositions 4–6.

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